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Readings in Mathematics

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Fernando Q. Gouvêa

# A Short Book on Long Sums

## Infinite Series for Calculus Students

Springer

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# Readings in Mathematics

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Fernando Q. Gouvêa

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## Infinite Series for Calculus Students

Springer

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# What This Book Is About

The goal of this short book is to introduce you to power series, a fundamental tool of pure and applied mathematics, and to infinite series in general. There are two major themes throughout, which might be described as *approximation* and *representation.* Throughout we will work with a function $f(x)$ of one variable. The book focuses on two questions:

a. Given enough information about the function at a reference point $x = a$, is it possible to know, at least approximately, what the function does when $x$ is near $a$?

b. Can we use these approximations to find a representation of the function $f(x)$ that is (perhaps) more useful than its original one?

To illustrate the two problems and what an answer would look like, think about the decimal expansion of a fraction. Suppose we have a fraction $f = 1/3$. Then we know that

$$f = \frac{1}{3} = 0.3333333\ldots$$

This answers both questions. If we want an *approximation* to the fraction $f$, we simply truncate the expansion at some point. So, 0.333333 is an approximation to $1/3$; counting the digits, we can even conclude that the approximation is off by no more than $10^{-6} = 0.000001$.

On the other hand, the decimal expansion, taken as an infinite whole, is also a *representation* of the fraction $1/3$. The decimal expansion represents a unique real number that is exactly our original $1/3$, so in a sense it "knows" everything the expression $1/3$ does. Its form easily reveals things to us (for example, that $1/3$ is between $3/10$ and $35/100$) that might have been harder to see from the fraction representation.

The decimal representation is in fact so useful that it sometimes replaces the "more exact" fractional representation. This is true for functions too!

In what follows we will work with a function $f(x)$ instead of a fraction. That makes things harder, of course, because a function has many values (one for each allowable $x$). So when we approximate it or represent it, we will have to worry about the *range of validity*: for which values of $x$ is it a good approximation or a valid representation? On the other hand, we can use the tools of calculus (especially derivatives) to study a function very closely, and that will help us find our approximations/representations.

We will focus almost entirely on one specific kind of approximation, the kind known as a *power series.* This is not the only kind that is important; it is simply the easiest and oldest kind. Our goal is to understand how to find a power series for a given function and how to use the series when we have it. Like the decimal representation of a fraction, our power series will usually be *infinite.* Indeed, one of the names for the subject that includes power series is "infinite series," even though most mathematicians use the word "series" by itself to mean the same thing.

One big contrast between fractions and functions is that every fraction has a decimal representation, but not every function has a power series representation. In fact, functions that do are some of the nicest and most well-behaved functions.

Power series are used a lot, by engineers, physicists, and mathematicians of all kinds. Anyone interested in applied mathematics needs to know and understand them. So do future mathematicians. Real understanding goes beyond the "here's how to do it" approach of many textbooks. I hope this book will get you started in that direction.

We begin with a careful discussion of the first-order approximation to a function given by its tangent line. (Of course, this assumes the function *has* a tangent line—a minimal form of "good behavior" for functions.) For some readers, this may be a review, but I hope that seeing it again will help prepare you to move to higher-order approximations.

As we go along, I will either prove things or tell you that certain things are true. In mathematics, something we have shown to be true is called a *theorem.* To help signpost these critical conclusions I have labeled them as theorems and given them numbers.

Mathematics today is done with the help of technology. This is especially true when we want to do computations, including algebraic manipulation. I will use a very powerful public-domain program called *SageMath.* See Appendix A for some basic information on how to get it and how to use it. Almost all the computations you will need to do can be done online on the SageMath Cell site. Of course, other mathematical software can do these things too, be it *Desmos*, *Geogebra*, *Mathematica*, *Maple*, or MATLAB. If you are used to one of those, I think you will have no problem translating my examples to your preferred tool. I like SageMath because it is free, very powerful, and can be used online without installing anything on your computer.

We often learn best, however, by first doing some computations by hand. So we will begin that way, and then I will show you how to do those computations with SageMath as well.

As in every mathematics book, there are problems. I have avoided including many "just do it" problems. You won't find 20 almost identical exercises. Instead, I have tried to provide problems that are interesting, that require some thought, like the problems you are likely to encounter in the real world.

This book has a narrow focus and does not do many proofs or go into technical issues. I have included references to other books that can provide these details if you want to see them. A reference like [4] points you to an item in the bibliography, in this case *Lion Hunting and Other Mathematical Pursuits*, by Ralph P. Boas, which gave me some interesting ideas I used in this book.

My way of introducing you to infinite series is different from the usual one. Professors reading this introduction and wanting to understand why I have done it the way I did it should read Appendix B.

Power series are critically important, but of course there are other kinds of series representations for functions. In the last chapter, I have tried to point you towards some of them. I hope that you will find learning about series useful and that you will be interested in going further.

Enjoy the ride!




# 1 Getting Close with Lines

Sunset yesterday was at 7:27. What time do you think sunset will be today?

Of course, that isn't enough information for an exact answer. We don't even know whether we expect sunset today to be earlier or later. (Is it Spring or Fall?) But we can still give a fairly good approximate answer. (Can you see what it is before I tell you?)

The sunset question is a typical approximation problem of the sort we will be considering. Here's how to translate it into a mathematics problem. We have a function we want to understand:

$$\text{ss}(x) = \text{ time of sunset on day } x.$$

(We could have called the function $f(x)$ as usual, but it's often better to use something more memorable; here ss stands for "sunset.") We know one value. If we decide that yesterday was day zero, we know that $\text{ss}(0) = 7{:}27$. And we want to approximate the value of $\text{ss}(1)$.

What can we say? The first guess to make is that sunset today will be at about the same time: $\text{ss}(1) \approx 7{:}27$. That is reasonable because we know that the time-of-sunset function doesn't change too fast: sunset today can't be too much later or earlier than yesterday. As long as we are trying to find an approximation for a value of $x$ that is *close* to our reference value $x = 0$, we expect there to be little change. Of course, what counts as close depends on the problem. For the time of sunset, one or two days is close; for a speeding car, we'd want one or two seconds.

Can we say more? Well, we can if we have a little more information. Suppose we know that it is Fall. Then we know that the length of the day is *decreasing*, so sunset is getting a little bit earlier each day. In math talk, we are saying that the derivative $\text{ss}'(x)$ is negative right now. So we might add to our approximation that 7:27 is likely an *overestimate*: the true value should be a little bit less. In the same way, if we knew it was Spring, we would know we have an *underestimate*.

Can we do even better? In order to be more precise, we would need to know even more. For example, if we knew the actual *value* of $\text{ss}'(0)$ (so that right now sunrise is getting earlier by $\text{ss}'(0)$ minutes per day), we might be able to make our guess a bit more precise, or at least get more information about how far off our guess might be.

Our goal in this chapter and the next is to push this idea as hard as we possibly can. We will always have a reference point and we will assume we know a lot of information about what is happening there. We will try to use this information to come up with a reasonable approximation for what is happening at some nearby point.

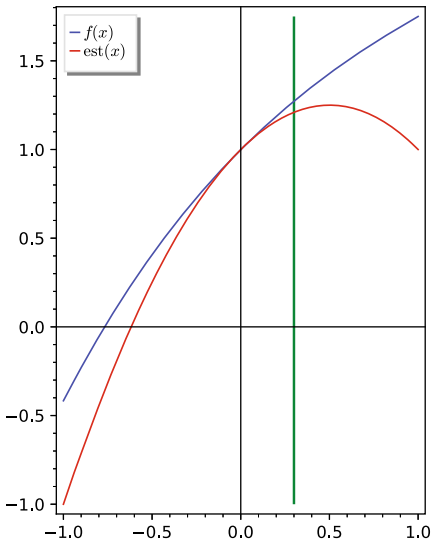
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## 1.1 Approximation: the general setup

Since we will be discussing approximations, it is good to have some vocabulary to talk about them. Let's create a general setup and give names to various quantities we will need to worry about.

- We are given a *function* $f(x)$ whose values we want to approximate. In the graph, I have shown a function $f(x)$ in blue.
- We have a *reference point* $x = a$ for which we know something. In the sunset example, we have $a = 0$ and we know the value ss(0). In the graph, I chose $a = 0$ and we can see that $f(0) = 1$. In chapter 2, we will sometimes refer to $a$ as the *center*, for reasons that will become clear.




- We have a *target point* $x$ for which we want to estimate $f(x)$. I sometimes use the more colloquial "plug-in point." In the sunset example, the target point was $x = 1$. In the plot, I have chosen $x = 0.3$ and drawn a vertical line in green.
- An important bit of information is the difference between our reference point and our target point, $x - a$. We will refer to this as the *increment*. In the sunset example, the increment was 1; in the plot, it is 0.3. It's pretty clear that this kind of approximation will be easier when the increment is small.
- Instead of finding $f(x)$ exactly (usually, we can't do this because we don't have a computable formula for $f(x)$), we will instead estimate it. That is, we will try to come up with some other function est$(x)$ such that $f(x) \approx \text{est}(x)$ when $x$ is near $a$. In the sunset example, we just made est$(x)$ be a constant function always equal to $f(a)$. In the plot, I have shown a possible est$(x)$ in red.
- Whenever we approximate there is an *error*: the difference between our estimate/guess and the real value of $f(x)$. So

$$\text{Err}(x) = f(x) - \text{est}(x).$$

  In the plot, this is the segment on the green line between the blue and red graphs. Often, we care more about the absolute value of the error than about its sign.

In most cases, the hard part is to *control* the error. That is, we want to somehow obtain a formula that says that the absolute value of the error can't be too big:

$$\left|\mathrm{Err(x)}\right| \leq B.$$

The number $B$ is called an *error bound*, because it sets a boundary on the error. In our examples, it will always depend on the increment $x - a$.

An everyday example of error bounds is the "margin of error" that is often given when some survey attempts to measure public opinion. When we are told that the margin of error is 3 percentage points, that means that the number reported has an error that is no bigger than 3. So when we are told that 95% of people like fountain pens, the true value might be as low as 92% or as high as 98%. We want to make sure that our estimates come with a margin of error.

The estimates we will be developing will depend on what we know about the function at the reference point $x = a$ and on the increment $x - a$. The error bounds will typically depend on knowing something about what happens *between* $a$ and $x$. That makes sense: if something dramatic happens as $a$ becomes $x$, there might be a huge change, so a big difference between $f(a)$ (which we know) and $f(x)$ (which we are estimating). Huge differences are likely to cause errors, and so lead to large values of $B$.

What makes an estimate good or bad? It all boils down to the error, which we hope is small. But we need to make that a little more precise as well. For one thing, the position of the vertical green line matters. In other words, the error depends on $x$, and we should take that into account.

If our estimate $\mathrm{est}(x)$ is any good, it should give $f(a)$ exactly when we plug in $a$. (If we can't predict the past, we are in trouble!) That is: we expect that $\mathrm{est}(a) = f(a)$, or, equivalently, that $\mathrm{Err}(a) = 0$. That happens in our graph.

We have the right to expect a bit more, I guess. As $x$ gets closer to $a$, we would expect (or at least prefer[1]) our estimate to get *better*. That happens in our example graph: imagine moving the green line closer to the $y$-axis.

We can express the behavior we want as a limit: we would like

$$\lim_{x\to a} \mathrm{Err}(x) = \lim_{x\to a} \big(f(x) - \mathrm{est}(x)\big) = 0.$$

In other words, we want our error to go to zero as $x$ approaches $a$. In fact, we will even be concerned to find out *how fast* the error tends to zero.

## 1.2 The horizontal line approximation

Let's look back at the "just guess it's constant" example and translate it into the general language we have just learned. We did it for the "time of sunset" function, but let's now do it for a general function $f(x)$ and a reference point $x = a$.

[1] This idea works well for the kind of approximations we have in mind, but there are other options too. For example, we might prefer an approximation that is "good on the average" even if it does not have this limit property. But for now we'll stick to this preference.

We know $f(a)$ and we decide to estimate by saying "it won't change," so we set $\text{est}(x) = f(a)$, a constant function whose graph is a horizontal line. So we have

$$\text{Err}(x) = f(x) - \text{est}(x) = f(x) - f(a).$$

Clearly, if we plug in $a$ for $x$, we get zero, so the first condition is true: we get the right answer when $x = a$.

How about the limit? Well, what we want is

$$\lim_{x \to a}(f(x) - f(a)) = 0,$$

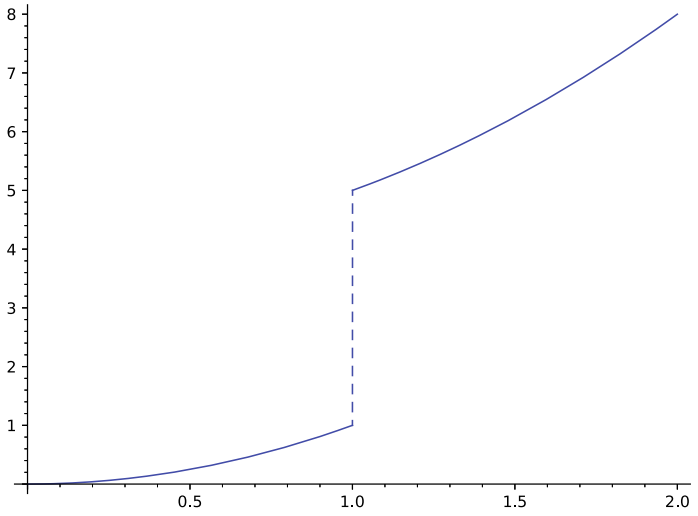
which is the same as

$$\lim_{x \to a} f(x) = f(a).$$

That will happen when $f(x)$ is *continuous*. (It's actually the definition of continuity.) So:

**Theorem 1.2.1.** *The "just assume it's constant" approximation is acceptable exactly when the function $f$ is continuous at $x = a$.*

If you think about some discontinuous functions you will see that if we don't assume continuity the situation is hopeless. For example, if we had a function that jumped at $x = a$ then knowing $f(a)$ would give little or no information about nearby values like $f(a + \text{small})$.




In the graph, $f(1) = 1$, but the value of $f(1.01)$ is close to 5. The jump could have been of any size, and just knowing what happens when $x = 1$ gives us no information at all about $f(x)$ when $x > 1$.

The fact that continuity is required for the "guess it is constant" estimate to work is a hint for the work we will do later: good estimates require good properties of the

function $f(x)$. The properties that matter are exactly the properties that calculus is all about: being continuous, having a derivative, and so on. In the case of the degree zero approximation we are talking about, continuity is what we need, but for better approximations, we will need more.

Can we control the error in our approximation? Well, not if all we know is that $f(x)$ is continuous. But if we know it is differentiable, we can. What we need is one of the big calculus theorems you learned but perhaps never saw the point of: the Mean Value Theorem. Remember what it says:

**Theorem 1.2.2** (Mean Value Theorem)**.** *If a function $f$ is continuous and differentiable between $a$ and $x$ then*

$$\frac{f(x) - f(a)}{x - a} = f'(c)$$

*for some value $c$ between $a$ and $x$.*

The thing in the numerator is exactly the error we want to understand, $\mathrm{Err}(x) = f(x) - f(a)$. So let's clear the denominator to get

$$\mathrm{Err}(x) = f(x) - f(a) = f'(c)(x - a).$$

That looks like an exact formula for the error, but the problem (as always when we use the MVT) is that we don't know what $c$ is. How can we get around that?

The key is to find a bound for what $f'(c)$ can be. Suppose we know that the absolute value of $f'(c)$ is never more than 5, so $|f'(c)| \leq 5$. Then, taking absolute values, we see that

$$|\mathrm{Err}(x)| = |f(x) - f(a)| = |f'(c)|\,|x - a| \leq 5|x - a|,$$

which tells us that the error is never more than five times the increment. That is very nice: the closer we get, the smaller the error, and we can actually give a margin of error for our estimate.

In general, if we can show that $|f'(c)| \leq M$ for *all* the relevant values of $c$ (that is, all values between $a$ and $x$; we don't care about any other values). Then we get

$$|\mathrm{Err}(x)| = |f(x) - f(a)| = |f'(c)|\,|x - a| \leq M|x - a|.$$

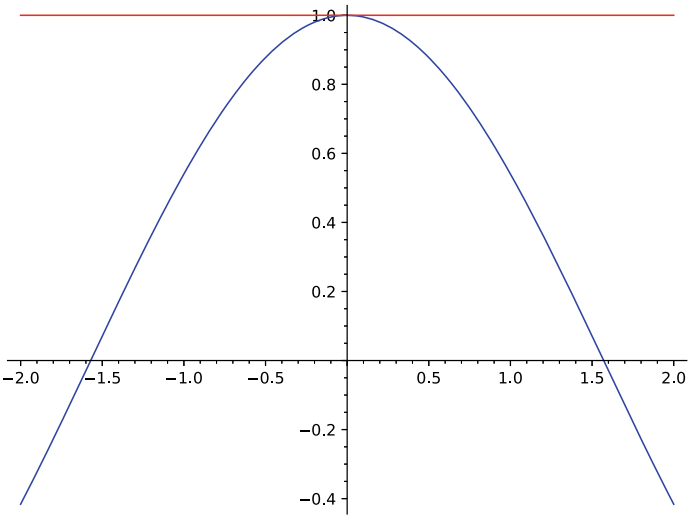
So the mean value theorem tells us that if we can control how big the derivative is between $x$ and $a$, then we can also control how big our error is, and the error bound will look like $M|x - a|$. This makes sense: knowing how fast the function changes will allow us to put a limit on how much it can change.

As an aside, this is the whole point of the Mean Value Theorem! It allows us to turn information about how the derivative behaves in a certain range into information about how much the function changes.

Let's finish with a couple of examples. For a convincing example, we want a function we don't actually know how to compute by hand, since for the ones we can compute we don't need to approximate. Trigonometric functions are like that: we can

get them with a calculator or computer, but most of us wouldn't want to (or know how to) compute them by hand. Suppose $f(x) = \cos(x)$. Then I know $f(0) = \cos(0) = 1$, but the only way I know to compute $\cos(0.1)$ is to ask a computer. But let's try to see what our "make believe it's constant" estimate tells us.

First a picture. Here is a graph of $y = \cos(x)$ (in blue) together with the constant function $y = 1$ (in red).




You can see from the picture that the approximation is not bad near the reference point $x = 0$, but gets worse fairly quickly as we move away from that point.

Our estimate would be that

$$\cos(0.1) \approx 1.$$

The error, then, will be

$$\mathrm{Err}(0.1) = \cos(0.1) - \cos(0) = \cos(0.1) - 1,$$

which we can see will be negative, because the horizontal line is above the graph. Can we control the size of the error?

The derivative of $\cos(x)$ is $\sin(x)$, which we know is never bigger than one. So

$$|f'(c)| = |\sin(c)| \leq 1;$$

that is, we can take $M = 1$. (The reason for choosing cosine as our first example is exactly to get an easy bound for the derivative.) Putting that into the MVT estimate tells us that the error is at most $1|0.1 - 0| = 0.1$. So we have shown that $\cos(0.1)$ is about equal to 1, with error at most 0.1. We could summarize it all in a single formula:

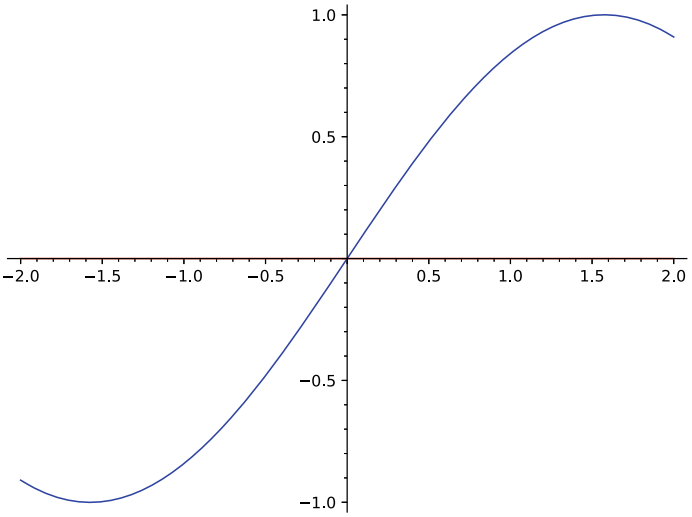
$$\cos(0.1) = 1 \pm 0.1.$$

Is that any good? Well, SageMath tells me

```
sage: cos(0.1)
0.995004165278026
sage: cos(0.1)-1
-0.00499583472197418
```

So, in fact, this worked very well: our value is off by only a small amount that is indeed smaller (a lot smaller!) than the bound we found.

That was sort of a success. What happens if we do sin(0.1) instead? Let $f(x) = \sin(x)$. This picture is very different: the red line is the $x$-axis now. The red is still there, but it's hard to see.




We would guess from the picture that the error will be a lot bigger this time. Let's see. Since $f(0) = \sin(0) = 0$, our "make believe it's constant" estimate would be $\sin(0.1) \approx 0$. The error, then, will be $\sin(0.1) - 0 = \sin(0.1)$.[2] The derivative of $\sin(x)$ is $\cos(x)$, which, as before, never bigger than one. So $|f'(c)| = |\cos(c)| \leq 1$; putting that into the MVT estimate tells us that the error is at most $1|0.1 - 0| = 0.1$, just as before. So we have shown that

$$\sin(0.1) = 0 \pm 0.1.$$

But this time, SageMath tells me that

$$\sin(0.1) = 0.0998334166468282,$$

which is also the error. So, while our estimate really is within 0.1 of the truth, the error is actually pretty big, very close to the maximum possible value 0.1. So we might want to do a bit better than this. We'll talk about how to do that next.

[2] It's a bad sign when your error is the same as the thing you are trying to compute, but let's plow on.

### 1.2.1 Problems

**Problem 1.2.1:** I said above that the "make believe it's constant" approximation doesn't work if the function $f(x)$ has jumps. Does the "time of sunset" function ever jump?

**Problem 1.2.2:** In the Maine portion of I-95, exits are labeled by mileage, from south to north. So exit 130 is three miles further north than exit 127. A car drives by exit 127 heading north at 70 miles per hour.

a. Why is it reasonable to estimate that the car will reach exit 130 in 2.6 minutes?

b. If, in fact, the car arrives there in 2 minutes, what would you conclude? What if it took 5 minutes?

c. What information would allow you to make a better prediction of the arrival time at exit 130?

## 1.3 Using the tangent line to approximate

As we pointed out in the previous section, we want a method for approximation that goes beyond the "just assume it's constant" estimate. The idea is to use the trend as well as the value at $a$. The trend is captured by the derivative, so it's easy to turn this idea into mathematics.

Suppose our function $f(x)$ is differentiable and we know not only the *value* $f(a)$ but also the *current rate of change* $f'(a)$. Then we can make a slightly better estimate by using that information.

To go back to our example, let's say that the sunset on August 26 was at 7:27. Suppose we also know that on that day the time of sunset was getting smaller (i.e., earlier) by about 2 minutes per day. Then if we want the time of sunset on August 28 we just subtract 2 minutes per day:

$$7{:}27 - (2 \text{ minutes per day}) \times (2 \text{ days}) = 7{:}23.$$

In math talk, that boils down to this: we know $f(a)$, the value at our reference point, and we also know $f'(a)$, the rate of change at our reference point. So the right way to approximate $f(x)$ is to start with $f(a)$ and then add the expected change as we move between $a$ and $x$. Since the rate of change is $f'(a)$, the result is

$$f(x) \approx f(a) + f'(a)(x - a).$$

It is really $\approx$ and not $=$, because the rate of change is $f'(a)$ *when* $x = a$, and for this estimate to be exactly right we would need to know that the rate is the same for all relevant values of $x$. So the formula really does give only an approximation: it would be exact if the rate of change were always the same (i.e., if the graph is a line), but that won't usually be the case.

We can phrase this idea in terms of increments: when we increment the input by $(x-a)$, we expect the corresponding change in the value of the function to be roughly $f'(a)(x-a)$. This amounts, of course, to assuming that the function is behaving like a straight line, with a constant rate of change. That may not be quite true, but we are hoping (implicitly) that it isn't too far from true, i.e., that the rate of change does not itself change too quickly.

Geometrically, the estimate we are making looks like this: we know a point in the graph of our function, namely, $(a, f(a))$. We also know the slope at that point: it is $f'(a)$. As you will remember, "slope at a point" means the slope of the tangent line at that point. So, we can use the point-slope formula to write down the equation of the tangent line:

$$y - f(a) = f'(a)(x - a),$$

or, rearranging a bit,

$$y = f(a) + f'(a)(x - a).$$

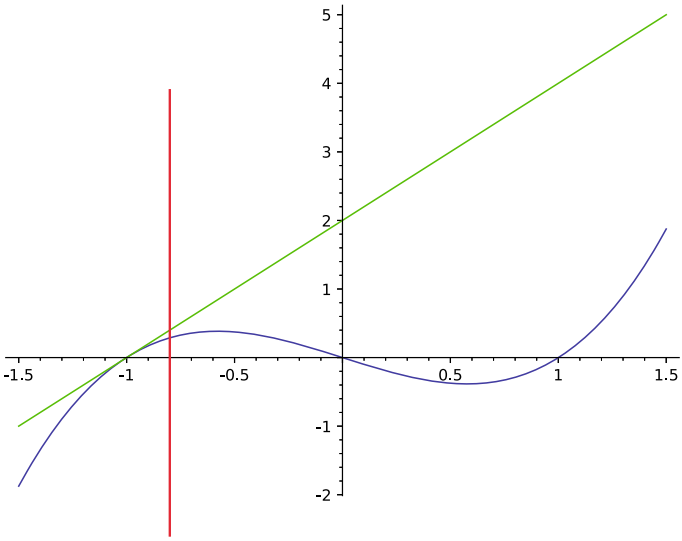
That's *exactly* our estimate!

So we can see our estimate easily in a picture: we are saying that the tangent line to the graph of a function stays close to the function, at least if we step away only a little from the reference point.

Let's graph an example. Take $f(x) = x^3 - x$ (in blue in the picture) and $a = -1$, so $f(a) = 0$ and $f'(a) = 2$. Suppose we want to estimate $f(-0.8)$. The equation of the tangent line (our estimate, in green in the picture) is

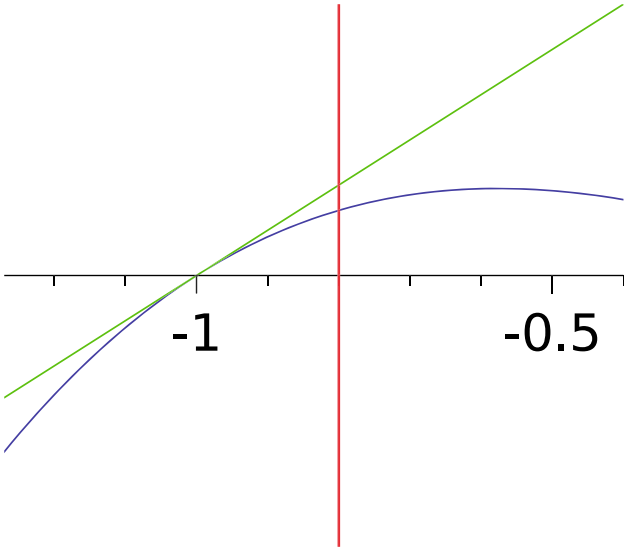
$$y - 0 = 2(x - (-1)),$$

which just gives $y = 2x + 2$.



I added a red vertical line at $x = -.8$. Our increment is $x - a = 0.2$, so our estimate is $f(-0.8) \approx 0 + 2(0.2) = 0.4$, the point where the green tangent line intersects the red line. The actual value (the point where the red line intersects the blue curve) is 0.288. The error is $-0.112$. Right now we don't have any way of controlling the error without actually computing it.

Let's zoom in on the portion of the picture that contains our approximation. The approximation is the green line. The actual value is given by the blue curve. The error is the line segment on the red line between the intersections with the green and blue graphs.




Here are some things to notice in the picture:

a. If we look at $x = -1$ the curve and the tangent line are at the same value. In other words, our estimate is exact when $x = a$, or $\mathrm{Err}(a) = 0$.

b. If we slide the red line closer to $x = -1$, the error gets smaller and smaller. In other words, $\lim_{x \to a} \mathrm{Err}(x) = 0$.

c. It is hard to decide from the picture how quickly the error goes to zero. We might try computing the error for various values of $x$ to see that.

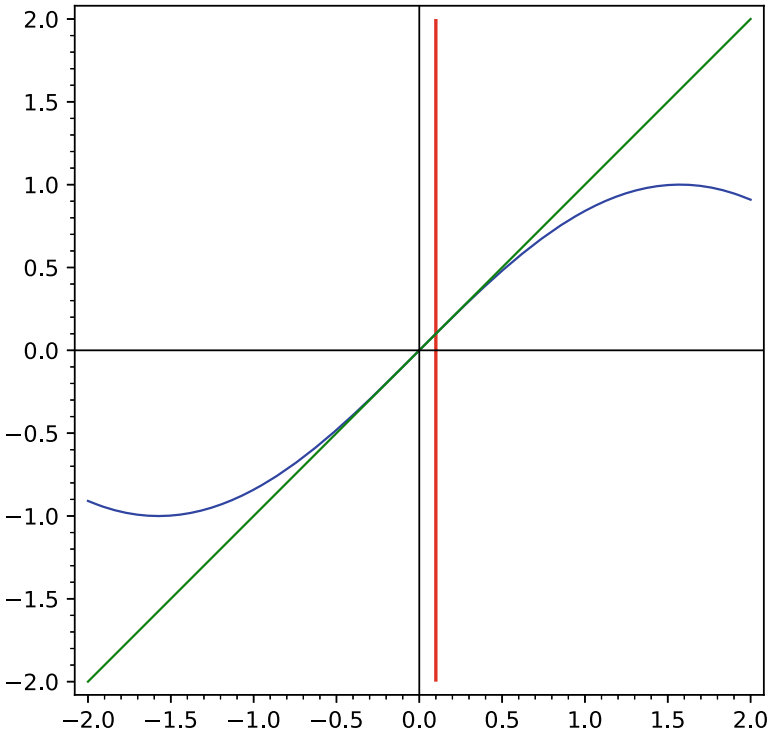
d. Our estimate is going to be too big, at least for $x$ near $-1$. That is because the tangent line we drew is *above* the blue graph. Can you see what feature of the blue graph causes that?

Of course, "noticing in the picture" is not proof that these things will always happen. We'll get to that.

Let's see what happens with the example $\sin(0.1)$ that we tried before. We know $\sin(0) = 0$ and $\cos(0) = 1$, so the tangent line approximation is

$$\sin(x) \approx 0 + 1(x - 0) = x.$$

Here is the picture, with the sine curve in blue, the tangent line in green, and the line $x = 0.1$ in red:




One can hardly see the error at this scale: along the red vertical line, the blue and green graphs seem to be at the same point. (I tried zooming in by a factor of ten, and still couldn't see a space between the two.)

Of course, we expect that there is a difference, even if it is very small. And indeed, the approximation we get is $\sin(0.1) \approx 0.1$, and the error is about $-0.000166$, much smaller than before. You should check that this picture also shows the three features we noted above.

To summarize:

**Tangent Line Approximation:** Suppose that we are given a differentiable function $f(x)$ and that, for a certain value $a$, we know $f(a)$ and $f'(a)$. The *tangent line approximation* to $f(x)$ near $a$ is

$$f(x) \approx f(a) + f'(a)(x - a).$$

If we want to take into account the error, we write instead

$$f(x) = f(a) + f'(a)(x - a) + \mathrm{Err}(x),$$

where we expect $\mathrm{Err}(x)$ to be small when $x$ is close to $a$.

The tangent line approximation is sometimes encoded into something called the *differential* of a function. It is tempting to include that here, but let's resist that temptation.

### 1.3.1 Problems

**Problem 1.3.1:** Use a tangent-line approximation to estimate the value of $\sqrt{4.1}$. Compare the result with what you get from your computer. Is your estimate larger or smaller than the computer's value? Can you explain why?

**Problem 1.3.2:** Suppose you put a yam in a hot oven, maintained at a constant temperature of 200°C. As the yam picks up heat from the oven, its temperature rises.

a. Draw a possible graph of the temperature $T$ of the yam against time $t$ (in minutes) since it is put into the oven. Explain any interesting features of the graph, and in particular, explain its concavity.

b. Suppose that, at $t = 30$, the temperature $T$ of the yam is 120°C and increasing at the rate of 2°C/min. Using this information, estimate the temperature at time $t = 40$.

c. Suppose in addition you are told that at time $t = 60$ the temperature of the yam is 165°C. Can you improve your estimate of the temperature at time $t = 40$?

d. Assuming all the data given so far, estimate the time at which the temperature of the yam is 150°C.

**Problem 1.3.3:** The acceleration due to gravity, $g$, is given by

$$g = \frac{GM}{r^2},$$

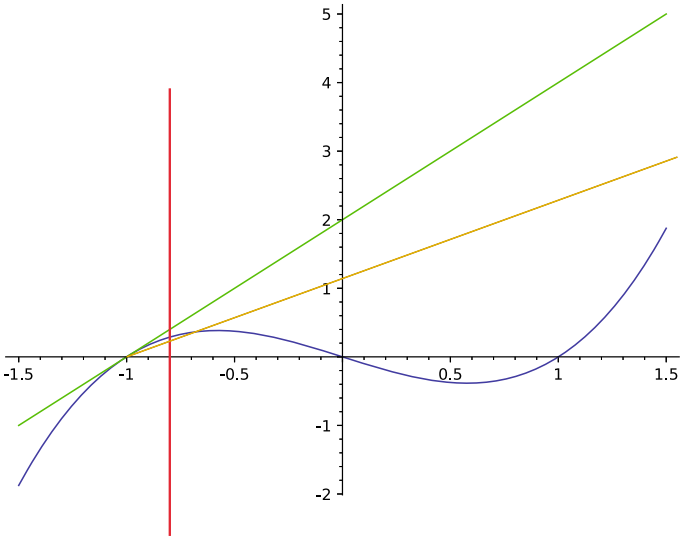
where $M$ is the mass of the Earth, $r$ is the distance from the center of the earth, and $G$ is a constant (the universal gravitational constant). Show that if we change $r$ by a small amount $\Delta r$, the change $\Delta g$ in the acceleration is approximately equal to $-2g\Delta r/r$. Use this approximation to compute the percentage change in $g$ when you move from sea level to the top of Pikes Peak, which is 4.315 km above sea level. (You'll need to know that the radius of the Earth is 6400 km.)

## 1.4 What's so good about the tangent line?

When we use the tangent line approximation, we get something that (we hope) gives us a reasonable value for the value of a function near a reference point $a$:

$$f(x) \approx f(a) + f'(a)(x - a).$$

This has our two required features: when $x = a$ it is exact, and when $x \to a$ the error goes to zero. But here's the mystery: any line through the point $(a, f(a))$ does the same thing! Let's go back to the picture from last time and add one more line:




The orange line is not the tangent line at $x = -1$, but it is also close to the blue curve. In fact, for our target value $x = -0.8$ it does a better job! So what's so good about the tangent line?

Looking at the picture very closely, you might (if you have very sharp eyes) notice this: if you slide the red line toward $x = -1$, both lines get closer to the curve, but the tangent line does it *faster*, especially as we get really close. This is very hard to see, but luckily mathematics is sharper than our eyes.

Let's translate the question. We have a (differentiable) function $f(x)$ and a reference point $a$. The tangent line approximation is

$$f(x) \approx f(a) + f'(a)(x - a).$$

The potential competitor, a secant line approximation, is

$$f(x) \approx f(a) + m(x - a),$$

where $m$ is the slope of the secant line. (Of course in the picture I chose a particularly tempting $m$!) The two error terms are

$$\mathrm{Err}(x) = f(x) - f(a) - f'(a)(x - a)$$

and

$$\mathrm{Err}_m(x) = f(x) - f(a) - m(x-a).$$

These are both the same formula, really. The first just chooses a special value for the constant $m$. If we plug in $x = a$, then both formulas just give us zero. And since $f(x)$ is continuous it's clear that when $x$ approaches $a$ both error terms go to zero in the limit.

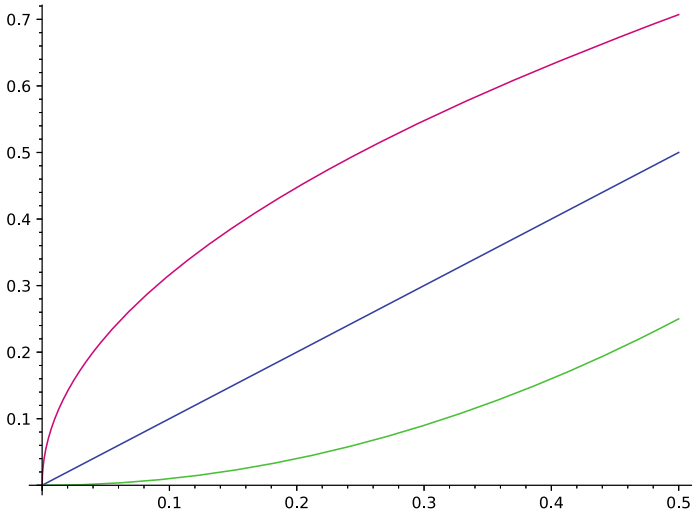
The question we want to ask is: how fast do they go to zero? What does that even mean?

**Detour: How fast does it go to zero?**

Well, how do we measure that? Suppose we take a number $h$. Clearly, as $h$ goes to zero, it goes to zero. (Doh!) What function $f(h)$ goes to zero when $h$ does, but does it faster/slower?

Let's graph some easy functions of $h$, namely $h$, $h^2$ and $\sqrt{h}$. I asked SageMath to do it, like this:

```
var('h')
plot([h,h^2,sqrt(h)],(h,0,.5))
```




The graph suggests that $h^2$ (green) goes to zero faster than $h$ (blue), while $\sqrt{h}$ (red) does it slower: for small positive values of $h$ we always have $h^2 < h < \sqrt{h}$. Notice that this is the opposite of our usual feeling that $h^2$ is bigger than $h$. That is true when $h > 1$, but not near zero!

One way to measure that is to look at the quotient. Suppose we have two functions $n(h)$ and $d(h)$, both of which have limit zero when $h \to 0$. What happens to the limit

$$\lim_{h\to 0} \frac{n(h)}{d(h)} \quad ?$$

The numerator $n(h)$ is getting closer to zero, so it wants to make the fraction get closer to zero. The denominator $d(h)$, however, is also going to zero, and that tends to make the fraction *bigger*. So the limit is like a race: the numerator wants to make the fraction be zero, while the denominator wants to make it infinitely large. Who wins?

Of course, the limit may not exist, but let's say it does. If the limit is zero, that means $n(h)$ has beaten $d(h)$. It approached zero faster. If the limit is infinity, then $d(h)$ has won the race. If it is some other thing (a number like 3), then we have some sort of almost-tie.

Thinking that way, we can confirm what we saw in the picture:

$$\lim_{h\to 0^+} \frac{h^2}{h} = \lim_{h\to 0^+} h = 0,$$

but

$$\lim_{h\to 0^+} \frac{\sqrt{h}}{h} = \lim_{h\to 0^+} \frac{1}{\sqrt{h}} = +\infty.$$

So $h^2$ approaches zero faster than $h$, while $\sqrt{h}$ goes to zero slower than $h$. It's easy to see that this is true in general: if $a > b$ then $h^a$ goes to zero faster than $h^b$.

In our situation, we will have $n(h) = \mathrm{Err}(a + h)$, which goes to zero when $h = x - a$ goes to zero, and $d(h) =$ some power of $h = |x - a|$. As we just saw, higher powers go to zero faster than lower powers. Taking the limit, we can decide how fast the error approaches zero.

**End detour**

Let's apply use this idea to figure out how quickly the error goes to zero in our situation with $f(x)$, the tangent line, and another line of slope $m$.

What serves as $h$ in our case is the difference $x - a$, which is what is getting closer to zero. To say $(x - a) \to 0$ is the same as $x \to a$. So we want to compute the limit

$$\lim_{x\to a} \frac{\mathrm{Err}_m(x)}{x - a} = \lim_{x\to a} \frac{f(x) - f(a) - m(x - a)}{x - a}$$

If we work this out for general $m$ then we can also know what happens when $m = f'(a)$. We compute the limit by breaking the fraction into two pieces:

$$\frac{f(x) - f(a) - m(x - a)}{x - a} = \frac{f(x) - f(a)}{x - a} - \frac{m(x - a)}{x - a} = \frac{f(x) - f(a)}{x - a} - m.$$

The number $m$ is fixed, so that part of the limit is easy. If we recognize the first part as exactly the thing that appears in the definition of the derivative, it's easy to take the limit:

$$\lim_{x\to a} \frac{\mathrm{Err}_m(x)}{x - a} = \lim_{x\to a} \left( \frac{f(x) - f(a)}{x - a} - m \right) = f'(a) - m.$$

Notice what that tells us: *the limit will be zero exactly when* $m = f'(a)$. So for a random line, we don't get zero, which means the error and $(x - a)$ go to zero at about

the same rate. But when we use the tangent line with $m = f'(a)$, the error goes to zero faster than the increment $(x - a)$. In other words, *the error in the tangent line approximation goes to zero faster than* $(x - a)$, *and the tangent line is the only line with this property.*

The usual slogan is: *the tangent line is the best linear approximation to* $f(x)$ *near* $a$. Out of all possible lines, it is the only one that has the property that the error gets small faster than $x - a$ does. In fact, looking at the limit we just computed we can see that this is pretty much the *definition* of the derivative $f'(a)$. Let me say it this way:

**Theorem 1.4.1.** *The derivative of* $f(x)$ *at* $a$ *is the only value of* $m$ *for which*

$$f(x) = f(a) + m(x - a) + \mathrm{Err}(x)$$

*with*

$$\lim_{x \to a} \frac{\mathrm{Err}(x)}{x - a} = 0.$$

The idea of a best linear approximation is one of the key ways to understand what the derivative of a function of *more* than one variable should be.

## 1.5 Examples of tangent line approximation

It's always good to see some examples. Let's look at three of them. In each case, I have chosen a function that we don't know how to evaluate by hand but our calculators and computers do. This allows me to create tables giving both the "correct" values (by which I mean the ones my computer gives, to some level of precision) and the approximated values. Subtracting, we get the error. We expect the error to be small, at least when the plug-in point $x$ is close to the reference point $a$. In fact, we have shown that it goes to zero faster than linearly as $x$ approaches $a$.

In order to be able to write down a computable formula for the tangent line approximation

$$f(a) + f'(a)(x - a),$$

I'll need to choose $a$ so that it is a number I can compute with (so it shouldn't be something like $\pi/2$) and also so that the values $f(a)$ and $f'(a)$ are easy to evaluate. As you will see, this severely restricts which $a$ we can use.

Of course, the theorem is true for any $a$. The only reason to choose "nice" values for $a$ is that we want to be able to compute the approximate value "by hand," just adding and multiplying. My tables were created with *Excel*.

I encourage you to choose some other function and make your own tables.

### 1.5.1 The sine function

Let $f(x) = \sin(x)$ and so $f'(x) = \cos(x)$. The good choice of reference point is $a = 0$, since we know that $\sin(0) = 0$ and $\cos(0) = 1$. The tangent line approximation is then

$$\sin(x) \approx 0 + 1(x - 0) = x,$$

| $x$ | $\sin(x)$ | $\sin(x) - x$ | $\dfrac{\sin(x) - x}{x}$ |
|---|---|---|---|
| point | function | error | relative error |
| 1.00 | 0.8414709848 | -0.1585290152 | -0.1585290152 |
| 0.90 | 0.7833269096 | -0.1166730904 | -0.1296367671 |
| 0.80 | 0.7173560909 | -0.0826439091 | -0.1033048864 |
| 0.70 | 0.6442176872 | -0.0557823128 | -0.0796890182 |
| 0.60 | 0.5646424734 | -0.0353575266 | -0.0589292110 |
| 0.50 | 0.4794255386 | -0.0205744614 | -0.0411489228 |
| 0.40 | 0.3894183423 | -0.0105816577 | -0.0264541442 |
| 0.30 | 0.2955202067 | -0.0044797933 | -0.0149326445 |
| 0.20 | 0.1986693308 | -0.0013306692 | -0.0066533460 |
| 0.10 | 0.0998334166 | -0.0001665834 | -0.0016658335 |
| 0.00 | 0.0000000000 | 0.0000000000 | undefined |

Table 1.1: $\sin(x)$ is close to $x$ when $x$ is close to 0

that is,

$$\sin(x) \approx x \qquad \text{when } x \approx 0.$$

We then expect that the limit of the error divided by $x - 0 = x$ should go to zero, and it does:

$$\lim_{x\to 0} \frac{\sin(x) - x}{x} = \lim_{x\to 0} \left( \frac{\sin(x)}{x} - 1 \right) = 1 - 1 = 0.$$

Look at Table 1.1. Notice that the error gets smaller and smaller, but the remarkable thing is the last column, which shows that the relative error is also approaching 0. Can you see why the errors are all negative? Why didn't I compute the values and errors for negative values of $x$?

### 1.5.2 The exponential function

If $f(x) = e^x$, then $f'(x) = e^x$ as well, and the convenient choice of reference point is $a = 0$ again, since the only value of $e^x$ that has a simple exponent and a simple answer: $e^0 = 1$. Then $f(0) = f'(0) = 1$, so the tangent line approximation is

$$e^x \approx 1 + 1(x - 0) = 1 + x \qquad \text{when } x \approx 0.$$

See Table 1.2. Notice that, while the errors and relative errors do approach zero, they do it far less enthusiastically than the ones for the sine function. We'll eventually figure out why that is.

### 1.5.3 The square root function

Now let $f(x) = \sqrt{x}$. This derivative is a bit less nice: $f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$. Notice that plugging in 0 is not a good idea! The function is not differentiable at $a = 0$, so nothing we have done so far would work.

| $x$ | $e^x$ | $e^x - (1+x)$ | $\frac{e^x-(1+x)}{x}$ |
|---|---|---|---|
| point | function | error | relative error |
| 1.00 | 2.7182818285 | 0.7182818285 | 0.7182818285 |
| 0.95 | 2.5857096593 | 0.6357096593 | 0.6691680624 |
| 0.90 | 2.4596031112 | 0.5596031112 | 0.6217812346 |
| 0.85 | 2.3396468519 | 0.4896468519 | 0.5760551199 |
| 0.80 | 2.2255409285 | 0.4255409285 | 0.5319261606 |
| 0.75 | 2.1170000166 | 0.3670000166 | 0.4893333555 |
| 0.70 | 2.0137527075 | 0.3137527075 | 0.4482181535 |
| 0.65 | 1.9155408290 | 0.2655408290 | 0.4085243523 |
| 0.60 | 1.8221188004 | 0.2221188004 | 0.3701980007 |
| 0.55 | 1.7332530179 | 0.1832530179 | 0.3331873052 |
| 0.50 | 1.6487212707 | 0.1487212707 | 0.2974425414 |
| 0.45 | 1.5683121855 | 0.1183121855 | 0.2629159678 |
| 0.40 | 1.4918246976 | 0.0918246976 | 0.2295617441 |
| 0.35 | 1.4190675486 | 0.0690675486 | 0.1973358531 |
| 0.30 | 1.3498588076 | 0.0498588076 | 0.1661960253 |
| 0.25 | 1.2840254167 | 0.0340254167 | 0.1361016668 |
| 0.20 | 1.2214027582 | 0.0214027582 | 0.1070137908 |
| 0.15 | 1.1618342427 | 0.0118342427 | 0.0788949515 |
| 0.10 | 1.1051709181 | 0.0051709181 | 0.0517091808 |
| 0.05 | 1.0512710964 | 0.0012710964 | 0.0254219275 |
| 0.00 | 1.0000000000 | 0.0000000000 | undefined |
| -0.05 | 0.9512294245 | 0.0012294245 | -0.0245884900 |
| -0.10 | 0.9048374180 | 0.0048374180 | -0.0483741804 |
| -0.15 | 0.8607079764 | 0.0107079764 | -0.0713865095 |
| -0.20 | 0.8187307531 | 0.0187307531 | -0.0936537654 |
| -0.25 | 0.7788007831 | 0.0288007831 | -0.1152031323 |
| -0.30 | 0.7408182207 | 0.0408182207 | -0.1360607356 |
| -0.35 | 0.7046880897 | 0.0546880897 | -0.1562516849 |
| -0.40 | 0.6703200460 | 0.0703200460 | -0.1758001151 |
| -0.45 | 0.6376281516 | 0.0876281516 | -0.1947292258 |
| -0.50 | 0.6065306597 | 0.1065306597 | -0.2130613194 |
| -0.55 | 0.5769498104 | 0.1269498104 | -0.2308178371 |
| -0.60 | 0.5488116361 | 0.1488116361 | -0.2480193935 |
| -0.65 | 0.5220457768 | 0.1720457768 | -0.2646858104 |
| -0.70 | 0.4965853038 | 0.1965853038 | -0.2808361483 |
| -0.75 | 0.4723665527 | 0.2223665527 | -0.2964887370 |
| -0.80 | 0.4493289641 | 0.2493289641 | -0.3116612051 |
| -0.85 | 0.4274149319 | 0.2774149319 | -0.3263705082 |
| -0.90 | 0.4065696597 | 0.3065696597 | -0.3406329553 |
| -0.95 | 0.3867410235 | 0.3367410235 | -0.3544642352 |
| -1.00 | 0.3678794412 | 0.3678794412 | -0.3678794412 |

Table 1.2: $e^x$ is close to $1 + x$ when $x$ is close to 0

| $x$ | $\sqrt{x}$ | $\sqrt{x} - (2 + (x-4)/4)$ | $\frac{\text{error}}{x-4}$ |
|---|---|---|---|
| point | function | error | relative error |
| 3.00 | 1.732050808 | -0.017949192 | 0.017949192 |
| 3.10 | 1.760681686 | -0.014318314 | 0.015909238 |
| 3.20 | 1.788854382 | -0.011145618 | 0.013932023 |
| 3.30 | 1.816590212 | -0.008409788 | 0.012013982 |
| 3.40 | 1.843908891 | -0.006091109 | 0.010151848 |
| 3.50 | 1.870828693 | -0.004171307 | 0.008342613 |
| 3.60 | 1.897366596 | -0.002633404 | 0.00658351 |
| 3.70 | 1.923538406 | -0.001461594 | 0.004871979 |
| 3.80 | 1.949358869 | -0.000641131 | 0.003205655 |
| 3.90 | 1.974841766 | -0.000158234 | 0.001582342 |
| 4.00 | 2 | 0 | undefined |
| 4.10 | 2.024845673 | -0.000154327 | -0.001543269 |
| 4.20 | 2.049390153 | -0.000609847 | -0.003049234 |
| 4.30 | 2.073644135 | -0.001355865 | -0.004519549 |
| 4.40 | 2.097617696 | -0.002382304 | -0.005955759 |
| 4.50 | 2.121320344 | -0.003679656 | -0.007359313 |
| 4.60 | 2.144761059 | -0.005238941 | -0.008731568 |
| 4.70 | 2.167948339 | -0.007051661 | -0.010073802 |
| 4.80 | 2.19089023 | -0.00910977 | -0.011387212 |
| 4.90 | 2.213594362 | -0.011405638 | -0.012672931 |
| 5.00 | 2.236067977 | -0.013932023 | -0.013932023 |

Table 1.3: $\sqrt{x}$ is close to $2 + \frac{1}{4}(x-4)$ when $x$ is close to 4

Since $\sqrt{x}$ appears in both the function and the derivative, what we need is a value of $a$ that is easy to work with (say, a positive integer) and whose square root is easy to find. So the best choice is $a = n^2$ for some positive integer $n$.

Let's choose $a = 4$. Then $f(4) = \sqrt{4} = 2$ and $f'(4) = \frac{1}{4}$. So the tangent line approximation is

$$\sqrt{x} \approx 2 + \frac{1}{4}(x-4) \qquad \text{when } x \approx 4.$$

Another way to write this that I kind of like is

$$\sqrt{4+h} \approx 2 + \frac{h}{4},$$

of course when $h \approx 0$. Look at Table 1.3 to see what happens.

**Do it yourself! (i.e., Problems)**

**Problem 1.5.1:** Let $f(x) = \ln(1 + x)$, and let $a = 0$. (Why is that the right value of $a$?) Find the tangent line approximation near $a$, and build a table like the ones we found above.

**Problem 1.5.2:** What other functions are worth doing?

**Problem 1.5.3:** Since we know the error goes to zero faster than $|x - a|$, we could try comparing it to $(x - a)^2$. For each of the examples above, use a spreadsheet to create a table that includes one more column with $\text{Err}(x)/(x - a)^2$. What happens?

## 1.6 Controlling the error in the tangent line approximation

So here's where we are. We are given a function $f(x)$ that we want to approximate and a reference point $x = a$. If we know the value $f(a)$ and the derivative $f'(a)$, we can approximate $f(x)$ by

$$f(x) \approx f(a) + f'(a)(x - a) + \text{Err}(x),$$

so that

$$\text{Err}(x) = f(x) - \big(f(a) + f'(a)(x - a)\big).$$

We'll keep using the notation $\text{Err}(x)$ to avoid having to write it out all the time.

We know that the error goes to zero faster than linearly, which just means that

$$\lim_{x \to a} \frac{\text{Err}(x)}{x - a} = 0.$$

The goal of this section is to find an actual bound for the error. That will, as usual, require us to know a little bit more about the function.

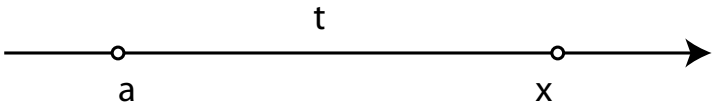
Remember that we can think of the tangent line approximation as "make believe it's a straight line." So if $f(x)$ actually were a straight line, there would be no error. That's because, on a straight line, the derivative $f'(x)$ is always the same: it's the slope of the line. So the source of the error is the fact that $f'(x)$ changes as $x$ varies. The rate of this change is of course the second derivative $f''(x)$. So it would be reasonable to expect that the bound for the error will depend on the second derivative.

(Well, at least if we assume that our function *has* a second derivative. For our argument, we'll assume that it does, and even that $f''(x)$ is a continuous function.)

There is one notational quirk we'll have to worry about. Since we have a reference point $a$ and a target point $x$, when we need to refer to some *other* point between the two we can't use either letter. So I'll use $t$ for a generic point, which will always be a point between $x$ and $a$.

When we looked at the "pretend it's constant" approximation, we saw that the error could be controlled if we could bound the size of the derivative $f'(t)$ when $t$ is between $a$ and $x$. So it's reasonable to guess that if we can bound the size of the *second* derivative in the relevant range (between $a$ and $x$) then we can control the error for the tangent line approximation. I'm going to give a proof of this, but let me say right up front that it isn't the world's most beautiful proof. (As far as I can tell there are no beautiful proofs of this, though there are trickier proofs that go faster than mine does.)

For this argument, let's assume that $x$ is bigger than $a$. The other case is exactly the same, but we want to write down integrals and need to make sure of the order of the values.




Our assumption is going to be that we can limit the size of the second derivative $f''(t)$ for all $t$ between $a$ and $x$. In other words, we assume that we know some number $M$ with the property that whenever $a \le t \le x$, we have

$$-M \le f''(t) \le M.$$

(Tricky bit: once we have this, notice that it is also true if we replace $x$ by a smaller number between $a$ and $x$: a bound that holds when $a \le t \le x$ also holds between $a$ and $t$. I'll point out where in the argument we need to use this.)

Start with the inequality

$$-M \le f''(t) \le M$$

and integrate from $a$ to $x$. The constants are easy:

$$\int_a^x M\,dt = M(x-a)$$

and the same for $-M$. For the middle term, we use the fundamental theorem of calculus: when $f''(t)$ is continuous, we have

$$\int_a^x f''(t)\,dt = f'(x) - f'(a).$$

Putting the results together, we get

$$-M(x-a) \le f'(x) - f'(a) \le M(x-a).$$

But remember (the tricky point) we could do the same thing for any other value between $a$ and $x$, so we actually know that whenever $a \le t \le x$, we have

$$-M(t-a) \le f'(t) - f'(a) \le M(t-a).$$

Well, it worked once, let's try one more time: integrate this between $a$ and $x$. This is a bit harder, but not much:

$$\int_a^x M(t-a)\,dt = \left[M\frac{(t-a)^2}{2}\right]_a^x = \frac{M}{2}(x-a)^2,$$

and similarly for $-M$.

For the middle term, remember that $f'(a)$ is just a constant, so it just gets multiplied by $x-a$. For the rest, use the fundamental theorem once more:

$$\int_a^x (f'(t)-f'(a))\,dt = \int_a^x f'(t)\,dt - \int_a^x f'(a)\,dt = f(x)-f(a)-f'(a)(x-a).$$

Putting together the inequalities, we end up with

$$-\frac{M}{2}(x-a)^2 \le f(x)-f(a)-f'(a)(x-a) \le \frac{M}{2}(x-a)^2.$$

The thing in the middle is exactly the error in the tangent line approximation, so this gives lower and upper bounds for the error. Very nice!

Let's write it out as a theorem. We have found out that:

**Theorem 1.6.1.** *Let $f(x)$ be a function whose first and second derivatives exist for all $t$ between $a$ and $x$. Suppose we know that $|f''(t)| \le M$ for any $t$ between $a$ and $x$. Then the absolute value of the error of the tangent line approximation is no bigger than $\frac{M}{2}(x-a)^2$.*

We actually proved this only when $a < x$, but a similar argument works for the case $a > x$. We also assumed that the second derivative exists and (when we invoked the fundamental theorem of calculus) that $f''(x)$ is *continuous*. This last assumption isn't actually necessary, but the proof gets harder if we don't make it. The only thing we really need is that the second derivative exists, so I stated the theorem only with that assumption.

This is the same kind of result we had before: an estimate on how big the next derivative can be allows us to estimate the error. The only surprise is that we get $M/2$ rather than just $M$. But we won't complain, since $M/2$ is smaller.

Let's try this on one of the examples above. Let $f(x) = e^x$, so that the tangent line approximation near 0 is $f(x) \approx 1 + x$. Let's suppose that we are interested in $x$ between 0 and 1. To use the error bound, we need to estimate the second derivative in that range.

Now, we know $f''(t) = e^t$ (the exponential is very friendly this way…). How big is that between $t = 0$ and $t = 1$? Students always find that a hard question, so let me spell out one way to figure it out. Notice first that $e^t$ is an *increasing* function, so the biggest possible value for the second derivative $e^t$ is $e^1 = e$. That's annoying because $e$ is an ugly number. But we don't need an *exact* bound for the derivative, because we are just going to get an error bound anyway. So remember that $e < 3$.

| $x$ | $e^x$ | $e^x - (1+x)$ | $\frac{3}{2}x^2$ |
|---|---|---|---|
| point | function | error | error bound |
| 1.00 | 2.7182818285 | 0.7182818285 | 1.5000000000 |
| 0.90 | 2.4596031112 | 0.5596031112 | 1.2150000000 |
| 0.80 | 2.2255409285 | 0.4255409285 | 0.9600000000 |
| 0.70 | 2.0137527075 | 0.3137527075 | 0.7350000000 |
| 0.60 | 1.8221188004 | 0.2221188004 | 0.5400000000 |
| 0.50 | 1.6487212707 | 0.1487212707 | 0.3750000000 |
| 0.40 | 1.4918246976 | 0.0918246976 | 0.2400000000 |
| 0.30 | 1.3498588076 | 0.0498588076 | 0.1350000000 |
| 0.20 | 1.2214027582 | 0.0214027582 | 0.0600000000 |
| 0.10 | 1.1051709181 | 0.0051709181 | 0.0150000000 |
| 0 | 1 | 0 | 0 |

Table 1.4: Comparing the error and the error bound for $f(x) = e^x$

Then we know that if $0 < t < 1$ then $f''(t) = e^t \leq e < 3$. Taking $M = 3$ in our formula, we get the error bound

$$|\text{Err}(x)| \leq \frac{3}{2}(x-0)^2 = \frac{3}{2}x^2.$$

The conclusion, then, is that

**Theorem 1.6.2.** *For any $x$ between 0 and 1, we have $e^x \approx 1 + x$ with error at most $\frac{3}{2}x^2$.*

This bound confirms what we already know: the error goes to zero faster than $x$. But it also tells us a bit more: the error goes to zero about as fast as $x^2$.

Table 1.4 is like the ones above, but the column for the relative error has been replaced by a column showing the error bound we just obtained. Notice that all the errors are (as expected) smaller in absolute value than the bound. In fact, they are quite a bit smaller.

### 1.6.1 Problems

**Problem 1.6.1:** The error estimate for the exponential is much easier for $-1 < t < 0$. Why?

**Problem 1.6.2:** Work out the error estimate for the tangent line approximation for $\sin(x)$ and construct a table like the one we made for $e^x$. Notice that the real error is *much smaller* than predicted.

**Problem 1.6.3:** Let $f(x) = \cos(x)$, whose tangent line approximation, as we saw, is $\cos(x) \approx 1$ when $x$ is close to 0. Find a bound on the error of the tangent line approximation that holds when $-1 \leq x \leq 1$.

**Problem 1.6.4:** Let $f(x) = \ln(1 + x)$.

a. Check that the tangent line approximation to $f(x)$ near $a = 0$ is $\ln(1 + x) \approx x$.

b. Suppose we want to approximate $\ln(1.5)$. (Notice that this means making $x = 0.5$ in our formula!) Will our estimate be larger or smaller than the true answer? Why?

c. Find a bound for the error of the approximation when $x = 0.5$.

**Challenge:** Here's a puzzle. For small values of $x$, the errors in Table 1.4 are smaller than the bounds by a factor of approximately 3. So the error is much closer to $x^2/2$ than to $3x^2/2$. Why?




# 2 Getting Closer with Polynomials

We have found a reasonably good way to approximate the value of a function $f(x)$ when $x$ is close to a reference value $a$. Can we do better? (Well, of course we can; why else would I ask the question?)

So far, we have created the approximations by choosing an approximating function that, at the point $a$ agreed, to some extent, with the function we want to approximate. So, in the constant line approximation, we used the function $\text{est}(x) = f(a)$ to approximate $f(x)$, and the two functions have the same *value* at $a$.

Then we went to the tangent line approximation. For that one, we chose the approximating function $\text{est}(x) = f(a) + f'(a)(x - a)$. This has the same value at $a$, i.e., $\text{est}(a) = f(a)$, but it also has the same *derivative* at $a$: $\text{est}'(a) = f'(a)$.

Well, there are second and higher derivatives. Perhaps we can use them? Of course, that means we have to assume that $f''(a)$, $f'''(a)$, … actually exist. So from now on, we will assume that our function has as many derivatives as we want. Technically, we are assuming that $f(x)$ is *infinitely differentiable*.

An astute student might ask, at this point, why we expect this to work. After all, to know the derivatives at $x = a$, we only need to know how the function $f(x)$ behaves very near $a$. The definitions involve limits, which only care about what happens when $x$ approaches $a$. So why should data that is very local to $a$ give information about other points, even if they are somewhat close to $a$?

The most honest answer is that I just know it works. People have tried it and got good results.

One might argue that knowing the value of several derivatives at $x = a$ tells us a lot of information about how the function is changing, and so might allow us to create something that agrees with our function at $x = a$ and changes similarly. We could also argue in reverse. Suppose we have found an approximation that does match $f(x)$ very closely near $x = a$. Then the approximation and the function should be very similar in a neighborhood of $a$, which implies that their derivatives at $a$ should be very similar as well. In other words, if we succeed in finding a good approximation, its derivatives should be closely related to the derivatives of the original function.

Either way, we are hoping for a miracle: that information very specific to the point $x = a$ will "spread" to nearby points. The extent to which this happens is one of the things we will need to work out.

So here is the method we will use: to find a good approximation, require that as many as possible of the derivatives of the approximation agree with those of the function. We'll start by showing how to do it for degree 2, then move to the general case.

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F. Q. Gouvêa, *A Short Book on Long Sums*, Reading in Mathematics,
https://doi.org/10.1007/978-3-031-37557-6_2

## 2.1 Approximations of degree 2

Since we have more derivatives to work with, let's try to set something up that agrees with value, derivative, and second derivative. Of course, that thing can't be a line any more, because the second derivative of any line is zero. So we will try something of degree *2* instead.

It's easiest to work with an example first. Let's put $f(x) = \sqrt{x}$ and choose $a = 4$ as before. We want to create an estimating function near $a = 4$. We found the tangent line approximation earlier: it looks like $2 + \frac{1}{4}(x-4)$. Since we now want something of degree 2 let's try $\text{est}(x) = 2 + \frac{1}{4}(x-4) + c(x-4)^2$ and try to figure out what $c$ should be equal to.

OK, so

$$a = 4, \qquad f(x) = \sqrt{x},$$

$$\text{est}(x) = 2 + \frac{1}{4}(x-4) + c(x-4)^2.$$

Notice first that $\text{est}(4) = 2 = \sqrt{4} = f(4)$; that's the neat thing about writing everything in terms of powers of $(x-4)$: they vanish when $x = 4$.

Now let's compute the derivative of our estimating function:

$$\text{est}'(x) = \frac{1}{4} + 2c(x-4).$$

So $\text{est}'(4) = 1/4$. On the other hand, $f'(x) = \frac{1}{2}x^{-1/2} = \frac{[}{2\sqrt{x}}$, so $f'(4) = 1/4$ as well. (No surprise, that's how we found the slope of the tangent line.)

The second derivative of $\text{est}(x)$ is easy to compute, because it is constant:

$$\text{est}''(x) = 2c.$$

The second derivative of $f(x)$ is a bit harder, but not much; since $f'(x) = \frac{1}{2}x^{-1/2}$,

$$f''(x) = \frac{1}{2}\frac{-1}{2}x^{-3/2} = -\frac{1}{4}x^{-3/2}.$$

That's a beast, but it isn't bad to plug in $x = 4$: since $4^{1/2} = 2$, we have $4^{3/2} = 8$, so $f''(4) = -\frac{1}{4}\frac{1}{8} = -\frac{1}{32}$. If we want this to agree with $\text{est}''(4) = 2c$, we have to choose $c = -1/64$.

Oooof. So, it seems that the second-degree approximation to $\sqrt{x}$ near $x = 4$ should be

$$\sqrt{x} \approx 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2.$$

Try graphing that to see that it does approximate $f(x) = \sqrt{x}$ quite well near $x = 4$.

Do we dare attempt to work that out in the general case?

Yes. We dare. [2]

We are given a function $f(x)$ and a reference point $a$. We assume that we know $f(a)$, $f'(a)$ and $f''(a)$. We want to use an estimating function of the form

$$\text{est}(x) = c_0 + c_1(x-a) + c_2(x-a)^2.$$

The rule for choosing the coefficients is that we want $\text{est}(x)$ and $f(x)$ to have the same value, derivative, and second derivative at $x = a$. Let's write them out:

$$\begin{aligned}\text{est}(x) &= c_0 + c_1(x-a) + c_2(x-a)^2\\ \text{est}'(x) &= c_1 + 2c_2(x-a)\\ \text{est}''(x) &= 2c_2\end{aligned}$$

Plugging in $x = a$ (see how neat the powers of $(x-a)$ are!), we get

$$\begin{aligned}\text{est}(a) &= c_0\\ \text{est}'(a) &= c_1\\ \text{est}''(a) &= 2c_2\end{aligned}$$

Since those are supposed to agree with $f(a)$, $f'(a)$, and $f''(a)$, we need to choose

$$c_0 = f(a) \qquad c_1 = f'(a) \qquad c_2 = \frac{f''(a)}{2}.$$

So, in general, the degree 2 approximating function to $f(x)$ near $a$ is

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2.$$

These approximating polynomials (of any degree) were considered by the British mathematician Brook Taylor, so they are known as Taylor polynomials.[1] In that language, we have

**The degree two Taylor polynomial for $f(x)$ centered at $a$** is

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2.$$

Notice that this degree 2 approximation is a refinement of the tangent line (degree 1) approximation: the first two terms are the same, and we have added a degree 2 term that (we hope!) makes the approximation better.

Things to think about:

a. For the tangent line approximation, we had both a theorem about the limit of the error as $x \to a$ and an error estimate using the second derivative. What do you think will be the case for the degree 2 approximation?

b. What would you do to get a degree 3 approximation?

The answers will soon come, but think about it on your own for a bit before you read on.

[1] It is rarely true that the person whose name is attached to a mathematical idea is actually the person who first created the idea. Taylor was certainly not the first person to study Taylor polynomials.

### 2.1.1 Problems

**Problem 2.1.1:** How good is the approximation

$$\sqrt{x} \approx 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2?$$

Try it!

**Problem 2.1.2:** Foreshadowing: following what we did for degree 2, work out a degree 4 polynomial that approximates $e^x$ near $a = 0$. (Remember that the goal is to find a polynomial so that the value, and the first, second, third, and fourth derivatives agree with those of the function when $x = 0$.)

## 2.2 Doing better: the general case

Let's do the general case now. We are given:

- A function $f(x)$ that we assume can be differentiated many times.
- A reference point $a$ where ALL IS KNOWN.
- A degree $n$.

We want to create a polynomial of degree $n$ that is a good approximation to $f(x)$ when $x \approx a$. Following the idea, we used before, we will write our polynomial in powers of $(x-a)$. There are two reasons to do this. First, it makes plugging in $x = a$ very easy. Second, when $x$ is close to $a$ the value of $x-a$ will be small, which means that powers $(x-a)^k$ get *smaller* as $k$ gets bigger.

The polynomials we are going to create are called *Taylor Polynomials*, so I will denote them by $T_n(x)$, assuming that the function $f(x)$ and the reference point (often called the "center") are understood (I really don't want to write $T_{f,a,n}(x)$. Do you?). So we want to write

$$T_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$$

and now we need to determine what the coefficients should be. The principle we will use is that the value and first $n$ derivatives of $f(x)$ and of $T_n(x)$ should agree when we plug in $x = a$. Since we are working with a general function $f(x)$, we can only write the derivatives symbolically:

$$f(a), \qquad f'(a), \qquad f''(a), \qquad \ldots \qquad f^{(n)}(a).$$

But for $T_n(x)$, we can work these out in terms of the coefficients $c_k$.

The value is easy: $T_n(a) = c_0$, so we must choose $c_0 = f(a)$. Notice that this is just the "make believe it's constant" approximation.

The derivative of $T_n(x)$ is

$$T_n'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 \cdots + nc_n(x-a)^{n-1}.$$

So when we make $x = a$, we get $T_n'(a) = c_1$. Since our goal is to have $T_n'(a) = f'(a)$, we need to choose $c_1 = f'(a)$. So our polynomial starts with

$$f(a) + f'(a)(x - a) + \dots,$$

reproducing the tangent line approximation.

One more: the second derivative of $T_n(x)$ is

$$T_n''(x) = 2c_2 + 2 \cdot 3c_3(x - a) + 3 \cdot 4c_4(x - a)^2 \cdots + (n - 1)nc_n(x - a)^{n-2}.$$

Plugging in $x = a$ we get $T_n''(a) = 2c_2$, and since we want $T_n''(a) = f''(a)$, we want $c_2 = \frac{1}{2}f''(a)$, just as in the previous section.

The general pattern is like this: the term $c_k(x - a)^k$ has derivative $kc_k(x - a)^{k-1}$, whose derivative is $(k - 1)kc_k(x - a)^{k-2}$, and so on. When we differentiate $k$ times we end up with

$$T_n^{(k)}(x) = 2 \cdot 3 \cdot 4 \cdot \cdots \cdot (k - 1)kc_k + \text{stuff in terms of } (x - a).$$

That product (all the integers from 1 to $k$ multiplied together) is denoted $k!$ and called "$k$ factorial." Let's rewrite using that notation:

$$T_n^{(k)}(x) = k!\, c_k + \text{stuff in terms of } (x - a).$$

Plugging in $x = a$ makes all the "stuff" be zero, so we get $T_n^{(k)}(a) = k!c_k$. Since our goal is to have $T_n^{(k)}(a) = f^{(k)}(a)$, we must choose

$$c_k = \frac{1}{k!}f^{(k)}(a) = \frac{f^{(k)}(a)}{k!}.$$

This awful formula is not so hard to understand: it says that the coefficient of $(x-a)^k$ in our polynomial should be the value of the $k$-th derivative at $a$ divided by $k!$.

Will that work when $k = 0$? It will as long as we decide that

- the "zero-th derivative" is just the function: $f^{(0)}(a) = f(a)$;
- the expression $(x - a)^0$ just means 1 no matter what the value of $x$ is;
- the factorial of zero is 1, so $0! = 1$.

Those are just choices to make the notation more convenient. Now let's grit our teeth and write the thing out.

> **The Taylor polynomial of degree $n$ centered at $a$ for the function $f(x)$** is
>
> $$T_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

The term of degree $k$ in the sum looks like

$$\frac{f^{(k)}(a)}{k!}(x-a)^k.$$

There is a convention to replace the name "Taylor" with the name "Maclaurin" when $a = 0$. Silly as that is, you should know about it, since people will use it. Colin Maclaurin was a more important mathematician than Brook Taylor, but there's really no point in assigning his name to a special kind of Taylor polynomials. Maclaurin polynomials are just Taylor polynomials centered at zero. The same comment applies to "Taylor series" and "Maclaurin series": the latter just means "centered at zero."

Notice that the formula for the term of degree $k$ is the same in all the Taylor polynomials whose degree is $k$ or more. For example, the coefficients of the polynomials of degree $n$ and of degree $n+1$ polynomial will be the same up to degree $n$; for $T_{n+1}$, we just add one extra term. That's good: it lets us think of each new term as a *correction* added to the previous approximation. Or at least we hope so.

The fact that $k!$ appears in the denominator is potentially very good. As the integer $k$ gets bigger, $k!$ grows very fast. A big number in the denominator makes things smaller. In the general term

$$\frac{f^{(k)}(a)}{k!}(x-a)^k,$$

the $k!$ makes things smaller and, when $|x-a| < 1$, the powers $(x-a)^k$ do as well. So unless the derivatives $f^{(k)}(a)$ grow very fast, each term is likely to be smaller than the previous ones, at least when $x$ is close to $a$.

As I already pointed out, if $n = 1$, we get $T_1(x) = f(a) + f'(a)(x-a)$, which is the tangent line approximation. So, the Taylor polynomials generalize what we have done so far.

Unfortunately, computing higher derivatives can be pretty hard unless there's an easy pattern. We will later work out ways to find these polynomials *without* computing the higher derivatives. Of course, SageMath is happy to compute derivatives for us, so it can help us find Taylor polynomials. What it *can't* do, however, is identify a pattern that the coefficients follow (even when there is a pattern).

Let's do an example. Suppose $f(x) = \dfrac{1}{1-x} = (1-x)^{-1}$ and choose $a = 0$. Let's work out the values of the derivatives.

- $f(0) = 1$.
- $f'(x) = (1-x)^{-2}$, so $f'(0) = 1$.
- $f''(x) = 2(1-x)^{-3}$, so $f''(0) = 2$.
- $f'''(x) = 3 \cdot 2(1-x)^{-4}$, so $f'''(0) = 6 = 3!$.
- In general, $f^{(k)}(x) = k!(1-x)^{-(k+1)}$, so $f^{(k)}(0) = k!$. (Can you convince yourself?)

And now, using the formula above, we see that all the coefficients are one: $c_k = \frac{k!}{k!} = 1$ for all $k$. (The easiest possible pattern!) So the degree $n$ Taylor polynomial for $\frac{1}{1-x}$ centered at 0 is

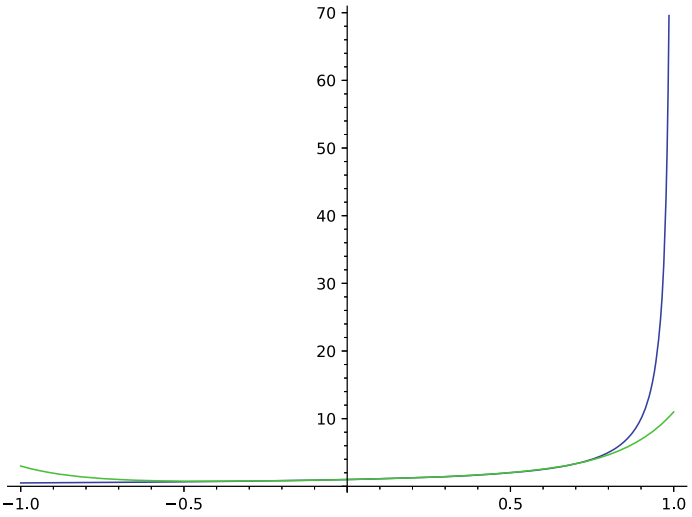
$$T_n(x) = 1 + x + x^2 + x^3 + \cdots + x^n.$$

We'll have to wait till a later section to see what we can prove about these polynomials. It's worth noticing, however, that in this case the derivatives do grow very fast and cancel out the big denominators.

For now, we can at least compare graphs when $n = 10$. Let's ask SageMath to do that:

```
sage: f(x)=1/(1-x)
sage: T10(x)=1+x+x^2+x^3+x^4+x^4+x^6+x^7+x^8+x^9+x^10
sage: plot([f(x),T10(x)],(-1,1))
```

The output is this:




SageMath always uses the sequence of colors blue, green red, etc., when we plot several functions. So blue graph is $f(x) = 1/(1-x)$; notice that it blows up as $x$ approaches 1. The green graph is the Taylor polynomial of degree 10. It does seem to give a good approximation near $x = 0$, but it gets pretty bad as we approach $x = 1$. It stays a bit closer on the negative side, but also drifts away eventually. This confirms the basic intuition that our approximations should work reasonably well *near* the reference point $x = a$, but there is no reason to expect them to work well for values of $x$ that are not near to $a$.

Here's another easy example. Let's work with $f(x) = \sin(x)$ and find some of its Taylor polynomials centered at zero. First, consider the derivatives of the sine function:

$$\begin{aligned} f(x) &= \sin(x) \\ f'(x) &= \cos(x) \\ f''(x) &= -\sin(x) \\ f'''(x) &= -\cos(x) \\ f^{(4)}(x) &= \sin(x) \\ &\text{and repeat.} \end{aligned}$$

Since it repeats after four steps, so do the values at $x = 0$. They are

$$0,\ 1,\ 0,\ -1,\ 0,\ 1,\ 0,\ -1\ \ldots$$

To get the coefficients, we divide these by the factorials, so the coefficients are

$$0\,1,\ 0,\ -\frac{1}{3!},\ 0,\ \frac{1}{5!},\ 0,\ -\frac{1}{7!}\ \ldots$$

So the Taylor polynomials look like

$$T_n(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

(Notice that since the even-numbered coefficients are all zero, we have $T_8(x) = T_7(x)$ and so on.)

I said SageMath can do it for you. Here's how. Given a function $f(x)$, we want SageMath to compute its Taylor polynomial of degree $n$ centered at $a$. The command for doing that is `taylor(f(x),x,a,n)`. In that command, `x` tells SageMath which variable you are working with (SageMath can work with more than one), `a` should be a number (the center), and `n` should be a positive integer (the degree). Of course, you can play around to see what happens if you break these rules: if you tell SageMath `a` is a variable and use it as a center, it seems to work!

For the example we just did,

```
sage: T10(x)=taylor(1/(1-x),x,0,10)
sage: T10(x)
x^10 + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1
```

That's what we got as well, but notice that (annoyingly!) SageMath writes the higher degree terms first.

That SageMath can't do, however, is figure out a pattern for you. In the case of $1/(1-x)$, it's fairly obvious that all the coefficients are equal to 1. But even for $\sin(x)$ it's not that easy: you would see only the odd powers, but SageMath writes out the values of the factorials, so it is not necessarily easy to recognize the pattern.

It can be even harder; here's an example:

```
sage: taylor(tan(x),x,0,11)
1382/155925*x^11 + 62/2835*x^9 + 17/315*x^7 + 2/15*x^5 + 1/3*x^3 + x
```

Aside from the fact that all the even degree terms are zero, I don't see any pattern in the coefficients. Do you?

### 2.2.1 Problems

**Problem 2.2.1:** Why doesn't $f(x) = x^{1/3}$ have a Taylor polynomial of degree 3 centered at $a = 0$?

**Problem 2.2.2:** Rodolfo attempted to use the formula

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

to find the fifth-degree Taylor polynomial for $f(x) = e^x$ near $a = 0$. He got

$$e^x \approx e^x + e^x x + \frac{e^x}{2}x^2 + \frac{e^x}{6}x^3 + \frac{e^x}{24}x^4 + \frac{e^x}{120}x^5.$$

That must be wrong, since it's not even a polynomial. What did Rodolfo do wrong?

**Problem 2.2.3:** We graphed $f(x) = 1/(1-x)$ and its degree 10 Taylor polynomial centered at zero $T_{10}(x)$. Make more graphs to see if you can find out more. What happens when $x > 1$? How about $x < -1$?

**Problem 2.2.4:** Try making a longer polynomial, say of degree 100, using

```
T100(x)=sum(x^n,n,0,100).
```

Compare that to $f(x)$ and to the degree 10 polynomial.

**Challenge:** Can you figure out a "closed" formula for the sum $1 + x + x^2 + \cdots + x^n$? (A "closed" formula is one that has no dots or summation signs.) We will do this soon, but give it a try now.

**Problem 2.2.5:** (Nothing to offer here but blood, sweat, tears, and toil.) In each item below I give a function $f(x)$, a reference point $a$, and a degree $n$. Compute enough derivatives and then use the general formula to find the Taylor polynomial of degree $n$ for the function $f(x)$ around $a$.

(Taking higher derivatives is usually not much fun. Use SageMath if you like!)

Once you have found all the series, use the `taylor` command in SageMath to check your answers.

a. $f(x) = \dfrac{1}{1+x}$, $a = 0$, $n = 4$.

b. $f(x) = \cos(x)$, $a = 0$, $n = 6$.

c. $f(x) = \ln(x)$, $a = 1$, $n = 4$.

d. $f(x) = (1+x)^5$, $a = 0$, $n = 4$.

e. $f(x) = \dfrac{1}{1+x}$, $a = 2$, $n = 4$.

f. $f(x) = \dfrac{x}{1-x^3}$, $a = 0$, $n = 4$.

**Problem 2.2.6:** Suppose you know that the Taylor polynomial of degree 10 of a function $f(x)$ is

$$\sum_{n=0}^{10} \frac{(x-1)^n}{n!}.$$

What is $f^{(5)}(1)$ equal to? (Remember that $f^{(5)}(1)$ means the value at $x = 1$ of the fifth derivative of $f(x)$.)

## 2.3 Taylor polynomials and derivatives

The way that Taylor polynomials are constructed, by taking many derivatives and then plugging in $x = a$, means that they interact well with taking derivatives. The degree $k$ term of a Taylor polynomial is

$$\frac{f^{(k)}(a)}{n!}(x-a)^k.$$

If we differentiate that, we get

$$\frac{f^{(k)}(a)}{k!}\, k(x-a)^{k-1} = \frac{f^{(k)}(a)}{(k-1)!}(x-a)^{k-1},$$

since $k! = k \cdot (k-1)!$. That looks a lot like the term of degree $k-1$ of a Taylor polynomial, but it has an $k$-th derivative instead. But (I know this is a brain-twister) *the $k$-th derivative of $f(x)$ is actually the $(k-1)$-st derivative of $f'(x)$*. So by differentiating the degree $k$ term of the Taylor polynomial for $f(x)$ we get exactly the degree $k-1$ term of the Taylor polynomial for $f'(x)$.

Since that works for each term, we conclude:

**Theorem 2.3.1.** *The derivative of the degree n Taylor polynomial for a function $f(x)$ is equal to the degree $n-1$ Taylor polynomial for $f'(x)$.*

That is very useful to know. For example, the derivative of the degree 3 Taylor polynomial for $\sin(x)$ is the degree 2 Taylor polynomial for $\cos(x)$. So if we know the Taylor polynomials for $\sin(x)$ we can find the Taylor polynomials for $\cos(x)$. And, of course, we can also read it in terms of antiderivatives, as long as we remember that taking antiderivatives introduces an arbitrary constant.

**Theorem 2.3.2.** *The antiderivative of the degree n Taylor polynomial for a function $f(x)$ is (up to an additive constant) equal to the degree $n+1$ Taylor polynomial for the antiderivative of $f(x)$.*

In any specific case, we can find the constant pretty easily by plugging in $x = a$, which makes all the terms vanish except that constant term.

### 2.3.1 Problems

**Problem 2.3.1:** Check it: find the degree 5 Taylor polynomial for $\sin(x)$ and the degree 4 Taylor polynomial for $\cos(x)$, then verify that the second is equal to the derivative of the first.

**Problem 2.3.2:** Check that the derivative of the degree $n$ Taylor polynomial for $e^x$ is the degree $n-1$ Taylor polynomial for $e^x$.

## 2.4 How close to the function is the Taylor polynomial of degree n?

Here's what we know so far. We are given a function $f(x)$, a reference point $a$, and a degree $n$. With that, we made up a polynomial that we hope gives a good approximation. We know the next question we must ask: what can we say about the error?

The Taylor polynomial of degree $n$ centered at $a$ for the function $f(x)$ is

$$T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

What can we say about this? Does it actually approximate $f(x)$?

Our model is the tangent line approximation, which in this notation is $T_1(x)$. Remember that in that case we had two good properties:

- We had a limit property:

$$\lim_{x \to a} \frac{f(x) - T_1(x)}{x-a} = 0$$

  and in fact $T_1(x)$ is the *only* linear function with this property. More precisely, if we replace $T_1(x)$ with any other linear function, this limit will not be zero. In other words, $T_1(x)$ is the best degree 1 approximation to $f(x)$ near $a$.

- We had an error estimate: if we know that $|f''(t)| \leq M$ for all $t$ between $a$ and $x$, then

$$|f(x) - T_1(x)| \leq \frac{M}{2}|x-a|^2.$$

Notice that the second property implies the first: something bounded by $Mh^2$, if we divide by $h$, becomes bounded by $Mh$, which goes to zero. But to get the second property we needed to know there was a second derivative, which the first property did not require.

Exactly the same two properties hold in general. (Indeed, the proofs, which I won't include, are pretty much the same. A good place to read the proofs, if you want to, is [19, chapter 20].)

**Theorem 2.4.1** (Limit Property) *If $f(x)$ is an $n$-times differentiable function and $T_n(x)$ is its degree $n$ Taylor polynomial centered at $a$, then*

$$\lim_{x\to a} \frac{f(x) - T_n(x)}{(x-a)^n} = 0$$

*and in fact $T_n(x)$ is the* only *polynomial of degree $n$ with this property; if we replace it with any other polynomial of degree $n$ the limit will not be zero. In other words, $T_n(x)$ is the best degree $n$ approximation to $f(x)$ near $a$.*

For the error estimate, we need to know how big the $(n+1)$-st derivative is. I will use $M_n$ for that estimate in order to highlight the fact that the inequality will depend on the value of $n$.

**Theorem 2.4.2** (Error Estimate) *Suppose $f(x)$ is an $(n+1)$-times differentiable function and $T_n(x)$ is its degree $n$ Taylor polynomial centered at $a$. If we know that the $(n+1)$st derivative satisfies*

$$|f^{(n+1)}(t)| \le M_n$$

*for all $t$ between $a$ and $x$, then*

$$|f(x) - T_n(x)| \le \frac{M_n}{(n+1)!}|x-a|^{n+1}.$$

As before, the second property is stronger than the first, but it needs one more derivative.

It's worth noticing that the error term is *almost* the same as the degree $n+1$ term of the next Taylor polynomial. The degree $n+1$ term would be

$$\frac{f^{(n+1)}(a)}{(n+1)!}(x-a)^{n+1}.$$

In the error term we replace that $(n+1)$-st derivative at $a$ with our bound for the $(n+1)$-st derivative between $a$ and $x$. This confirms the intuition that each extra term is a correction.

The upshot is that these polynomials provide a good way to find approximate values of $f(x)$, at least if $x$ is near the center point $a$. Of course, in the limit property, we don't know *how* close it needs to be. The error estimate can tell us that more precisely, at least if we can find a good bound $M_n$.

One problem here is that the bound $M_n$ depends on the $(n+1)$-st derivative, and so it can change when $n$ changes. We'd *like* the approximation to get better as $n$ grows, but we can't guarantee that in general: it's always possible that $M_n$ is huge, after all, and grows even faster than $(n+1)!$.

We should note that Theorem 2.4.2 is only one way to bound the error when we use the degree $n$ Taylor polynomial to approximate a function. (You can find two others in [19, ch. 20].) The dependence on finding a bound for the value of the $(n+1)$-st derivative makes it hard to use. There is no guarantee that it will work well

in any particular case. It sometimes fails. But, as the next example shows, for some especially nice functions it all works.

Let's go back to $f(x) = \sin(x)$. We found that the Taylor polynomials look like

$$T_n(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

(Notice that since the even-numbered coefficients are all zero, we have $T_8(x) = T_7(x)$ and so on.) Now we can estimate the error too. Since all the derivatives are either $\pm\sin(x)$ or $\pm\cos(x)$, we have $|f^{(n+1)}(t)| \leq 1$ for any $t$. So, we can take $M_n = 1$ for every $n$ to get

$$\left|\sin(x) - T_n(x)\right| \leq \frac{|x|^{n+1}}{(n+1)!}.$$

Since our estimate for the derivative is true for any $t$, this error bound is true for all values of $x$. Of course, powers of $x$ are only small if $|x| < 1$, but maybe those factorials in the denominator make up for that.

We noticed above that for the sine function we have $T_{2n+1}(x) = T_{2n+2}(x)$, since the extra term in the latter is zero. So perhaps we should use the error estimate for $T_{2n+2}(x)$, which has a bigger factorial in the denominator. In other words,

$$\left|\sin(x) - T_{2n+1}(x)\right| \leq \frac{|x|^{2n+3}}{(2n+3)!}.$$

This explains the special behavior of the tangent line approximation to sine function that you noticed in Problems 1.5.3 and 1.6.2. If we think of $T_1(x) = x$, we get

$$\left|\sin(x) - x\right| \leq \frac{1}{2}|x|^2,$$

but if we use $T_2(x) = x$, we get

$$\left|\sin(x) - x\right| \leq \frac{1}{6}|x|^3.$$

When $|x| < 1$, as it was in our tables, the second estimate is much smaller. Essentially the same thing works for the cosine, with "odd" and "even" switched around.

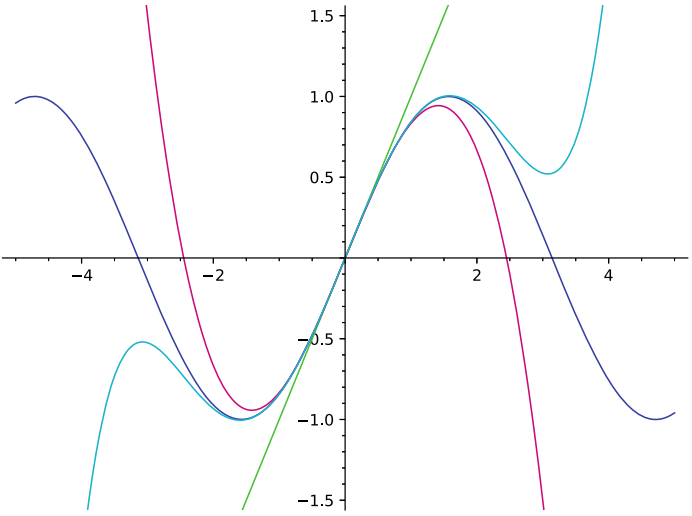
## 2.5 What happens as $n$ grows?

Our hope has been that when the degree of the approximating polynomial gets bigger, the approximation will improve. So far, we don't know that this is true. In fact, it will not be true for all functions.

Let's continue pushing on the example we just computed and do some plotting, asking SageMath to plot the sine function and its Taylor polynomials of degree 1, 3, and 5. The commands would be

```
sage: f(x)=sin(x)
sage: T1(x)=taylor(sin(x),x,0,1)
sage: T3(x)=taylor(sin(x),x,0,3)
sage: T5(x)=taylor(sin(x),x,0,5)
sage: plot([f(x),T1(x),T3(x),T5(x)],(x,-5,5),ymin=-1.5,ymax=1.5)
```

We get




The sine function is the dark blue curve, $T_1(x)$ is green, $T_3(x)$ is red, and $T_5(x)$ is light blue. Notice that we are actually seeing two things: better and better approximation for a given $x$ and a growing range of validity.

From the first point of view, $x$ is fixed and $n$ is changing: if we look at a small number $x$, we see that $T_3(x)$ gives a better approximation that $T_1(x)$, though they are all so good that near zero we can't see much difference in the graph. But the difference is there:

```
sage: T1(0.1)
0.100000000000000
sage: T3(0.1)
0.0998333333333333
sage: T5(0.1)
0.0998334166666667
sage: sin(0.1)
0.0998334166468282
```

Notice that $T_1(0.1)$ is a bit too big, $T_3(0.1)$ a little too small, then $T_5(0.1)$ is too big again… but the last agrees with sin(0.1) to ten decimal places. (Can you see why the "too big"/"too small" alternation is happening?) This is what we expected: for $x$ near zero, we are getting better and better approximations.

The second has to do with varying $x$: $T_3(x)$ seems to stay near $\sin(x)$ for more values of $x$ than $T_1(x)$, and $T_5(x)$ does it longer than $T_3(x)$. This is a surprise: as we have mentioned before, there is no reason to expect our approximations to work for large $x$. Nevertheless, it seems that they do. What happens if you try degree 7 or 9?

We can frame two questions, then. First, the standard approximation question: *for a fixed value of $x$, is it true that the error $|f(x)-T_n(x)|$ goes to zero as n grows?* This amounts to asking whether longer polynomials give better approximations. Our hope is that the answer is yes, at least when $x$ is close to the center $a$. Our error estimate will sometimes allow us to answer this question.

The second question is this. Suppose we know that the error goes to zero for some value of $x$; we can now consider all the values of $x$ for which this happens. What does this set of values of $x$ look like? For example, could it be that the sine series we just found converges for some $x$ and not for others?

We can try to use the error bound to answer both questions. For example, if $f(x) = \sin(x)$, we know that

$$\left|\sin(x) - T_n(x)\right| \leq \frac{1}{(n+1)!}|x|^{n+1}.$$

For some values of $x$, it's easy to see that this is very small, and gets smaller as $n$ grows. Suppose, for example, that $|x| < 1$. Then $|x|^k < 1$ for all exponents $k$. So the error is less than $1/(n+1)!$, which is a very small number that goes to zero as $n$ gets big.

### 2.5.1 Problems

**Problem 2.5.1:** What about the other values of $x$? After all, when $x$ is big a power $x^n$ gets very big too. We'll check soon that the factorials beat the powers, so that indeed it works for *every* $x$. Can you convince yourself without my help?

**Problem 2.5.2:** Use the degree 2 Taylor polynomial to estimate $\cos(0.1)$. Find a bound for the error in your approximation. Compare your bound with the error according to SageMath.

**Problem 2.5.3:** During an encounter with an extraterrestrial, it is revealed to you that the answer to life, the universe, and everything is $f(1)$ for some function $f$. The extraterrestrial, when pressed, refuses to tell you what $f(x)$ is, but it does let you know that

- $f(0) = 26$, $f'(0) = 22$, $f''(0) = -16$, and $f'''(0) = 12$,
- For any $x$ such that $|x| \leq 3$, we have $|f^{4)}(x)| \leq 7x^4$.

Find upper and lower bounds for the value of $f(1)$.

## 2.6 Dots, sigmas, and general terms

So far, we have mostly been writing sums using dots to indicate terms we do not write down. For example,

$$1 + x + x^2 + x^3 + \cdots + x^{15}$$

That is fine in examples like this one because the pattern is fairly clear. But it's more honest to tell the reader what the pattern is, so I really should write

$$1 + x + x^2 + x^3 + \cdots + x^n + \cdots + x^{15}$$

The $x^n$ term is called a *general term*. It gives a recipe for the terms: take $n = 0$ to get the 1, then $n = 1$ to get $x$, then $n = 2$, etc. That way no one can say "I don't see the pattern."

Once we have a general term, we can also use summation notation, writing

$$\sum_{n=0}^{15} x^n = 1 + x + x^2 + x^3 + \cdots + x^{15}.$$

The big sigma $\Sigma$ stands for "sum." At the bottom, we give the first value of $n$ to consider, and at the top, we indicate the last value of $n$ to use. To translate into a sum, plug in each value in turn and stick + signs between them.

If we want to go on forever (and we soon will), we will put $\infty$ at the top instead:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots + x^n + \ldots$$

Summation notation takes much less space than dots and for some computations it is actually easier to use, because you can work directly with the general term.

Alas, summation notation is also easier to get confused with, so I'll continue to prefer writing things out with dots, but you should practice translating back and forth between the two versions.

For now, here is the formula for the Taylor polynomial of degree $n$ written out in summation notation:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Here, we have used our definitions (see page 29) for how to understand $f^{(k)}(a)$, $k!$, and $(x-a)^k$ when $k = 0$.

Notice that since I wanted to use $n$ for the stopping point I need to use a different counting variable for the general term. It doesn't matter what we call that variable, because its only job is to be replaced by the numbers 0, 1, 2, ... .

### 2.6.1 Problems

**Problem 2.6.1:** Unpack each of these expressions by writing out the sums:

a. $\sum_{n=0}^{10} n^2 x^n$

b. $\sum_{n=0}^{5} (-1)^n \frac{x^{2n}}{(2n)!}$

c. $\sum_{n=1}^{5} \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)\dots(\frac{1}{2}-n-1)}{n!} x^n$

d. $\sum_{n=0}^{5} (-1)^{n+1} \frac{x^{2n+1}}{(2n+1)!}$

e. $\sum_{n=1}^{6} (-1)^{n+1} \frac{x^n}{n}$

## 2.7 The easiest examples: sine and cosine

There are several functions whose Taylor polynomials are easy to compute, either directly by finding derivatives or by using other tricks. Furthermore, in some cases, we can estimate the error so well that we can conclude the approximation gets better and better (or not) as the degree $n$ grows. Let's work through some of them.

### 2.7.1 The sine function

We have already worked out that

$$T_n(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

where $2k+1$ is the largest odd number smaller than $n$. We worked out that the error is controlled by

$$|\sin(x) - T_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!}.$$

We can actually do a little bit better by remembering that in this case $T_{2n+1}(x) = T_{2n+2}(x)$, since the coefficients of even powers of $x$ are always zero. So we can say

$$|\sin(x) - T_{2n+1}(x)| \leq \frac{|x|^{2n+3}}{(2n+3)!}.$$

As you may know, factorials grow much faster than powers. Let's prove it. The key point is to notice that

$$\frac{|x|^k}{k!} = \frac{|x| \cdot |x| \cdot |x| \cdot |x| \cdots |x|}{1 \cdot 2 \cdot 3 \cdot 4 \cdots k} = \frac{|x|}{1}\,\frac{|x|}{2}\,\frac{|x|}{3}\,\frac{|x|}{4} \cdots \frac{|x|}{k}.$$

As we go along, the number $x$ is fixed, but the denominators keep growing, so we are multiplying by ever-smaller fractions.

To estimate the product, we focus on when the fractions get smaller than some fixed number; say $1/2$. Let $N$ be the smallest integer bigger than $2|x|$. Then if $i > N$ the fraction $|x|/i$ is less than $1/2$. Assuming we have a long enough product so that $N < k$, we can split the product into two chunks: the fractions with $i \leq N$ and those with $i > N$.

$$\frac{|x|^k}{k!} = \left(\frac{|x|}{1}\,\frac{|x|}{2}\,\frac{|x|}{3}\,\frac{|x|}{4} \cdots \frac{|x|}{N}\right)\left(\frac{|x|}{N+1}\,\frac{|x|}{N+2} \cdots \frac{|x|}{k}\right).$$

Since $N$ is fixed, the first chunk is just some fixed number. That number is to be multiplied by the second chunk, which is the product of a whole bunch of fractions (well, $k - N$ fractions), each of which is less than $1/2$. So the product is smaller than that:

$$\frac{|x|^k}{k!} < (\text{first chunk})\left(\frac{1}{2}\right)^{k-N}.$$

The first factor is fixed, but the second factor goes to 0 as $k \to \infty$: multiplying by $1/2$ over and over makes the product arbitrarily small.

The upshot is that no matter what $x$ is

$$\frac{|x|^k}{k!} \to 0 \qquad \text{as } x \to \infty.$$

So no matter how big $x$ is, the fraction $|x|^k/k!$ can be made to be as close to zero as we like by taking $k$ large enough.

Applying this to the error bound for the Taylor polynomial, we see that, no matter what $x$ is, the error

$$\left|\sin(x) - T_n(x)\right|$$

can be made as small as desired by taking $n$ large enough. That is,

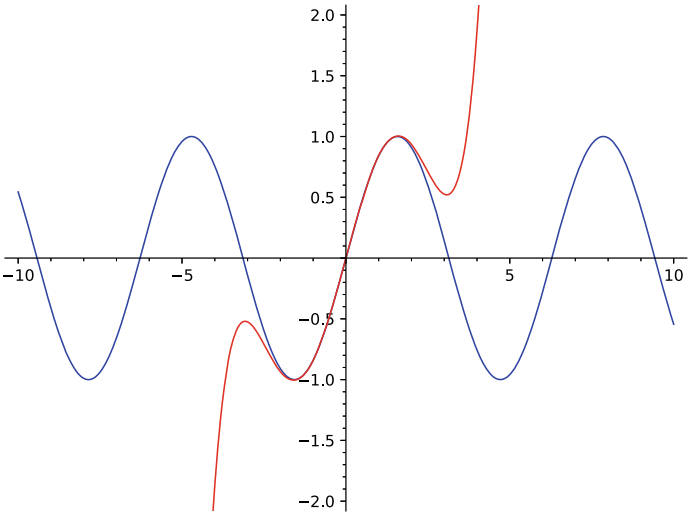
$$\lim_{n\to\infty} \left|\sin(x) - T_n(x)\right| = 0.$$

This is true for any $x$. Of course, for larger $x$ one needs to let $n$ become quite big to see the error become small.

We've seen this in graphs, but let's do it again even more carefully. I'll ask SageMath to plot the function and some of the polynomials

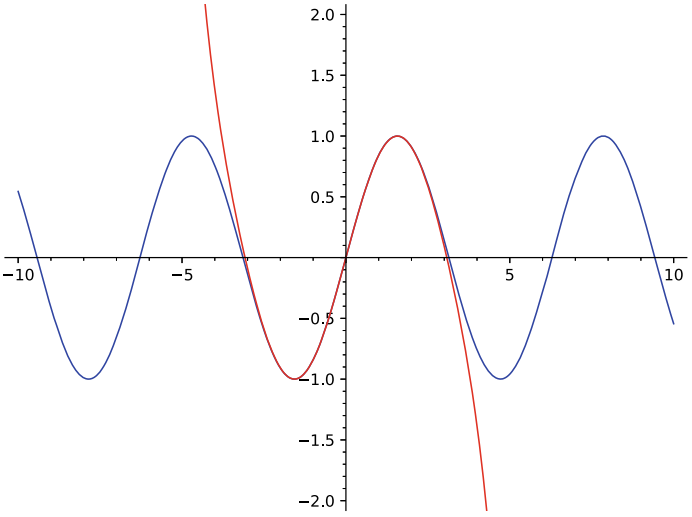
```
sage: P1=plot(sin(x),(-10,10),color='blue')
sage: P2=plot(sin(x),taylor(x,0,5),(-10,10),ymin=-2,ymax=2,color='red')
sage: show(P1+P2)
```

I get this:



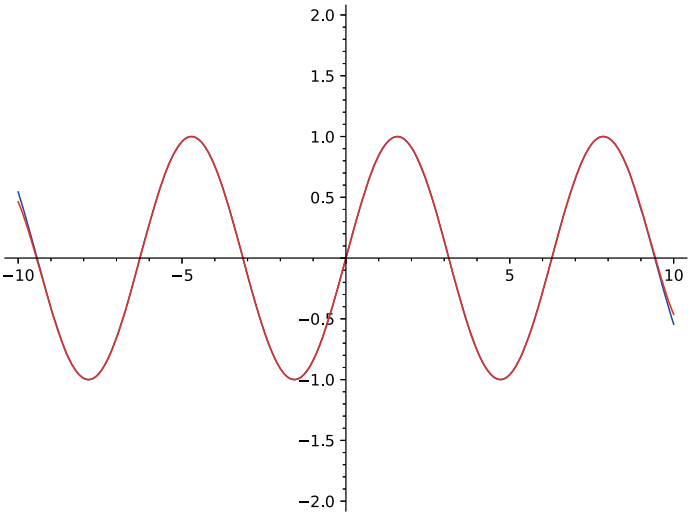

So $T_5(x)$ gives a good approximation when $-\pi/2 < x < \pi/2$ but it starts being bad when $|x| > \pi/2$. At this scale, we can hardly tell the difference between the two graphs when $x$ is close to 0.

Now do the same replacing the 5 by 7:




As you can see, the degree 7 approximation is closer than the degree 5 approximation, *both* in the sense that it gets closer when $x$ is small *and* in the sense that "small" includes a bigger range of choices for $x$. This seems to match the sine fairly well when $-\pi < x < \pi$.

Let's go the whole hog and replace 7 with 25:




In this range, one can hardly tell the difference. It takes quite a bit of zooming in to see that the graphs are not quite identical.

### 2.7.2 The cosine function

Everything we did works just as well for the cosine, except that the Taylor polynomials now only have *even* powers of $x$. We can find them by taking the derivative of the polynomials for the sine. Notice that

$$\frac{d}{dx}\left((-1)^k \frac{x^{2k+1}}{(2k+1)!}\right) = (-1)^k \frac{(2k+1)x^{2k}}{(2k+1)!} = (-1)^k \frac{x^{2k}}{(2k)!},$$

where in the last step we used the fact that $(2k+1)!$ differs from $(2k)!$ by a factor of $(2k+1)$, which gets canceled out.

So, we see that the $n$-th Taylor polynomial for $\cos(x)$ centered at zero is

$$T_n(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^k \frac{x^{2k}}{(2k)!}$$

where $2k$ is the largest even number less than $n$. Just as before, the error estimate is

$$\left|\cos(x) - T_{2n}(x)\right| \leq \frac{|x|^{2n+2}}{(2n+2)!},$$

which goes to zero when $n \to \infty$ no matter what $x$ is. I'll let you play the SageMath games with plotting.

### 2.7.3 The language of convergence

In each of these cases, we have Taylor polynomials $T_n(x)$ that approximate the function $f(x)$ arbitrarily well as we make the sum longer. The word used for this is *convergence*. We will give general definitions later, but let's say it all in this case.

Think about the infinite sum

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots$$

where the dots at the end indicate we go on forever.[2] To avoid saying "infinite sum" the standard term is "series" or (redundantly) "infinite series." The $k$-th *partial sum* is what you get by stopping when $n = k$, that is

$$k\text{-th partial sum} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

The difference between the $k$-th partial sum and the sine function is what we have been calling "the error." Since the error goes to zero for any value of $x$, these partial sums always get closer and closer to $\sin(x)$ as $k \to \infty$ We say the infinite series *converges* to $\sin(x)$ for every $x$ and write that as an equality:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots \qquad \text{for all } x.$$

Similarly, we have

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \ldots \qquad \text{for all } x.$$

These two series should be thought of as *different ways of writing the function*, just as $0.3333\ldots$ is another way to write $1/3$. The series give a way of approximating the values of the sine and the cosine (just take a long enough chunk), but they *also* give us a new way of saying what the sine and cosine are: $\sin(x)$ is whatever number the partial sums converge to. In other words, saying

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots \qquad \text{for all } x$$

is a way of saying what $\sin(x)$ means. And it has some advantages over other ways, the most obvious one being that we can actually compute $\sin(x)$ to any desired precision using it.

One should stop for a moment and contemplate how amazing this is. The information that went into constructing the series for sine and cosine was obtained by looking *very near* $x = 0$. All we used, after all, were the values of the derivatives at 0. Yet somehow that is enough to recover sine and cosine for *every* value of $x$. In other words, if we know a small bit of the graph near $x = 0$, the entire graph is determined!

[2] Of course, we *can't* literally go on forever, just as we can't write infinitely many 3s in the decimal expansion of $1/3$. But, as in that case, we know how to go on as long as necessary. We'll need to define what we mean when we talk of infinite sums and convergence. We give the definition informally in this section, then return to it in the next chapter.

### 2.7.4 Problems

**Problem 2.7.1:** To construct a table of sines, the ancient astronomer Ptolemy used an approximation method to compute the sine of one degree, which is the same as $\pi/180 \approx 0.017453$ radians. Use a Taylor polynomial to compute the sine to five decimal places.

**Problem 2.7.2:** We proved that for any value of $x$ the limit of $x^k/k!$ as $k$ goes to infinity is zero. So if we want $\left|x^k/k!\right| < 10^{-7}$ for some fixed $x$, it is possible to choose $k$ large enough to make that happen. Suppose $x = 10$. How large does $k$ need to be? (You can either use our proof to get an estimate or use a computer and brute force.)

## 2.8 The exponential

The second easiest example is the exponential function $f(x) = e^x$, again with $a = 0$. The best part is that in this case we know all the derivatives: $f^{(k)}(x) = e^x$ for every $k$, and so $f^{(k)}(0) = 1$ for every $k$. So, the coefficient of $x^k$ in the Taylor polynomial is just $1/k!$.

**Theorem 2.8.1.** *The Taylor polynomial of degree n centered at $a = 0$ for the function $f(x) = e^x$ is*

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}.$$

To see whether these polynomials approach $e^x$, we'll have to look at the error term, which involves finding a bound for the $(n+1)$-st derivative. The $(n+1)$-st derivative is of course $e^x$ again, but now we don't just want to plug in zero, but rather to consider $e^t$ for all $t$ between 0 and $x$. What we are looking for is a number that we can guarantee is larger than the largest possible value of $e^t$ in that range. The key is to remember that $e^t$ is an *increasing* function, so its highest value happens when we are furthest to the right.

**Case 1:** $x < 0$.

If $x < 0$ and $t$ is between 0 and $x$ then $t < 0$ too, so $e^t < 1$. So we can use $M_n = 1$ as our error bound on the $(n+1)$st derivative and we get

$$|e^x - T_n(x)| < \frac{|x|^{n+1}}{(n+1)!}.$$

As before, factorials beat powers, so we see that the error will converge to zero as $n \to \infty$ and we get arbitrarily good approximations as $n$ gets larger.

**Case 2:** $x > 0$.

If $x > 0$, then we want to consider $e^t$ for $0 < t < x$, and so we have $1 < e^t < e^x$. We don't know what $e^x$ is equal to, but it isn't infinite, so there is some number $M$

that is bigger (for example, we could take $M_n = M = 3^m$, where $m$ is some integer larger than $x$). And since all the derivatives are the same, this same $M_n = M$ works for every $n$. (This is key. If we couldn't use the same $M_n$ for every $n$, the argument that follows would fail.) So, we get

$$|e^x - T_n(x)| < M\frac{|x|^{n+1}}{(n+1)!}.$$

If $M$ is big, this will take a bit longer to converge to zero, but a fixed $M$ times something that goes to zero is still going to zero. So, in the limit, the error goes to zero, and so we get arbitrarily good approximations as the degree $n$ grows.

In the language we introduced before, what we have proved is that the series for the exponential always converges to $e^x$:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \qquad \text{for all } x.$$

When we computed approximations and errors for $e^x$ above, I asked you to check that the process was easier for $x < 0$. That's what we just saw: when $x < 0$, we can take $M_n = 1$ and get an easy bound. When $x > 0$ it is harder to come up with a bound, and in fact, for each $x$, the bound $M$ is a different number (which luckily does not depend on $n$).

Let's try to use the series and our knowledge of how the error behaves to compute $e$ itself to five decimal places. We know

$$e = e^1 = 1 + 1 + \frac{1}{2} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \ldots$$

The question is how far we have to go, and for that we need a bound on the error.

We found a formula above: the error is bounded by $M_n\frac{|x|^{n+1}}{(n+1)!}$. If $0 < t < 1$ we know $e^t < e < 3$, so we take $M_n = 3$ for every $n$. Since we want $x = 1$, the factor $|x|^{n+1}$ is just equal to 1. So, our error bound is

$$|e - T_n(1)| < 3\frac{1}{(n+1)!}.$$

We want five correct decimal places, so let's make sure the error is small enough for that. If we make sure the error is less than $10^{-6}$, we will know for sure that the five decimal places are correct. So now we need to choose $n$ so that will happen. For example, if $n = 4$, that's not enough, because our error bound would be $3\frac{1}{5!} = \frac{1}{40}$, which is far too big. So we experiment: make $n$ bigger until the error becomes small enough. Since I'm lazy, I defined a function and just plugged in some values of $n$.

```
sage: err(n)=3.0/factorial(n+1)
sage: err(7)
0.0000744047619047619
sage: err(8)
```

```
8.26719576719577e-6
sage: err(9)
8.26719576719577e-7
```

The `e-5` notation means "times $10^{-5}$." So the smallest $n$ that makes the error small enough is $n = 9$. That means we should estimate $e$ by adding up to degree 9:

$$e \approx 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \frac{1}{9!} = 2.71828152557...$$

(Of course I didn't add that up by hand.) How close is that to $e$? The difference, according to SageMath, is $3.0289 \times 10^{-7}$, so we have done better than we expected.

### 2.8.1 Problems

**Problem 2.8.1:** Use a Taylor polynomial for $f(x) = e^x$ near $a = 0$ to approximate $1/\sqrt[3]{e}$ to three decimal places. (The key here is to decide which $n$ you need to use.)

## 2.9 The geometric series

This example is known as the *geometric series*. It is very famous for two reasons. First, it occurs in a lot of practical applications. Second, we can actually compute the *exact* error and check when it becomes small. (This is a miracle: usually, the exact error is not accessible.)

We start with the function $f(x) = \dfrac{1}{1-x}$. We worked out its Taylor polynomial of degree $n$ centered at 0 in section 2.2. We got

$$T_n(x) = 1 + x + x^2 + x^3 + \cdots + x^n.$$

To compute the error, we have to work out the difference

$$\frac{1}{1-x} - \left(1 + x + x^2 + x^3 + \cdots + x^n\right).$$

We could use the error estimate, but this is just algebra, so we can avoid doing the hard work of computing and estimating derivatives. Let's just do the algebra.

To subtract

$$\frac{1}{1-x} - \left(1 + x + x^2 + x^3 + \cdots + x^n\right),$$

I need to have a common denominator, which will have to be $1 - x$ itself. That will look like this:

$$\frac{1}{1-x} - \frac{(1-x)(1 + x + x^2 + x^3 + \cdots + x^n)}{1-x}.$$

Let's work out the numerator of the second fraction:

$$\begin{aligned}
&(1-x)(1+x+x^2+x^3+\cdots+x^n)\\
&=1(1+x+x^2+x^3+\cdots+x^n)-x(1+x+x^2+x^3+\cdots+x^n)\\
&=1+x+x^2+x^3+\cdots+x^n-(x+x^2+x^3+x^4+\cdots+x^{n+1})\\
&=1-x^{n+1}
\end{aligned}$$

(That's an amazing amount of cancellation!) So the error is

$$\frac{1}{1-x}-\left(1+x+x^2+x^3+\cdots+x^n\right)=\frac{1}{1-x}-\frac{1-x^{n+1}}{1-x}=\frac{x^{n+1}}{1-x}.$$

So we know the error *exactly* for every $x$ and every $n$.

How big is the error? Well, the denominator is fixed once we know $x$, and the numerator is just $x$ to a big power. If $|x|>1$ that tells us that the error is *large*. For example, if we take $x=2$, then the error is $-2^{n+1}$, which gets larger and larger (in absolute value) as $n$ grows. But if $|x|<1$, then we are good: powers of $x$ get very small, and larger $n$ means smaller error.

To summarize: if $f(x)=\dfrac{1}{1-x}$, then

$$T_n(x)=1+x+x^2+\cdots+x^n$$

and we have

$$|f(x)-T_n(x)|=\frac{|x|^{n+1}}{|1-x|}.$$

If $|x|<1$, the error goes to 0 as $n\to\infty$. If $|x|>1$, the absolute value of the error goes to infinity as $n\to\infty$.

It's always good to remember that when $A=B$ then also $B=A$. In some practical applications what we actually want to compute is the finite sum of powers of $x$. So let's rearrange our formula and record what we have proved about that.

$$1+x+x^2+\cdots+x^n=\frac{1}{1-x}-\frac{x^{n+1}}{1-x}=\frac{1-x^{n+1}}{1-x}.$$

This is true for every $x\neq 1$. (It's just algebra.)

The upshot is that we have proved that

$$\frac{1}{1-x}=1+x+x^2+\cdots+x^n+\ldots \qquad \text{if } |x|<1.$$

In that equation, the equals sign has a technical meaning: it means that, for those values of $x$, the series converges to $1/(1-x)$. If $|x|\geq 1$, the series does not converge. We say then that the series *diverges*.

This is very different from the other examples. For sine, cosine, and exponential, we ended up with a series that represented the function for all values of $x$. That did not happen here. Instead, we get a series that matches the function in a certain range,

when $-1 < x < 1$, and does not converge at all for other values of $x$. This makes the equals sign even more tricky than before: not only does "equals" really mean "converges to" but also comes with the caveat "but only for these values of $x$."

In books, one often finds this series in a slightly different form in which we multiply both sides by some constant $a$. We can write the finite geometric sum as

$$a + ax + ax^2 + \cdots + ax^n = \frac{a - ax^{n+1}}{1 - x}$$

and the geometric series as

$$a + ax + ax^2 + \cdots + ax^n + \cdots = \frac{a}{1 - x} \qquad \text{if } |x| < 1.$$

Some books will use $r$ instead of $x$, mainly for historical reasons.

We'll go back to the geometric series and why it is useful later on. For now, let me just comment that it is pretty much the *only* case in which we can compute the error explicitly, without needing to approximate at all.

### 2.9.1 Problems

**Challenge:** If we take $x = 1/2$ our formulas say

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots = 2.$$

Can you find a way to draw a picture that shows this?

**Problem 2.9.1:** Of course, we don't expect

$$\frac{1}{1 - x} = 1 + x + x^2 + \cdots + x^n + \ldots$$

to work when $x = 1$, because $f(1)$ is undefined. But we could take $x = -1$, and then $f(-1) = 1/2$. Check that in that case the error is always equal to $\pm\frac{1}{2}$, so that it does not go to zero as $n$ grows, i.e., the series does not converge.

**Problem 2.9.2:** When we write $\frac{1}{3} = 0.33333\ldots$ it may not really be clear what the right-hand side means.

a. Explain how to interpret the infinite decimal as a geometric series, and check that the equation is indeed true.

b. What is $0.999999\ldots$ equal to?

**Problem 2.9.3:** Suppose we are rolling a six-sided die. We know that each of the numbers 1, 2, 3, 4, 5, 6 occurs with the same probability, $1/6$.

a. What is the probability $P(1)$ that we will get a six in the first roll?

b. What is the probability $P(2)$ that we will not get a six in the first roll, then get one on the second roll?

c. Fix an integer $n > 0$. What is the probability $P(n)$ that we first time we get a six is on the $n$th roll?

d. It is possible that we will get a six only after a very large number of rolls, so $P(n)$ is never zero. Show, however, that $\lim_{n\to\infty} P(n) = 0$.

e. The sum of the probabilities of all possible events should be 1. Check that this is true for our situation.

**Problem 2.9.4:** (A bit of mathematical folklore.) Two cyclists, 120 miles apart, approach each other, each pedaling at 10 miles per hour. A fly starts at one cyclist and flies back and forth between the cyclists at 15 miles per hour. When the cyclists come together (and squash the fly between them), how far has the fly flown?

**Problem 2.9.5:** For which values of $x$ do these series converge?

a. $\displaystyle\sum_{n=0}^{\infty} \left(\frac{3}{x}\right)^n$ b. $\displaystyle\sum_{n=1}^{\infty} (\sin x)^n$

**Challenge:** We avoided using our error estimate for Taylor polynomials in this section. What happens if we try to do it that way?

## 2.10 The (natural) logarithm

The logarithm is one of the most useful functions we have, but we have no direct way to compute it. In that way, it is much like the exponential, sine, and cosine. In this section, we will find a series for it and (without proof) its range of convergence. Naturally, we will work with $\ln(x)$, which is in every way the "right" logarithm function. Notice that SageMath uses `log` for the natural logarithm (so do all mathematicians, except in calculus class).

One issue we have to deal with is that we can't plug in zero: $\ln(0)$ is undefined. We could instead use $a = 1$, but we prefer to use series centered at 0, so instead we will work with the function $f(x) = \ln(1 + x)$.

The key thing to remember is that, if $f(x) = \ln(1 + x)$, then $f'(x) = 1/(1 + x)$. But we already know a series for that! If we change $x$ to $-x$ in the geometric series, we get

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \ldots \qquad \text{if } |x| < 1.$$

Since Taylor polynomials play nice with derivatives (i.e., using Theorem 2.3.2), we can just integrate this to get a candidate series for the logarithm:

$$\ln(1+x) = C + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \ldots.$$

Here $C$ is the arbitrary constant that we always end up with when we take an antiderivative, but we can easily figure out what the right choice of constant is: if we plug in $x = 0$ on both sides, we get $0 = C + 0$, so we need $C = 0$. So the series for $\ln(1 + x)$ should be

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \dots .$$

This is the right series; how about convergence? Well, we could try to use the error estimate and check, but instead, I'll mention a theorem that settles the issue: integrating a Taylor series preserves convergence (for more detail, see section 4.3). So we conclude

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \dots \qquad \text{if } |x| < 1.$$

**Challenge:** "We could try to use the error estimate," he said, but would we succeed?

## 2.11 The binomial series

Suppose $p$ (for "power") is a real number and $f(x) = (1 + x)^p$. For this to make sense for general powers, $1 + x$ needs to be positive, so we need to assume $x > -1$.

Why would we want a series for this function? Well, consider some special values for $p$:

- If $p = \frac{1}{2}$, we have $f(x) = \sqrt{1+x}$, which is a function we know and love.
- If $p = n$ is a positive integer, we get $(1 + x)^n$, which is a polynomial of degree $n$. Its Taylor series will be the polynomial itself.
- If $p = -1$ we get $f(x) = 1/(1+x)$, and we already know that one: it's basically a geometric series. But if $p = -2$ we get $1/(1 + x)^2$, for which we don't yet have a series. So taking $p$ to be a negative integer can be interesting.
- Another common choice is $p = -\frac{1}{2}$, which is relevant because we often have to deal with things like $1/\sqrt{1+x}$.
- When we deal with circles, we often run into expressions involving $\sqrt{1-x^2}$. We can get a series for this by starting from a binomial series, as we will see later.

Usually, then, $p$ will either be an integer of a fraction.

To find the Taylor polynomials, we use brute force and compute many derivatives:

a. First we plug in to find the degree 0 term: $f(0) = (1 + 0)^p = 1$.

b. Next we need the derivative: $f'(x) = p(1+x)^{p-1}$, so $f'(0) = p$.

c. Now the second derivative: $f''(x) = p(p-1)(1+x)^{p-2}$, so $f''(0) = p(p-1)$.

d. Each time we take another derivative, we get another factor in front and the power goes down. The factors start at $p$ and then get smaller by integers: $p-1$, $p-2$, and so on. After $k$ steps it looks like

$$f^{(k)}(x) = p(p-1)(p-2)\cdots(p-(k-1))(1+x)^{p-k},$$

(make sure you check that). So plugging $x = 0$ into the $k$-th derivative gives

$$f^{(k)}(0) = p(p-1)(p-2)\cdots(p-(k-1)).$$

That's a little like a (descending) factorial, except that $p$ doesn't need to be a whole number.

To get the coefficients of the Taylor polynomials we now divide by the factorials. So the coefficient of $x^k$ term will look like

$$\frac{p(p-1)(p-2)\cdots(p-(k-1))}{k!}.$$

That ugly mess is sometimes abbreviated using the notation

$$\binom{p}{k} = \frac{p(p-1)(p-2)\cdots(p-(k-1))}{k!}.$$

This is called a generalized binomial coefficient. If you have met binomial coefficients before, check that when $p$ is a positive integer this gives the usual answer.

An example, for sanity's sake. Say $p = 1/2$ and $k = 5$. Then

$$\begin{aligned}\binom{1/2}{5} &= \frac{1/2\cdot(1/2-1)\cdot(1/2-2)\cdot(1/2-3)\cdot(1/2-4)}{5!}\\ &= \frac{(1/2)\cdot(-1/2)\cdot(-3/2)\cdot(-5/2)\cdot(-7/2)}{120}\\ &= \frac{3\cdot5\cdot7}{2^5\cdot120} = \frac{7}{256}.\end{aligned}$$

Using that notation, the Taylor polynomial of degree $n$ for $(1+x)^p$ is

$$T_n(x) = 1 + px + \frac{p(p-1)}{2!} + \cdots + \binom{p}{n}x^n.$$

Estimating the error $(1+x)^p - T_n(x)$ is pretty difficult in this case, so instead I'll just tell you the easy part of the answer: if $|x| < 1$ the error goes to zero as $n$ goes to infinity, so we get convergence. In other words,

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \cdots + \binom{p}{n}x^n + \ldots \qquad \text{if } |x| < 1,$$

where you should remember that

$$\binom{p}{n} = \frac{p\,(p-1)(p-2)\cdots(p-(n-1))}{n!}.$$

As the example with $p = 1/2$ suggests, it can be quite difficult to work out a general term without the binomial notation, but given a specific $p$ you should definitely try to do it.

### 2.11.1 Problems

**Problem 2.11.1:** Just to practice, find the Taylor series for $\sqrt{1+x}$. (This is $p = 1/2$, of course. I told you that you should try!) Try to work out the general term; if that is difficult, at least find the first few terms.

**Problem 2.11.2:** Suppose $p$ is a positive integer. Then $(1 + x)^p$ should just be a polynomial of degree $p$. For example,

$$(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1.$$

Check that the formula for the binomial series does what it should.

**Problem 2.11.3:** We have two ways to compute the Taylor series for the function

$$f(x) = \frac{1}{1+x}.$$

We can take the geometric series and replace $x$ with $-x$, or we can use the binomial series with $p = -1$. Check that both methods give the same answer.

## 2.12 Two monsters

All these pleasant examples might give you the impression that our method always works. Alas, this is not true. Here are two cases where things go badly wrong in two different ways. In both cases, I won't try to work out all the details. The goal instead is to alert you to the fact that bad things can and do happen.

Luckily, it takes a little bit of work to find really bad examples. Or maybe it's not luck, but rather that the reason we know and love the functions we know and love is that things do work out well for them. Our "everyday functions" are made out of polynomials, roots, exponential and logarithm, sine and cosine, all of which have Taylor series that behave nicely.

### 2.12.1 Convergence, but to the wrong answer

The first example is the rather simple-looking function $f(x) = e^{-1/x^2}$. We want to study its behavior near $x = 0$.

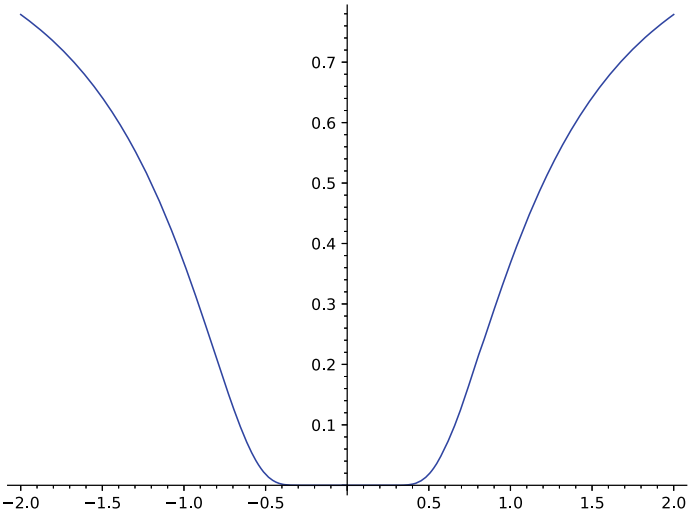
Of course, as written we can't put $x = 0$ into the formula, since $1/0$ is not defined. But as $x \to 0$ the exponent $-1/x^2$ goes to $-\infty$, and

$$\lim_{k \to -\infty} e^k = 0.$$

That says that the value $e^{-1/x^2}$ goes to 0 as $x$ goes to 0. So we define $f(0) = 0$. If you like, the function we are really working with is

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

The graph of $f(x)$ looks like this:




As you can see, the graph is very flat near the origin. It's straightforward (if you are good at using L'Hospital's rule) but annoying (even if you're good at using L'Hospital's rule) to check that our function is infinitely differentiable at $x = 0$. In fact,

$$f'(0) = 0,\ f''(0) = 0.\ f'''(0) = 0,\ \ldots\ f^{(n)}(0) = 0, \ldots$$

*All the higher derivatives are* 0 at $x = 0$. In other words, the differential calculus can't distinguish what this function does at $x = 0$ from what the constant function does. Of course, when $x \neq 0$ the derivatives are not zero since $f(x)$ is not actually constant.

What does that mean for the Taylor polynomials? Well, the coefficients are given by $f^{(n)}(0)/n!$, so they are all zero. So

$$T_n(x) = 0 + 0x + 0x^2 + 0x^3 + \cdots + 0x^n$$

is the zero polynomial and the Taylor series is the zero series. So the error is

$$f(x) - T_n(x) = f(x) - 0 = f(x).$$

The error doesn't depend on $n$, so it doesn't get smaller as $n$ grows.

Of course, it is still true that $f(x) \approx 0$ when $x$ is near enough to zero and our estimates are all still correct, but the polynomials don't say much more than "the graph is really very flat."

The annoying part is this: adding lots of zeros just gives zero, so the $T_n(x)$ "converge" as $n \to \infty$, but they converge to the zero function $g(x) = 0$, not to our starting function $f(x)$. So in this case we get a convergent series, but it converges to the wrong thing!

So this example shows that for some functions the Taylor polynomials get closer and closer to something, but that something is not the original function. And once we have one example we have many: consider, for example, $\sin(x) + e^{-1/x^2}$ and its Taylor series.

### 2.12.2 No convergence at all

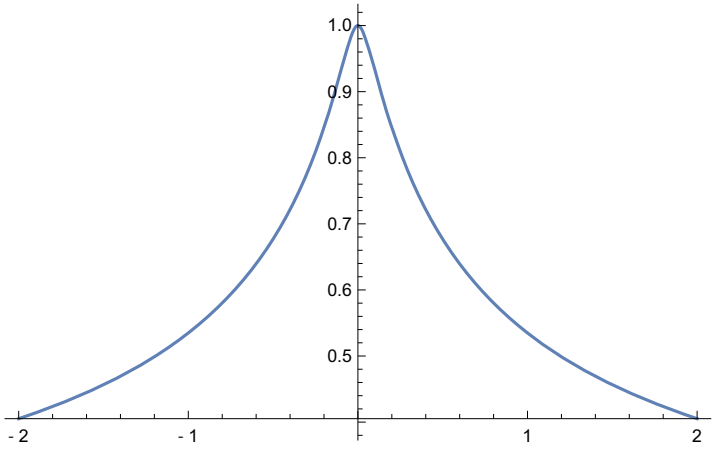
This situation is at the other extreme. It can happen that the Taylor series does not converge at any point except the center.

Here's an example.[3] The function is this:

$$f(x) = \int_0^\infty e^{-t} \cos(t^2 x)\, dt.$$

Since the thing we are integrating depends on $x$, the answer is a function of $x$. It seems hard to work with a function that is defined as the result of computing an integral, but it's actually straightforward. (Another reason to take a course in real analysis!)

The graph of that function looks like this:




[3] Taken from [5, Section 24].

It turns very quickly at the maximum point, but it's still a smooth curve.

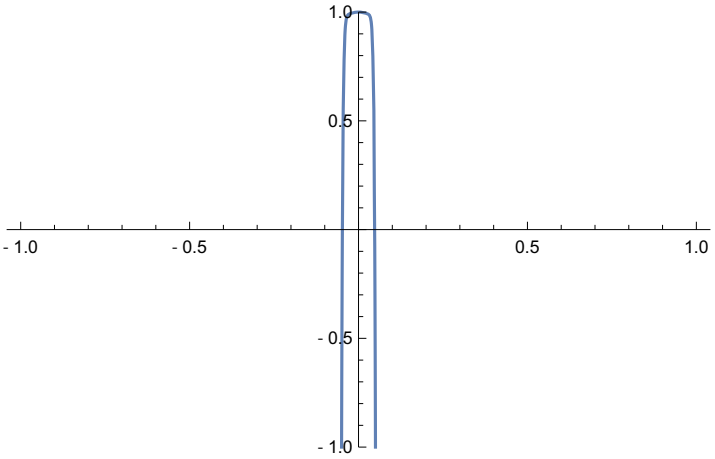
When we compute the derivatives at $x = 0$, we get this. The odd-numbered derivatives are all 0, while the even-numbered derivatives are

$$f^{(2n)}(0) = (-1)^n(4n)!$$

So the coefficient of $x^{2n}$ in the Taylor polynomials is $\pm(4n)!/(2n)!$. Here is the degree 14 polynomial:

$$T_{14}(x) = 1 - 12x^2 + 1680x^4 - 665280x^6 + 518918400x^8 - 670442572800x^{10} + 1295295050649600x^{12} - 3497296636753920000x^{14}$$

The coefficients grow very fast, since $(4n)!$ is enormously bigger than $(2n)!$. If $x \neq 0$, it certainly doesn't seem as if these polynomials will converge to anything. In fact, plotting the degree 14 polynomial (and showing only $-1 \leq y \leq 1$, since otherwise we'd only see a vertical line) gives this:




The value at $x = 0$ will always be 1, but as the degree of the polynomial goes up the curve gets narrower and narrower. It also flips from positive to negative: the polynomial for $n = 16$ would go upwards from 1 instead.

It's not hard to show that for any fixed $x \neq 0$ the absolute value of the sum gets bigger and bigger as we make the degree larger, so the polynomials are not giving us better and better approximations. In other words, there are Taylor polynomials, but if $x \neq 0$ they do not converge to anything at all as $n \to \infty$

### 2.12.3 Banning the monsters

Monsters of both kinds are scary and in a technical sense plentiful.[4] But in everyday mathematics, we run into them only rarely. It is good to know they are there, however, so that we can be ready for them if we are unlucky.

[4] A good place to look for the technical information on this, including what I mean by "plentiful," is [5, Section 24].

From now on, however, we will focus[5] on what happens when we do have convergence, at least for some values of $x$. Functions that are not monsters, so that their Taylor series exist and converge to them at least for some values of $x$, are called *analytic*. What our monsters show is that there exist functions that are infinitely differentiable but are not analytic, either because the Taylor series doesn't converge at all or because it converges to the wrong thing.

## 2.13 Series to know in your sleep

To conclude this chapter, let's collect our positive results. We have found several cases where the Taylor polynomials do give better and better approximations for our starting function, at least for a certain range of values of $x$. The basic ones are so important that I like to call them the series you should know in your sleep. Should someone get into your dorm and shake you in the middle of the night, demanding you tell them the series for the sine, you should be able to answer! Knowing a series includes knowing the range in which it is valid, so I have included that each time.

**Exponential**

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \qquad \text{for all } x.$$

**Sine**

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots \qquad \text{for all } x.$$

**Cosine**

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \ldots \qquad \text{for all } x.$$

**Geometric**

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \ldots \qquad \text{if } |x| < 1.$$

**Logarithm**

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^n \frac{x^{n+1}}{n+1} + \ldots \qquad \text{if } |x| < 1.$$

[5] We are "banning the monsters," to use language due to Imre Lakatos [16].

**Binomial**

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \cdots + \binom{p}{n}x^n + \ldots \qquad \text{if } |x| < 1,$$

where

$$\binom{p}{n} = \frac{p(p-1)(p-2)\cdots(p-(n-1))}{n!}.$$




# 3 Going All the Way: Convergence

We have just found several examples of infinitely long sums which we can plausibly say are equal to something. Or course, there is some kind of limit involved, since we cannot actually add infinitely many numbers. In this chapter, we will see the formal definitions and look for criteria that let us decide when convergence happens and when it doesn't. That will allow us to explore, in the next chapter, the convergence behavior of power series.

To see how naturally infinite sums appear, consider this very old example. Imagine a bar two feet long. We can cut it in half. The left piece will be one unit long:

$$2 = 1 + 1.$$

Now split the second piece in half to get a piece of size one half:

$$2 = 1 + \frac{1}{2} + \frac{1}{2}.$$

Keep doing it...

$$\begin{aligned} 2 &= 1 + 1 \\ &= 1 + \frac{1}{2} + \frac{1}{2} \\ &= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \\ &= \dots \\ &= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \frac{1}{2^n} \end{aligned}$$

for every $n$. Does it make sense to do it infinitely many times and end up with

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots = 2?$$

Well, not if we actually think of adding infinitely many things; we don't have that much time. But yes if we are willing to define the equality in terms of increasingly close approximations. Indeed, this equality is what we get when we take $x = 1/2$ in the geometric series.

## 3.1 Definitions

So far we have only looked at the convergence of Taylor series, for which there is the complicating factor of the variable $x$. Sometimes a Taylor series converges for

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F. Q. Gouvêa, *A Short Book on Long Sums*, Reading in Mathematics,
https://doi.org/10.1007/978-3-031-37557-6_3

some values of $x$ and diverges for others. For the general theory, let's just assume that we have something that looks like the sum of infinitely many *numbers*. (That is the situation once we have plugged a number in for $x$, after all.)

To help understand the formal definitions, let's use as a running example the sum we just wrote down:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots = 2.$$

This section is mostly about giving a precise meaning to that equals sign.

A *series* is an infinite sum

$$a_0 + a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

The numbers being summed, $a_1$, $a_2$, etc., are called the *terms* of the series. So in our geometric example, the terms are $a_n = 1/2^n$. A formula for the terms that is valid for all $n$ is often called the *general term*.

The numbering of the terms is arbitrary. We could, for example, have called the first term $b_1$ instead. But then the general term will change: $b_n = 1/2^{n-1}$. We could even be perverse and call the first term $c_{17}$ if we wanted to.

Given a series, we look at the *partial sums* that we get by adding up to a point:

$$S_k = a_0 + a_1 + a_2 + \cdots + a_k.$$

In general it will be hard to work out what exactly the partial sums are equal to, but, as we saw above, geometric series are miraculous. So, in our geometric example, we know that

$$S_3 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{15}{8}$$

and even

$$S_k = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^k} = 2 - \frac{1}{2^k},$$

either by the computation we just did or by using our formula for the sum of a finite geometric sum.

For non-geometric series we won't be able to compute $S_k$, and we certainly won't be able to compute an error term. So we will need to come up with ways to figure out what the $S_k$ do even when we cannot work them out exactly.

One caution: we have to take the terms in order when making partial sums. So, for example, if we add only some even-numbered terms as in $a_0 + a_2 + a_4$ then this is *not* a partial sum in the mathematical sense, even though it's a sum of part of the terms. Our partial sums always include the entire initial chunk of the series.

Now we consider what happens to the partial sums as they get longer, i.e., as $k \to \infty$. If they converge to a limit, we say that the series *converges* and that the limit is the *sum* of the series. So if

$$\lim_{k \to \infty} S_k = S$$

then we say the series converges to $S$ and write

$$a_0 + a_1 + a_2 + a_3 + \cdots + a_n + \cdots = S.$$

In our geometric example, we know that $S_k = 2 - \frac{1}{2^k}$, which converges to 2.

The limits that appear here are just like the ones you have met before. What they mean is that the difference between the partial sums $S_k$ and the limit $S$ can be made as small as I want by choosing $k$ large enough. So when I claim the limit is $S$ what I am really claiming is that, no matter what your error tolerance is, I can satisfy it by making $k$ large enough, i.e., by proving something like "if you choose $k$ bigger than…, then …."

To see what can go wrong, look first at

$$1 + 1 + 1 + 1 + \ldots,$$

that is, a series with terms $a_n = 1$ for every $n$. In this case $S_k$ just keeps getting bigger as $k$ gets bigger, and doesn't converge to anything. The series diverges. (Since we do know the partial sums get bigger and bigger, people sometimes say it "diverges to infinity.")

Another example is to have $a_n = (-1)^n$. Then the series is

$$1 - 1 + 1 - 1 + 1 - 1 + \ldots$$

The partial sums are all equal either to 1 or to 0, so again they don't have a limit as $k \to \infty$. Suppose, for example, we try $S = 1/2$. Then the difference between $S_k$ and $S$ is always either $-1/2$ (if $S_k = 0$) or $1/2$ (if $S_k = 1$). No matter how large I make $k$ that distance is always $1/2$. It will be that way for any potential limit $S$, so this series diverges too, oscillating instead of going to infinity.

These are all very simple examples, but things can and do get more subtle. For example, look at this one:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \ldots$$

If you don't already know the answer, it is fairly difficult to decide whether this converges or not.

Of course the Taylor series for sine, cosine, and exponential converge to the sine, cosine, and exponential for any input $x$, so they provide many examples of convergent series: just choose a value for $x$. More interesting, perhaps, is the geometric series. For example, take $x = 1/10$ to get

$$1 + \frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + \cdots = \frac{1}{1 - \frac{1}{10}} = \frac{10}{9}.$$

Now multiply by $\frac{9}{10}$. We get

$$\frac{9}{10} + \frac{9}{10^2} + \frac{9}{10^3} + \frac{9}{10^4} \cdots = 1,$$

that is,

$$0.99999\cdots = 1.$$

As you see, there really is a link between decimal expansions and series.

### 3.1.1 Problems

**Problem 3.1.1:** Does the series

$$1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} + \dots$$

converge or diverge? If it converges, what is the sum?

**Problem 3.1.2:** Does the series

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} + \dots$$

converge or diverge? If it converges, what is the sum?

## 3.2 Some basic properties of convergence

We defined convergence using partial sums and taking a limit, so the usual properties of limits can be used. Suppose, for example, that the series

$$a_0 + a_1 + a_2 + \cdots + a_n + \dots$$

converges. Then we can conclude that the series

$$5a_0 + 5a_1 + 5a_2 + \cdots + 5a_n + \dots$$

also converges, since the partial sums of the second series are always equal to five times the partial sums of the first. Since

$$\lim_{n\to\infty} 5S_n = 5 \lim_{n\to\infty} S_n,$$

we can even conclude that the sum of the second series is five times the sum of the first. Scaling does not affect convergence, though of course it changes what the sum is.

Something similar works for adding. Suppose we have two convergent series

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots = A$$

and

$$b_0 + b_1 + b_2 + \cdots + b_n + \cdots = B,$$

where of course the equal signs mean "converges to." What can we say about the series whose terms are the sums $a_n + b_n$? Well, if we look at partial sums then

$$(a_0 + b_0) + (a_1 + b_1) + \cdots + (a_k + b_k) = (a_0 + a_1 + \cdots + a_k) + (b_0 + b_1 + \cdots + b_k),$$

since these are just finite sums. Now remember that the limit of a sum is just the sum of the limits (as long as the limits exist). So we see that

$$(a_0 + b_0) + (a_1 + b_1) + (a_2 + b_2)\cdots + (a_n + b_n) + \cdots = A + B.$$

Of course this only works when both series converge.

Here's a more interesting observation. Suppose we have a series

$$a_0 + a_1 + a_2 + \cdots + a_n + \ldots$$

and we somehow know that if we lop off the first two terms the resulting thing converges:

$$a_2 + \cdots + a_n + \cdots = S.$$

The partial sums of the complete series are

$$a_0 + a_1 + a_2 + \cdots + a_k = (a_0 + a_1) + (a_2 + \cdots + a_k),$$

so we are just adding a fixed number $(a_0 + a_1)$ to each partial sum. So the sum of the whole series is

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots = a_0 + a_1 + S.$$

The slogan for this is "convergence depends only on the tail." If we remove the first two (or, for that matter, the first one million) terms, we change the limit but we don't change the convergence behavior.

## 3.3 Convergence: an easy "no"

There is an easy condition that lets us say for sure that a series does *not* converge. Here it is.

Suppose you have a series

$$a_0 + a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

that does converge to a limit $S$. Then when $k$ is large enough the partial sums $S_k$ are very close to $S$. So we have

$$S - (a_0 + a_1 + a_2 + a_3 + \cdots + a_{k-1} + a_k) = \text{small}$$

and indeed the small difference goes to zero as $k$ increases. We also have

$$S - (a_0 + a_1 + a_2 + a_3 + \cdots + a_{k-1}) = \text{small}.$$

Two things that are both close to $S$ will be close to each other, so

$$\left(a_0 + a_1 + a_2 + \cdots + a_{k-1} + a_k\right) - \left(a_0 + a_1 + a_2 + \cdots + a_{k-1}\right) = \text{small}.$$

(If I wanted to be annoyingly precise I would say that the difference is at most 2 times small; do you see why?) But the difference between those sums is exactly $a_k$. So we see that:

**Theorem 3.3.1.** *If the partial sums get arbitrarily close to $S$ then the terms $a_k$ get arbitrarily close to zero.*

Or, more formally,

**Theorem 3.3.2.** *If a series*

$$a_0 + a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

*converges then*

$$\lim_{n\to\infty} a_n = 0.$$

This gives us a negative criterion:

**Theorem 3.3.3.** *If $\lim_{n\to\infty} a_n$ does not exist or exists and is not equal to zero, then the series $a_0 + a_1 + a_2 + \cdots + a_n + \ldots$ does not converge.*

We have already seen two examples of this: both

$$1 + 1 + 1 + \cdots + 1 + \ldots$$

and

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots + (-1)^n + \ldots$$

diverge, since in the first case the limit of the terms is 1 and in the second case the limit of the terms does not exist.

But be careful! That tells us nothing at all about what happens when the limit of the terms *is* zero. In fact, as we will show, in that case the series might or might not converge. Just understanding the limit of the terms isn't enough to settle the issue.

A small warning: in this situation (and in others soon to come up) there are two limits going on. The first is the limit of the terms $a_n$. The second is the limit of the partial sums, which, if it exists, is the sum of the series. Students are always tempted to use phrases like "it converges." The problem is that here there are two different "it"s! Our last theorem says "if the terms do not converge to zero, then the series does not converge to anything." Never use "it" unless it's clear what "it" is. In the following, I will try to use "converge" and "diverge" *only in reference to series*, and say "the limit exists" or "does not exist" when I am talking about some other limit.

## 3.4 Convergence: when we can compute the partial sums

There are very few cases where we can just compute the partial sums exactly, in which case it is often easy to decide whether a series converges. One of them we have already met: the geometric series

$$1 + x + x^2 + \cdots + x^n + \ldots$$

converges to $1/(1-x)$ as long as $|x| < 1$. The reason we know that is that we can actually compute the partial sums explicitly. We saw that

$$1 + x + x^2 + \cdots + x^n = \frac{1}{1-x} - \frac{x^{n+1}}{1-x},$$

so we can see the convergence happen when $x^{n+1} \to 0$.

There are only a few other examples where we can compute partial sums explicitly. The most common one is when there is cancellation. For example, suppose we have the divergent series

$$1 - 1 + 1 - 1 + 1 - 1 + \ldots$$

Then we see at once that if we add an odd number of terms we get 1 and if we add an even number of terms we get 0.

Of course, we always have to work with *finite* sums in order to find partial sums. Indeed, we should always keep in mind that infinite sums are not really sums at all, they are limits, and so weird things can happen.

Here's a more interesting example of (nicely hidden) cancellation. Look at the series

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} + \ldots$$

All the terms are positive, so there's no cancellation to consider yet. But suppose we notice something. (Leibniz was the one who first did this, I think, so I'm not suggesting it's easy to think of it!)

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Let's use that on a partial sum:

$$\begin{aligned} S_k &= \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1}\right) \end{aligned}$$

almost everything cancels, so we get

$$= 1 - \frac{1}{k+1}.$$

Now it's easy to see that the limit as $k \to \infty$ is 1. So we have shown that

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} + \cdots = 1.$$

This situation, when each term turns out to be equal to a difference and we get cancellation of all the middle terms, is called "telescoping." So this is an example of a telescoping series. They are somewhat artificial, but every so often a useful one shows up. Leibniz was interested in this one because the denominators are exactly twice the triangular numbers.

## 3.5 Alternating series: a slightly harder "yes"

The example we want to consider here is of a kind of series for which we can prove convergence even though we don't know what the limit is. In other words, we are going to consider certain series for which we can prove $\lim S_k$ exists even though we do not know what the sum $S$ should be.

Let's look at the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^n \frac{1}{n+1} + \ldots$$

The general term looks that way because my labeling starts at $a_0 = 1$, forcing me to say $a_n = \dfrac{(-1)^n}{n+1}$. We could also start at 1; that would change the formula for the general term, but not the overall series.

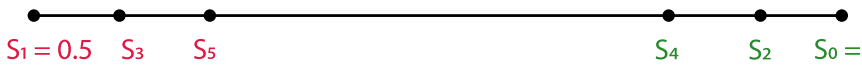
The features we will be using are three:

- The terms alternate in sign.
- The absolute values of the terms are decreasing.
- The limit of the terms is zero.

Any series with these properties will behave the same way, but we'll use our specific series to make things more concrete.

As we know, deciding on convergence means looking at the partial sums. So let's.

- $S_0 = 1$ and $S_1 = 1 - \frac{1}{2} = \frac{1}{2} = 0.5$.
- $S_2 = 1 - \frac{1}{2} + \frac{1}{3} = 0.8333$ and $S_3 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = 0.583333$.
- $S_4 = 0.783333$ and $S_5 = 0.616666$.




What do we see? It seems that the even-numbered sums (the ones that end with adding something) are getting smaller. It seems that the odd-numbered sums (the ones that end with a subtraction) are getting bigger. This is promising, since it seems that there must be a limit in the middle there somewhere. Of course, we have to convince ourselves that the pattern will continue forever.

**The even-numbered partial sums:** We know that

$$\begin{aligned} S_0 &= 1 \\ S_2 &= 1 - \frac{1}{2} + \frac{1}{3} = S_0 - \frac{1}{2} + \frac{1}{3} = 0.833 \\ S_4 &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = S_2 - \frac{1}{4} + \frac{1}{5} = 0.78333. \end{aligned}$$

How do we get $S_6$? Well, we take $S_4$ and add two more terms to it. So

$$S_6 = S_4 - \frac{1}{6} + \frac{1}{7}.$$

Notice that *we subtract more than we add*, so the answer gets smaller. This always happens, because each term has smaller absolute value than the previous one.

In general,

$$S_{2n} = S_{2n-2} - \frac{1}{2n} + \frac{1}{2n+1} < S_{2n-2}$$

because $1/(2n+1)$ is smaller than $1/2n$. So *the even-numbered partial sums form a decreasing sequence*

$$S_0 > S_2 > S_4 > \ldots$$

**The odd-numbered sums:** This is the same, but we add first, then subtract, so *we add more than we subtract*, since we know that the absolute values of the terms are getting smaller. The general formula is

$$S_{2n+1} = S_{2n-1} + \frac{1}{2n+1} - \frac{1}{2n+2} > S_{2n-1}.$$

In other words, *the odd-numbered partial sums form an increasing sequence*

$$S_1 < S_3 < S_5 < \ldots$$

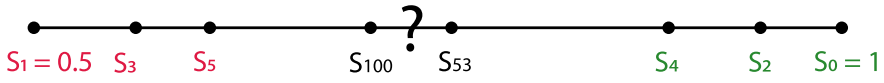
**Do they ever overlap?** We know that we have

$$S_0 > S_2 > S_4 > \ldots$$

and

$$S_1 < S_3 < S_5 < \ldots$$

But we also have, at the beginning at least, that $S_1 < S_0$. So the picture is that we have a large starting point $S_0$ after which the even ones get smaller and a small starting point $S_1$ after which the odd ones get bigger. So at first evens are bigger than odds but they are marching toward each other. Do they ever overlap? For example, could it happen that by the time we get to $S_{100}$ the evens have gotten past one of the odds? Could $S_{100}$ be smaller than $S_{53}$, for example?

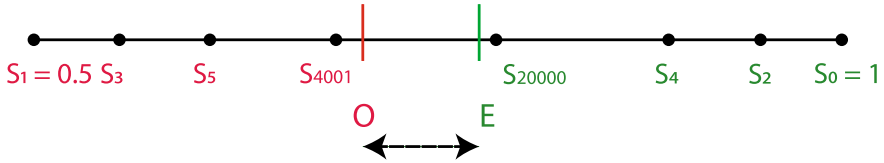



Here's how to see that it can't happen. (This is tricky; watch my hands!) When we add up to get $S_{100}$, the last thing we do is add something (we add $\frac{1}{101}$, but that doesn't matter). So to get $S_{101}$ we will subtract. So we know that $S_{101} < S_{100}$. But

we *also* know that the odd-numbered sums are increasing, so $S_{53} < S_{101}$. The two inequalities together show that $S_{53} < S_{100}$. So it cannot happen that $S_{100}$ is smaller.

Generalizing, we see that *any odd-numbered sum is smaller than any of the even-numbered sums.*

**How far apart are they?** Now we have a very interesting picture. The odd-numbered sums start small and get bigger, but never bigger than any of the even-numbered ones. The even-numbered sums start big and get smaller, but never smaller than any of the odd-numbered sums. So only two things can happen: either both sequences converge to the same limit in the middle or each converges to a separate limit. We want to convince ourselves that the latter is impossible.



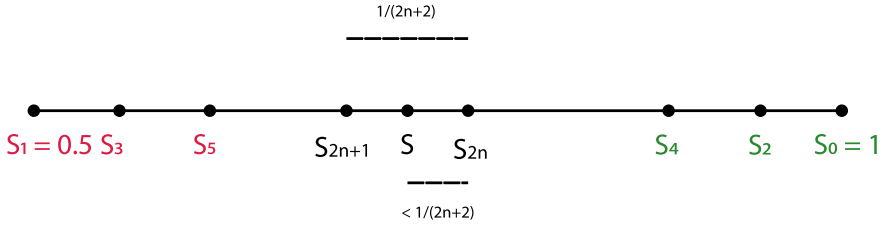

Suppose the even-numbered sums converge to a limit $E$ and the odd-numbered ones converge to a different limit $O$. Of course we must have $O < E$. All the even sums will be above $E$, approaching it from above, while all the odd sums will be below $O$, approaching it from below. That means that the distance between an odd sum and an even sum is always at least $E-O$. But that can't be! The distance between $S_{2n}$ and $S_{2n+1}$ is exactly $1/(2n+2)$, which goes to zero as $n$ gets bigger. So it is impossible to have two different limits $O$ and $E$. Instead, the two sums must converge to the same limit, which means all the sums get close to the same number $S$, and the series converges.

This argument has shown that if a series has the three properties we highlighted (alternating, terms decreasing in absolute value with limit zero), then it converges.

**And one more thing!** We actually get a bit more information that is worth recording. For any $n$ the limit $S$ must be between $S_n$ and $S_{n+1}$, because one is an odd sum and the other is an even sum. Since $S_{n+1} = S_n \pm \dfrac{1}{n+2}$, we see that

$$|S - S_n| < \frac{1}{n+2}.$$

The picture shows the situation when $n$ is even.



Or, to make it more general, if the terms $a_n$ have the properties we are assuming, then the series converges to a sum $S$, and we have

$$|S - S_n| < |a_{n+1}|.$$

In English, *the difference between the limit and the n-th partial sum is smaller than the first unused term.*

In our example, we have shown that the series

$$1 - \frac{1}{2} + \frac{1}{3} - \dots$$

converges. We don't know what the sum is. But we do know that if we take a partial sum, say

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{100} = 0.688172,$$

then the difference between that number and the real value of the series is at most equal to the first term we did not use, which is $1/101 \approx 0.0099$. That tells us that this partial sum should agree with the final sum to at least two decimal places. In fact, the true sum is approximately 0.693147, and the error we make by stopping at the 100th term is about 0.005.

Of course, it's very unsatisfying to just say "the series converges to something" if we can't decide what that something is. Can you guess what it is in our example? (Hint: take one of the series you know in your sleep and push the boundaries.)

OK, here's the upshot, then:

**Theorem 3.5.1** (Alternating Series). *Suppose we have a series*

$$a_0 + a_1 + a_2 + \dots$$

*such that*

*a. The terms alternate in sign.*

*b. The absolute values $|a_n|$ of the terms are decreasing.*

*c. The limit of the terms is zero, i.e., $\lim_{n\to\infty} a_n = 0$.*

*Then we can conclude*

*a. The series converges to a limit $S$.*

*b. The difference between $S$ and the partial sum $a_0 + a_1 + \dots + a_n$ is smaller than the absolute value of the first term we did not use, i.e.,*

$$\left| S - (a_0 + a_1 + \dots + a_n) \right| < |a_{n+1}|.$$

This is particularly nice because many of the series we know in our sleep are alternating! The series for $\sin(x)$, $\cos(x)$, and $\ln(1+x)$ all alternate in sign (the last one only if $x > 0$) and, since they converge, the terms tend to zero. Because we did all that work using the Taylor error bound, we know what they converge to. As long as the terms are decreasing, we can use this theorem to conclude that the error is bounded by the first unused term. That is *much* easier than trying to find the darned $M_n$.

### 3.5.1 Problems

**Problem 3.5.1:** There is a big difference between "converges" and "converges quickly"! The series in this section provides an example. We saw that after adding 100 terms all we could say was that the partial sum

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{100} = 0.688172$$

was no more than 0.005 away from the true answer. All we got from adding 100 terms were two decimal places.

To see that it is not just a matter of our error estimate being bad, we can use the fact that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln(2).$$

Suppose we want the answer with error less than $10^{-6}$. How many terms do we need to add?

**Problem 3.5.2:** One can prove that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36} + \cdots = \frac{\pi^2}{12}.$$

How many terms does one have to add in order to be sure to get $\pi^2/12$ with error less than $10^{-6}$?

## 3.6 The harmonic series: a not-so-easy "no"

The easy way to tell that a series will not converge is to see that its terms do not go to zero. But, as we emphasized, that only works to show something does *not* converge. In this section, we will look at an example of a series whose terms do go to zero, but which diverges anyway. This famous example is known as the harmonic series.

Let's take the very simple series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \ldots$$

This is just like the example we studied in the previous section, but now all the terms are positive, so we don't get any of the cancellation we saw before. In fact, now the partial sums do not oscillate.

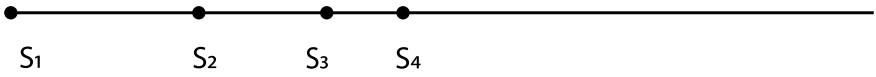
Let's compute a few partial sums:

$$S_1 = 1, \qquad S_2 = 1.5, \qquad S_3 = 1.833, \qquad S_4 = 2.0833, \qquad S_5 = 2.2833\ldots$$

They keep getting bigger. That is to be expected, since each time we are adding a positive number $1/n$. That number gets very small as we keep going, but it is always positive, so each sum is a little bigger than the previous one. In other words, the partial sums are always increasing:

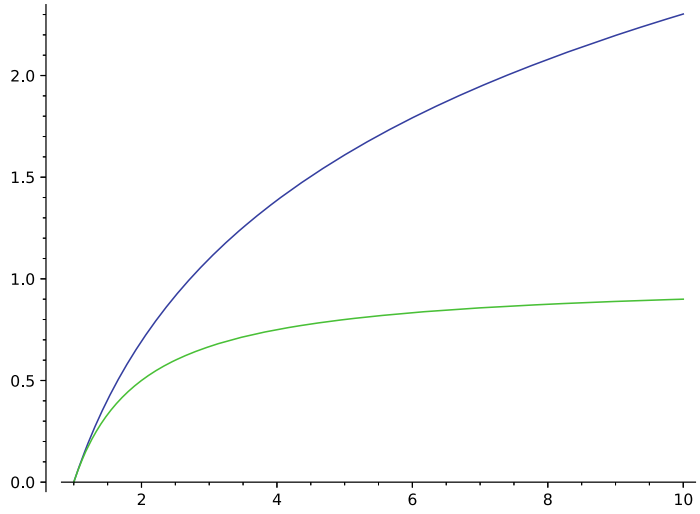
$$S_1 < S_2 < S_3 < S_4 < \cdots < S_n < S_{n+1} < \ldots$$

We can make a picture like the ones in the previous section:




The partial sums increase all the time; the gap between two of them is equal to the last term we added, which in this case is just $1/n$. So the gaps get smaller as we go along.

What can happen? It might help to look at an example from single-variable calculus first. Let's graph two increasing functions. In blue, we have $f(x) = \ln(x)$, while in green we have $g(x) = 1 - 1/x$.



Both functions increase as $x$ grows, and both increase less and less, which is like our partial sums that always increase but with gaps that get smaller. But the logarithm increases forever, while $1 - 1/x$ is asymptotic to $y = 1$:

$$\lim_{x\to+\infty} \ln(x) = +\infty, \qquad \lim_{x\to+\infty} \left(1 - \frac{1}{x}\right) = 1.$$

Those are the only two things that can happen for an increasing function.

The same thing is true for increasing partial sums: either they approach a limit from below, like the asymptotic function, or they keep growing and go to infinity. In the latter case, there is no limit and the series does not converge.

To show that for the harmonic series the partial sums do keep getting bigger. We'll use an idea that goes back to Nicole Oresme, writing around 1350. He grouped the series into chunks ending at a power of 2:

$$\begin{aligned}
&1 + \frac{1}{2} && && = \frac{3}{2}\\
&\frac{1}{3} + \frac{1}{4} && > \frac{1}{4} + \frac{1}{4} && = \frac{1}{2}\\
&\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} && > \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} && = \frac{1}{2}\\
&\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \cdots + \frac{1}{15} + \frac{1}{16} && > 8\,\frac{1}{16} && = \frac{1}{2}\\
&\frac{1}{17} + \frac{1}{18} + \frac{1}{19} + \frac{1}{20} + \cdots + \frac{1}{31} + \frac{1}{32} && > 16\,\frac{1}{32} && = \frac{1}{2}.
\end{aligned}$$

Adding up, what this tells us about the partial sums is

$$\begin{aligned}
S_1 &= 1,\\
S_2 &= 1 + \frac{1}{2},\\
S_4 &> 1 + \frac{2}{2},\\
S_8 &> 1 + \frac{3}{2},\\
S_{16} &> 1 + \frac{4}{2},\\
&\text{etc.}
\end{aligned}$$

and in general, for every $n \geq 2$,

$$S_{2^n} > 1 + \frac{n}{2}.$$

This shows that the partial sums get arbitrarily big, since $\frac{n}{2} \to \infty$. In other words,

**Theorem 3.6.1.** *The harmonic series diverges.*

On the other hand, the estimate $S_{2^n} > 1 + \frac{n}{2}$ suggests that it takes quite a while, since it says that doubling the number of terms adds at least $1/2$ to the partial sum. That means that the estimate is growing like the logarithm of the number of terms.

For example, if we want a partial sum that is bigger than $10 = 1 + \frac{18}{2}$, our estimate says we need to add up to $n = 2^{18} = 262,144$.

Notice, however, that all we know is that adding $2^{18}$ terms is enough. Maybe our estimate for $n$ is too big, and we can get a partial sum greater than 10 much sooner than that. Let's use SageMath to check.

```
sage: var('n')
n
sage: sum(1.0/n,n,1,2^18)
13.053866822328144
```

So we overshot; this partial sum is more than 10, but we could have taken fewer terms. What this means is each partial sum is actually bigger than our estimate suggests, but it's hard to tell by how much. That makes them diverge even faster, of course, but it would be nice to have a more precise estimate. The exercises below pick up on this.

Meanwhile, the thing to remember is that the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \ldots$$

does not converge even though $\frac{1}{n} \to 0$. Looking at the general term is not enough to decide on convergence.

### 3.6.1 Problems

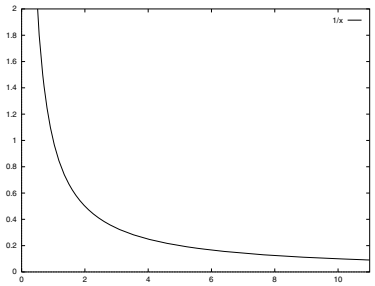
**Problem 3.6.1:** Our attempt, with $n = 2^{18} = 262{,}144$ terms gave much too big an answer. Play around with SageMath to find out how many terms are actually needed to make $S_n > 10$. Is the answer surprising?

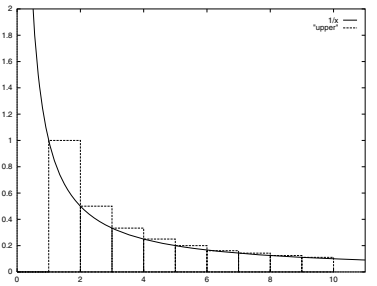
**Problem 3.6.2:** To get a more precise error term, let's look more carefully at the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$$

We'll use a graphical approach to get an approximation to the partial sums.

First, notice that $1/n$ is what you get when you plug in $x = n$ into the function $f(x) = 1/x$, whose graph is drawn on the left below:




Now, for each $n$, draw a box of height $1/n$ and width 1, with its bottom left corner at the point $x = n$, as in the second graph.

a. What is the area of the $n$-th box? What is the sum of the areas of the first $k$ boxes?

b. Use your answer to write down an integral that is smaller than the $k$-th partial sum of the harmonic series. That is, it gives a *lower bound* for the $k$-th partial sum.

c. Compute your integral, and check that its limit, as $k$ goes to infinity, is infinity. Since the partial sums of the series are larger, this means they go to infinity too.

d. Can you use a similar picture to find an *upper* bound for the $k$-th partial sum?

e. Explain why $\ln(k)$ is a pretty good estimate for the $k$-th partial sum.

## 3.7 Series with positive terms, part 1

There is one observation we made when analyzing the harmonic series that generalizes and is useful. It applies whenever we have a series where all of the terms are *positive* and leads to another way to decide whether such a series converges or not, known as the comparison test.

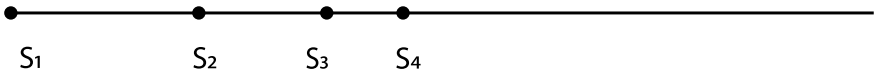
If we have a series

$$a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

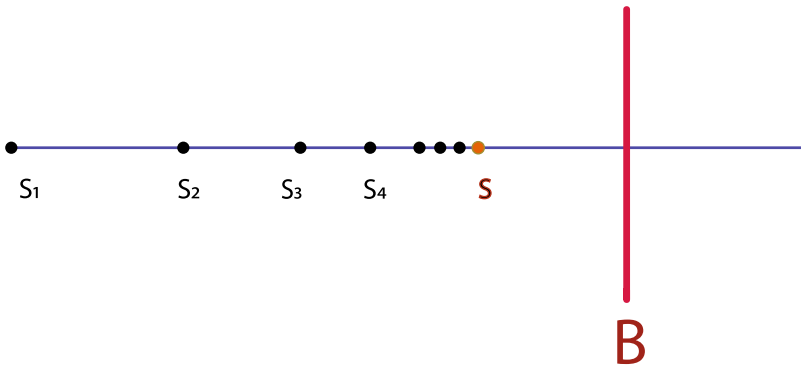
with all of the terms positive, that is, $a_n > 0$ for all $n$, then the partial sums will be an *increasing* sequence:

$$S_1 < S_2 < S_3 < S_4 < \cdots < S_n < S_{n+1} < \ldots$$

That's easy to see, since to get $S_{n+1}$ we add $a_{n+1} > 0$ to $S_n$: adding positive terms makes things bigger. So the partial sums look sort of like this:




Now, as we pointed out when studying the harmonic series, an increasing sequence can only do two things as $n \to \infty$. Either it keeps growing arbitrarily large, so $S_n \to +\infty$, or it converges to a limit. That's because[1] if we have a bound $S_n \leq B$ then it acts as a barrier: the partial sums can't grow bigger than $B$ and can't backtrack. They will have to approximate some number (it could be $B$ itself or some smaller number). Since they are increasing, there is no chance of any other behavior.




So that tells us:

**Theorem 3.7.1.** *In a series with all of the terms positive*

$$a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

*either the partial sums go to infinity or the series converges.*

For the harmonic series, we used the bad side of this: we showed that the partial sums were not bounded and concluded that the harmonic series did not converge. Here we highlight the good side:

**Theorem 3.7.2.** *If a series with all of the terms positive*

$$a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

*has bounded partial sums, then it converges.*

The point is that we only need to know the partial sums are bounded, which is usually easier than trying to get at what they are actually equal to.

One way this gets used is as follows: suppose we have two series with all terms positive and that we happen to know that $a_n \leq b_n$ for all $n$ *and* that the series $b_1 + b_2 + \ldots$ converges. Then we know:

[1] The claim that bounded increasing sequences must converge, which we also used when studying the alternating series, is just one way of saying that the real line does not have any holes. (This property is known as "completeness.") One can think of it as one of our base assumptions about the real numbers. If you are curious about this, plan to take a course in Real Analysis.

- The partial sums $T_n = b_1 + b_2 + \cdots + b_n$ are bounded.
- We have $S_n = a_1 + a_2 + \cdots + a_n \leq b_1 + b_2 + \cdots + b_n = T_n$ for every $n$.

So the $S_n$ must be bounded too: the bound on the $T_n$ serves as a bound for the $S_n$ as well. From the theorem, we can then conclude that the smaller series $a_1 + a_2 + \ldots$ also converges. (This is the *comparison test.*) We can also conclude that

$$\lim_{n\to\infty} S_n \leq \lim_{n\to\infty} T_n,$$

so if we know the sum of the bigger series we get a little bit of information about the sum of the smaller series.

**Theorem 3.7.3** (Comparison Test). *Suppose we have two series with positive terms,*

$$a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

*and*

$$b_1 + b_2 + b_3 + \cdots + b_n + \ldots$$

*for which we can show that*

*a. The series $b_1 + b_2 + b_3 + \cdots + b_n + \ldots$ converges.*

*b. For every n, $a_n \leq b_n$.*

*Then the series $a_1 + a_2 + a_3 + \cdots + a_n + \ldots$ converges*

Here's an example of how we can use this. Take the series from section 3.4

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \cdots + \frac{1}{n(n+1)} + \ldots$$

We showed that it converges to 1. Now compare it to

$$\frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots + \frac{1}{(n+1)^2} + \ldots$$

Since for all $n \geq 1$

$$\frac{1}{(n+1)^2} < \frac{1}{n(n+1)}$$

(bigger denominator = smaller fraction) and we know the first series converges, we can conclude that the second series also converges. We can also conclude that the sum is less than 1, which is the sum of the bigger series.

Once we know the series starting with $1/4$ converges, we can add 1 in front without changing the fact that it converges, since that just adds 1 to all the partial sums. So we conclude that the series of reciprocals of squares converges:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots + \frac{1}{n^2} + \cdots = \text{ something }.$$

But all we know is that the sum is less than 2 (adding one to what we knew before). (This is where Leibniz and the Bernoulli brothers were stuck until 1736 or so, when Euler discovered that the sum is actually $\pi^2/6$. That surprised everyone: why would $\pi$ show up?)

The comparison only needs to hold for large enough $n$, since the behavior of the partial sums when $n \to \infty$ only depends on the tail, when $n$ is large. The little trick of lopping off a term then putting it back after we know the rest converges also tells us something important, namely, that we can even handle a shift like $a_n \le b_{n+1}$. Since scaling also doesn't change convergence, we can use that as well: if $a_n \le 12b_{n+5}$ for all $n \ge 10^6$ and the $b$-series converges, then so does the $a$-series. If you push this you will end up with a convergence condition called "limit comparison," which I will let you look up if you are curious.

The big problem with this is that in general we only have the series in which we are interested (the $a$-series) and it's not always easy to find a $b$-series that we can use. That requires a creative idea. In the next section, we will come up with a test (called the *ratio test*) that avoids that problem.

Meanwhile, get some practice.

### 3.7.1 Problems

**Problem 3.7.1:** Use the comparison test to show that

$$1 + \frac{1}{8} + \frac{1}{27} + \cdots + \frac{1}{n^3} + \ldots$$

converges.

**Problem 3.7.2:** Generalize: show that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges when $s > 2$.

**Problem 3.7.3:** We can also use the comparison test to show a series does not converge. Show that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^s}$ diverges when $s < 1$.

**Problem 3.7.4:** Show that $\displaystyle\sum_{n=0}^{\infty} \frac{1}{2n+1}$ diverges.

**Challenge:** In this set of exercises, we have shown that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \ldots$$

diverges if $s \le 1$ and converges if $s \ge 2$. Can you settle the remaining cases, when $1 < s < 2$? (One method is to follow the ideas in Problem 3.6.2.)

## 3.8 Series with positive terms: part 2

The goal of this section is to introduce one of the more useful tests used to determine whether a series converges. One of its best features is being quite simple, but it is particularly useful for us because it is well suited to the kind of series we are really most interested in, the power series which involve powers of $x$ or of $x - a$.

The test involves a comparison, but one that is hidden from view. Here's the basic idea. Look at a geometric series

$$a + ar + ar^2 + ar^3 + \cdots + ar^n + \ldots$$

(this is the series you know in your sleep except that we have made $x = r$ (because we want to save $x$ for something else) and multiplied through by $a$). It's easy to see that the ratio between each term and the previous one is exactly $r$:

$$r = \frac{ar}{a} = \frac{ar^2}{ar} = \frac{ar^3}{ar^2} = \cdots = \frac{ar^{n+1}}{ar^n} = \ldots$$

We can't expect that in a general case

$$a_0 + a_1 + a_2 + \cdots + a_n + \ldots,$$

but suppose it is *almost* true, at least for large $n$. To write down something (a little) more precise, suppose that there is some large number $k$ for which we have

$$r \approx \frac{a_{k+1}}{a_k} \approx \frac{a_{k+2}}{a_{k+1}} \approx \frac{a_{k+3}}{a_{k+2}} \approx \ldots$$

Then it seems like we could say that the series looks, in the long run, a lot like a geometric series, and so should converge as long as $r < 1$.

I'll give a precise proof below, but let me state the conclusion first. If that handwaving suffices for you, feel free to ignore the proof, which depends on a subtle (but interesting) trick.

**Theorem 3.8.1** (The Ratio Test). *Suppose*

$$a_0 + a_1 + a_2 + \cdots + a_n + \ldots$$

*is a series with positive terms, so* $a_n > 0$ *for all n. Suppose also that we have*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = L.$$

*Then*

- *If* $L > 1$ *the series diverges.*
- *If* $L < 1$ *the series converges.*
- *If* $L = 1$ *or the limit does not exist, we cannot conclude either way.*

This is very useful, because it does not involve finding another series for a comparison, just computing the limit. Of course, it is not infallible: the limit may fail to exist or be equal to 1.

**Proof of the ratio test:**

Suppose first that $L > 1$. Then for large enough $n$ the ratio $\frac{a_{n+1}}{a_n}$ will be close to $L$, hence will be greater than 1. That says $a_{n+1} > a_n$ for all large enough $n$. But then (since all terms are positive) we cannot have $a_n \to 0$ and the series cannot converge.

Suppose $L < 1$. (Here's the subtle trick part.) Choose a number $r$ that is slightly larger than $L$ but still less than 1, so that $L < r < 1$. Since the ratio $\frac{a_{n+1}}{a_n}$ converges to $L$, it will eventually have to be smaller than $r$. Let $N$ be large enough so that $\frac{a_{n+1}}{a_n} < r$ for all $n \geq N$. Then we have $a_{N+1} < ra_N$. But we also have $\frac{a_{N+2}}{a_{N+1}} < r$, so, plugging in, $a_{N+2} < ra_{N+1} < r^2 a_N$. Going on like that we see

$$a_{N+k} < r^k a_N \qquad \text{for all } k \geq N.$$

That means that the series

$$a_N + a_{N+1} + a_{N+2} + \dots$$

has terms that are smaller than those of the series

$$a_N + ra_N + r^2 a_N + \dots$$

This last one is *geometric*, and since $r < 1$ it converges. By the comparison test, the former series converges. Adding the initial segment back in, we conclude that the original series

$$a_0 + a_1 + a_2 + \cdots + a_n + \dots$$

converges.

So we have proved the first two claims. To show things are inconclusive when $L = 1$, I'll let you check that for both the divergent harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \dots$$

and the convergent

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} + \dots$$

we will have $L = 1$. So if $L = 1$ we really don't know what will happen.

Finally, if there is no limit we have nothing to work with, so there's no conclusion in this case either.

So, end of proof. Notice that the main tool here was a comparison, so all of this works only for series whose terms are *positive*. But wait, absolute values will rescue us (sort of).

### 3.8.1 Problems

**Problem 3.8.1:** Let's try this on some series. In each case apply the ratio test and decide what it tells us. (The "what it tells us" will always be either "the series converges," "the series converges," or "the ratio test is inconclusive.") I've given examples both in summation notation and written out.

a. $1 + 3 + \frac{3^2}{2!} + \frac{3^3}{3!} + \cdots + \frac{3^n}{n!} + \ldots$

b. $\sum_{n=0}^{\infty} \frac{(4n)!}{(2n)!}$

c. $\sum_{n=1}^{\infty} \frac{n^2+1}{2^n}$

d. $1 + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \cdots + \frac{n}{n+1} + \ldots$

e. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

## 3.9 One way to handle series whose terms are not all positive

As usual, I'm going to consider an example, but everything I will do applies in general. I invite you to check that as we go along.

Let's consider this series:

$$1 - \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \frac{1}{25} - \frac{1}{36} + \frac{1}{49} - \cdots \pm \frac{1}{n^2} \pm \ldots$$

where the pattern of signs is $+\ -\ -$ throughout. We can't use any of our results for series whose terms are positive, because these aren't. And we can't use our alternating series theorem because the signs don't alternate. So what can we do?

Well, notice first that if we replaced each term by its absolute value, making all the signs plus, then we would get

$$\sum_{n=1}^{\infty} \frac{1}{n^2},$$

which is a convergent series, as we showed above. Adding signs creates cancellation, and that should *help* with convergence. Can we prove that?

Let's consider four different series. We'll call the original one, with $+\ -\ -$ signs, **O** (for "original"). We'll call the one with all plus signs **A** (for "absolute value"). Next

we'll create a series **P** (for "positive") like this: when the term from **O** is positive, keep it; when it is negative, write down 0. Finally, we create **N** the obvious way, with a twist: when the term in **O** is negative, write down its absolute value, and write down 0 otherwise. Here's what it looks like:

$$\begin{array}{lccccccc}
\mathbf{O} & 1 & -\frac{1}{4} & -\frac{1}{9} & +\frac{1}{16} & -\frac{1}{25} & -\frac{1}{36} & +\frac{1}{49} & \dots \\
\mathbf{P} & 1 & +0 & +0 & +\frac{1}{16} & +0 & +0 & +\frac{1}{49} & \dots \\
\mathbf{N} & 0 & +\frac{1}{4} & +\frac{1}{9} & +0 & +\frac{1}{25} & +\frac{1}{36} & +0 & \dots \\
\mathbf{A} & 1 & +\frac{1}{4} & +\frac{1}{9} & +\frac{1}{16} & +\frac{1}{25} & +\frac{1}{36} & +\frac{1}{49} & \dots
\end{array}$$

Now notice that the last three series all have positive terms, so we can use the comparison test on them. First compare **P** and **A**: every term in **P** is either equal to the corresponding term in **A** or it is zero, which is less than the corresponding term in **A**. Since **A** converges, we can conclude that **P** converges as well. Similarly, every term in **N** is either equal to or less than the corresponding term in **A**, so we can conclude that **N** converges as well.

Suppose we write $N_k$ for the $k$-th partial sum of **N** and $P_k$ for the $k$-th partial sum of **P**. Since both series converge, these partial sums converge to something. Let's say

$$\lim_{k\to\infty} N_k = N \qquad \text{and} \qquad \lim_{k\to\infty} P_k = P.$$

The $k$-th partial sum of the original series is $O_k = P_k - N_k$ (for finite sums we can rearrange, so we can add all the + terms and then subtract all the − terms). So the limit of the partial sums of the original series is

$$\lim_{k\to\infty} S_k = \lim_{k\to\infty} (P_k - N_k) = \lim_{k\to\infty} P_k - \lim_{k\to\infty} N_k = P - N.$$

Since the last two limits exist, so does the limit of the $S_k$, and the original series **O** will converge.

The upshot is this:

**Theorem 3.9.1** (Absolute Convergence). *If the series* $\sum_{n=0}^{\infty} |a_n|$ *converges, then so does the series* $\sum_{n=0}^{\infty} a_n$. *When this happens we will say that the series* $\sum_{n=0}^{\infty} |a_n|$ converges absolutely.

We already know an example that shows that not all series that converge do it

absolutely. The alternating series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

converges, but if we take absolute values we get the harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

which we know does *not* converge. But our result is still useful, because it will allow us to use tests that apply only to series with positive terms (especially the ratio test) to more general series.

**Two notes:**

1) Series that converge but whose absolute value series do not converge are said to converge *conditionally*, since their convergence depends on the way the signs are arranged.[2] In your real analysis course (you're going to take one, right?) you will learn that absolute convergence is much friendlier than conditional convergence. For example, if a series is absolutely convergent then reordering the terms will not change convergence nor the sum. On the other hand, a conditionally convergent series can be reordered to give a different sum, or to fail to converge at all. So the commutative law is false for "infinite addition" unless the convergence is absolute.

2) The proof above is based on a fact about the real numbers: the only solutions to $|x| = b$ are $x = b$ and $x = -b$. That is what allowed us to split our series into **P** and **N**. This is not true for complex numbers, but the theorem is still true. That tells me that this is not the best proof of the theorem. The best proof uses the Cauchy Criterion, however, and that is a bit harder to understand, so the easier proof will do for us.

There is much more I could say about infinite series in general, but that would divert us from our real goal, which is to understand power series. If you want to see more, the natural place to start would be a real analysis textbook. You might also look at [6] and/or [15].

[2] That's the standard explanation, but I'm willing to bet that it's also a punning reference to the absolute/conditional distinction in philosophy.




# 4 Power Series

Now that we know the basics of convergence, let's go back to the thing we are really interested in, namely, series in increasing powers of $x$ or of $x - a$. We know that for any (infinitely differentiable) function $f(x)$ we can find a Taylor series for $f(x)$, which will look like

$$c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \ldots$$

for some list of numbers $c_0, c_1, c_2, \ldots$. We want to know whether this series converges and if so (a separate question!) whether the sum is actually $f(x)$. If this happens, then we can think of the series as another way to write down the function $f(x)$, one that is sometimes easier to work with.

We also want to reverse the process, however. Suppose we start from some list of numbers $c_0, c_1, c_2, \ldots$. Then we can make a series from them:

$$c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \ldots .$$

If for some value $x$ this series converges, we get a number that depends on $x$. If we call that number $f(x)$ we have just *defined* a new function, one whose Taylor series is the one we wrote down.

This is again like decimal expansions. If we start with a number like $1/3$, we get a decimal expansion $0.3333\ldots$. But we can do it backward too: if we write $0.12345\ldots$, that is some real number. (The implied series always converges; can you check that?) And I really know nothing about the number $0.12345\ldots$ except that expansion.

The first thing we need to do is decide when a power series converges (to something). Then we will consider how to manipulate known series to find new series. Finally, we will look at some of the things we can do with power series: create new functions, solve differential equations, and study numerical sequences via generating functions.

There is a name for series of the kind we want to study: power series. To be careful, here's a definition.

> A **power series** centered at $a$ is a series whose terms look like $c_n(x-a)^n$, where each $c_n$ is a number.

All of the Taylor series we have written down are power series. But of course there are series that do not look like this. For example, this is not a power series:

$$1 + 2(x-1) + 3(x-2)^2 + 4(x-3)^3 + \cdots + (n+1)(x-n)^n + \ldots$$

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F. Q. Gouvêa, *A Short Book on Long Sums*, Reading in Mathematics,
https://doi.org/10.1007/978-3-031-37557-6_4

There is a clear pattern involving powers, but there isn't a center as there must be for power series.

There are two big reasons to like power series. First, they are the ones we get out of the Taylor polynomials. In other words, they arise in a fairly "natural" way from the theory of derivatives. Second, and much more important, power series behave really well. That is the main message of this chapter.

## 4.1 Convergence of power series

Let's look at when a power series converges. If it is a series centered at some $a \neq 0$, we can always let $u = x - a$ to get a power series in powers of $u$ centered at 0. Whatever we prove about that series we can translate back replacing $u$ by $x - a$. So we will first do our analysis for series centered at 0, then translate to the general case.

Suppose we have a power series centered at $a = 0$:

$$c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \ldots$$

The first thing to note is that when $x = 0$ this is just equal to $c_0$, so there is always at least one value of $x$ for which the series converges. This is good to know but not very interesting.[1]

What about other values of $x$? Let's work out what happens in a special case, when none of the coefficients $c_n$ are zero. One way to check for convergence of a series (almost the only one we know) is to take absolute values (to get a series with positive terms) and then apply the ratio test. For that, we need to compute

$$\lim_{n\to\infty} \frac{|c_{n+1}x^{n+1}|}{|c_n x^n|} = \lim_{n\to\infty} \frac{|c_{n+1}|}{|c_n|} \frac{|x|^{n+1}}{|x|^n} = \lim_{n\to\infty} \frac{|c_{n+1}|}{|c_n|} |x| = L\,|x|$$

as long as the limit $\lim_{n\to\infty} \dfrac{|c_{n+1}|}{|c_n|}$ exists and is equal to $L$.

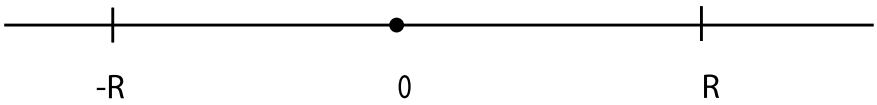
If $L = 0$, then $L\,|x| = 0$ is less than 1 no matter what $x$ is, and so no matter what $x$ is we can conclude that the series converges. So in this case the series converges for every $x$. If $L = \infty$, then $L\,|x| = \infty$ unless $x = 0$, so the series converges only when $x = 0$.

For the other cases, the answer will depend on $x$:

- The series converges when $L\,|x| < 1$, that is, when $|x| < \frac{1}{L}$.
- The series diverges when $L\,|x| > 1$, that is, when $|x| > \frac{1}{L}$.

We usually write $R = 1/L$ and call it the *radius of convergence*. Since saying $|x| < R$ means $-R < x < R$, this translates nicely to a picture:

[1] Still, it's nice; the non-power series above does not converge for any value of $x$.




Here $a = 0$ is at the center, and we know the series converges *inside* the range from $-R$ to $R$ and diverges *outside* that range. When $x = \pm R$ the limit in the ratio test is 1, and that tells us nothing, so we don't know whether the series converges or not at those points. If we want to know exactly for which $x$ there is convergence, those two values need to be considered separately.

The cases of $L = 0$ and $L = +\infty$ are usually included, like this:

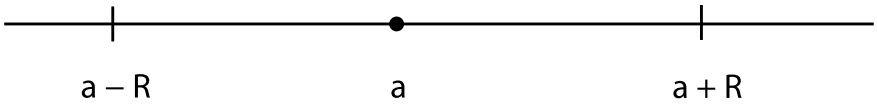
- If the series always converges, i.e., converges for every $x$, we say $R = \infty$. (Picture $\pm R$ moving farther and farther from zero.)
- If the series converges only for $x = 0$, then we say $R = 0$.

That way (as long as the ratio test limit exists or is $+\infty$) we always have a radius of convergence.

If instead of a series centered at zero we have one in powers of $x - a$, the result will be similar. The series will always converge when $x = a$. For other $x$, we use the ratio test; the limit looks like

$$\begin{aligned}\lim_{n\to\infty} \frac{|c_{n+1}(x-a)^{n+1}|}{|c_n(x-a)^n|} &= \lim_{n\to\infty} \frac{|c_{n+1}|}{|c_n|} \frac{|x-a|^{n+1}}{|x-a|^n} \\ &= \lim_{n\to\infty} \frac{|c_{n+1}|}{|c_n|} |x-a| \\ &= L\,|x-a|.\end{aligned}$$

Again we want to see if this is less than or greater than 1, and we end up with a condition that looks like $|x - a| < R$. This translates to $-R < x - a < R$ or, if you like, $a - R < x < a + R$. It's the same picture again: $a$ is at the center, and convergence happens as long as we are less than $R$ units away from $a$.




We did these computations assuming that the limit existed and that none of the coefficients were zero (which isn't true for the sine series!) and that the limit in the ratio test exists. But in fact (though we won't give a proof) it is the case that all power series behave this way.

**Theorem 4.1.1.** *Consider a power series*

$$c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \ldots$$

*Then one of the following things is true:*

*a. The series converges only if $x = a$. In that case we say that the radius of convergence of the series is $R = 0$.*

*b. The series converges for every $x$. In that case we say that the radius of convergence of the series is $R = \infty$.*

*c. There exists a number $R > 0$ such that the series converges when $|x - a| < R$ and diverges when $|x - a| > R$. We call $R$ the radius of convergence of the series.*

*In the third case, the series may or may not converge at the endpoints $x = a - R$ and $x = a + R$.*

There are two ways to prove this theorem. In the first, we use the comparison test to show that if the series converges when $|x - a| = b$ then it also converges when $|x - a| < b$. That implies that $R$ must exist, but doesn't find it. The second method actually finds the radius of convergence $R$ using a stronger result called the "root test" and a generalized notion of limit.

Given a power series, we can often find the radius of convergence $R$ by using the ratio test. Of course, this never tells us what the series converges to, but at least it may tell us that it converges to something. The ratio test never tells us what happens at the endpoints, when $|x - a| = R$, which have to be investigated directly.

## 4.2 Examples of finding the radius of convergence

We now know that power series always have a radius of convergence. The argument in the previous section suggested that we can find the radius of convergence by using the fact that $R = 1/L$ where

$$L = \lim_{n\to\infty} \frac{|c_{n+1}|}{|c_n|}.$$

That's actually not the best way to do it, because many power series have $c_n = 0$ for some values of $n$. Then the limit above won't exist, because many of the fractions have zero in the denominator. But when there are coefficients that are zero, there is usually a pattern (as in the sine function, whose even-numbered coefficients are zero), and it turns out we can still do the computation by working directly with the ratio test. So we'll do that in all our examples too.

First let's look at some of the series we know in our sleep.

**1. Exponential.** We know that for all $x$ we have

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots$$

So the radius of convergence of this series must be infinity. Let's use the ratio test to

see that it is. We need to compute this monster:

$$\lim_{n\to\infty} \frac{\dfrac{|x|^{n+1}}{(n+1)!}}{\dfrac{|x|^n}{n!}}.$$

In the denominator we just put the absolute value of general term of the series, while in the numerator we replace $n$ by $n+1$ everywhere. When factors are already positive, we don't need to worry about absolute values, of course. Then we rearrange and cancel as must as we can. (Factorials and powers of $|x|$ are great for canceling!)

So:

$$\begin{aligned}\lim_{n\to\infty} \frac{\dfrac{|x|^{n+1}}{(n+1)!}}{\dfrac{|x|^n}{n!}} &= \lim_{n\to\infty} \frac{|x|^{n+1}\, n!}{|x|^n\,(n+1)!} \\ &= \lim_{n\to\infty} \frac{|x|^{n+1}}{|x|^n}\,\frac{n!}{(n+1)!} \\ &= \lim_{n\to\infty} |x|\frac{1}{n+1} = 0,\end{aligned}$$

since $x$ is fixed and $n$ is going to infinity.

The limit is 0 no matter what $x$ is, so it is less than 1 no matter what $x$ is, which means that the series always converges no matter what $x$ is, i.e., the radius of convergence is infinity.

Notice that the ratio test does *not* tell us what the sum is, just that there is one. We know it is equal to $e^x$ because we estimated the difference between the partial sums and $e^x$ using our formula for the error in the approximation given by the Taylor polynomials. Finding the sum always takes a lot more work than just checking convergence.

**2. Sine and Cosine.** Now let's do the series for $\sin(x)$, which is very similar except for the fact that only odd powers of $x$ appear. We know

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots$$

Since this is true for every $x$, we already know $R = \infty$ but let's check it anyway. The main difficulty is getting the algebra right. So here we go.

Notice that replacing $n$ by $n+1$ changes $2n+1$ into $2n+3$, the next odd number. So the ratio test limit looks like

$$\lim_{n\to\infty} \frac{\dfrac{|x|^{2n+3}}{(2n+3)!}}{\dfrac{|x|^{2n+1}}{(2n+1)!}} = \lim_{n\to\infty} \frac{|x|^{2n+3}\,(2n+1)!}{|x|^{2n+1}\,(2n+3)!}.$$

Now notice that in the numerator there are two extra factors of $|x|$ , since $|x|^{2n+3} = |x|^2\,|x|^{2n+1}$; similarly,

$$(2n+3)! = (2n+3)(2n+2)\,(2n+1)!$$

So we have

$$\lim_{n\to\infty} \frac{\dfrac{|x|^{2n+3}}{(2n+3)!}}{\dfrac{|x|^{2n+1}}{(2n+1)!}} = \lim_{n\to\infty} \frac{|x|^{2n+3}\,(2n+1)!}{|x|^{2n+1}(2n+3)!}$$

$$= \lim_{n\to\infty} \frac{|x|^2}{(2n+3)(2n+2)} = 0.$$

Since the answer is less than 1 no matter what $x$ is, we can conclude $R = \infty$ again.

**3. One for which we don't know the answer yet.** Let's consider

$$\sum_{n=1}^{\infty} nx^{2n} = x^2 + 2x^4 + 3x^6 + \cdots + nx^{2n} + \ldots$$

For the ratio test, we need to consider the limit

$$\lim_{n\to\infty} \frac{(n+1)|x|^{2n+2}}{n|x|^{2n}}.$$

Notice again how we build that: the denominator is the general term; the numerator just takes the absolute value of the general term and replaces $n$ by $n+1$. Since $n$ and $n+1$ are positive, they don't need absolute values. As always, the key is to notice that most of the powers of $|x|$ cancel out. We get

$$\lim_{n\to\infty} \frac{(n+1)|x|^{2n+2}}{n|x|^{2n}} = \lim_{n\to\infty} \frac{n+1}{n}|x|^2 = |x|^2.$$

So the ratio test tells us that the series converges when $|x|^2 < 1$ and diverges when $|x|^2 > 1$. Taking square roots (everything is positive!), we see that the radius of convergence is $R = 1$.

For the previous examples we didn't need to worry about endpoints, because the series converged for every value of $x$. In this case, we know the series converges when $|x| < 1$ and diverges when $|x| > 1$, but we need to test what happens when $|x| = 1$. For that, we just plug in $x = \pm 1$. Since we have even powers throughout, in both cases we get

$$1 + 2 + 3 + \cdots + n + \ldots$$

which just gets bigger and bigger and so does not converge. So the series converges for $|x| < 1$ and diverges for $|x| \geq 1$.

**4. One final example.** Look at

$$\sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots .$$

The ratio test is easy:

$$\lim_{n\to\infty} \frac{\dfrac{|x|^{n+1}}{n+1}}{\dfrac{|x|^n}{n}} = \lim_{n\to\infty} \frac{n}{n+1}|x| = |x|,$$

so again we have convergence for $|x| < 1$ and divergence for $|x| > 1$, and the radius is $R = 1$.

What is interesting are the endpoints. When $x = 1$ we get

$$1 + \frac{1}{2} + \frac{1}{3} + \dots$$

That is the harmonic series, which we know diverges. But if we make $x = -1$ we get

$$-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots ,$$

which is a convergent alternating series. So $R = 1$, but we have some extra information: the series converges when $-1 \le x < 1$ and diverges when $x < -1$ or $x \ge 1$. This emphasizes that knowing $R$ doesn't tell us what will happen at the endpoints of the interval of convergence.

### 4.2.1 Problems

**Problem 4.2.1:** What is the radius of convergence for the rest of the series you know in your sleep?

**Problem 4.2.2:** Use the ratio test to decide for which values of $x$ the series below converge.

a. $\displaystyle\sum_{n=0}^{\infty} nx^n.$

b. $\displaystyle\sum_{n=0}^{\infty} \frac{8^n x^n}{n!}.$

c. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 2^n x^n}{n^2+7}.$

d. $\displaystyle\sum_{n=0}^{\infty} n(x-3)^n.$

e. $\displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2n}.$

f. $\displaystyle\sum_{n=0}^{\infty} \frac{n!x^n}{2^{3n+2}}.$

**Problem 4.2.3:** Find the radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{n^2 x^{2n}}{4^n}.$$

Can you decide what happens at the endpoints of the region of convergence?

**Problem 4.2.4:** Consider a power series $\sum_{n=0}^{\infty} c_n(x-3)^n$. Suppose you know that the series converges when $x = -1$ and that it diverges when $x = 9$. For each of the statements below, decide whether the truth or falsity of the statement can be determined from this information, and if so decide which it is.

a. The series converges when $x = 3$.

b. The series diverges when $x = 8$.

c. The series converges when $x = -9$.

d. The series converges for all $x$.

e. The series converges when $x = 1$.

**Problem 4.2.5:** Suppose you are writing an exam and want to include a problem where students use the ratio test to find a radius of convergence. You want it to be fairly interesting, so you don't want $R$ to be one of the standard trio 0, 1, $\infty$. You also do not want the center to be 0. Come up with a good example and write out the solution.

## 4.3 Power series freedom

Playing around with series as if they were finite sums can lead to mistakes. For example, if you change the order of the terms in a conditionally convergent series, you can change the sum. "Infinite addition," as opposed to ordinary finite addition, is not commutative.

But power series are special. This section summarizes several operations that can be done with power series without messing up convergence nor giving unexpected results. Some of these are fairly easy to prove, while others are much harder. (In the list below, the proofs get harder and harder as we go on.)

We will just state the results without trying to give proofs, but let's say a word (OK, two words) about why it all works. Two properties of power series are critical. The first you already know about: since we find the radius of convergence using the ratio test, we will always have *absolute* convergence, because that is what the ratio test tests for. (To be precise, the convergence is absolute except maybe at the endpoints.) The second property you haven't heard about yet (but see Section 5.2

if you are curious): on a closed interval inside the region of convergence, we will always have *uniform* convergence.

Taken together, these two properties guarantee that power series behave very nicely under all sorts of algebraic and calculus procedures. And that is wonderful: it allows us to find power series for complicated functions by using the series we know in our sleep.

**Plugging in:** We can always replace $x$ by something else; the resulting series will converge when the thing we plugged in satisfies the convergence condition. Start, for example, with the geometric series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \dots,$$

valid when $|x| < 1$. We can replace $x$ by $u - 1$ to get

$$\frac{1}{2-u} = 1 + (u-1) + (u-1)^2 + (u-1)^3 + \cdots + (u-1)^n + \dots,$$

which will converge when $|u-1| < 1$. Notice that the result is a power series centered at 1. We changed the variable to $u$, but variable names are arbitrary, so we could have used $x$.

A similar computation can be used to shift the center of a series. For example, since $1 - x = 3 - x - 2 = 3 - (x+2) = 3(1 - \frac{1}{3}(x+2))$, we can use the geometric series to compute

$$\frac{1}{1-x} = \frac{1}{3}\frac{1}{1-\frac{x+2}{3}} = \frac{1}{3}\sum_{n=0}^{\infty}\frac{(x+2)^n}{3^n} = \sum_{n=0}^{\infty}\frac{(x+2)^n}{3^{n+1}},$$

which is the Taylor series for $1/(1-x)$ centered at $a = -2$.

**Adding and Subtracting:** We can *add* or *subtract* two power series term by term. The region of convergence of the resulting series is the region where both series converge. So, for example, if we start with the series for $e^x$ and plug in $-x$ for $x$, we get a series for $e^{-x}$. Adding the two, we have

$$e^x + e^{-x} = \sum_{n=0}^{\infty}\frac{x^n}{n!} + \sum_{n=0}^{\infty}\frac{(-x)^n}{n!} = \sum_{n=0}^{\infty}\frac{x^n + (-x)^n}{n}.$$

Since the series for $e^x$ is valid for all $x$, so is this one. Now we can simplify. When $n$ is odd, $(-x)^n = -x^n$ and that term cancels out. When $n$ is even we have $(-x)^n = x^n$. So only the even terms survive; we can write the final series like this:

$$e^x + e^{-x} = \sum_{k=0}^{\infty}\frac{2x^{2k}}{(2k)!}.$$

If the summation notation confuses you, write out the series to see. The key thing is that we are adding term by term, pairing together those terms that have the same degree.

**Multiplying and Dividing:** We can also *multiply*, and again the convergence will happen wherever both series converge. But remember that when you multiply two sums you need to distribute one sum over the other (i.e., "foil it"), but there are infinitely many things to add. Dividing also works as long as the series we are dividing by is nonzero, but is even harder to work out the general term.

But you don't always want to multiply or divide by a whole series. Multiplying by $x$ is particularly easy and doesn't affect the convergence because the series $x$ isn't really a series. Dividing by $x$ is also safe as long as the series we start with has no term of degree zero. So, for example,

$$\frac{\sin(x)}{x} = \frac{1}{x}\sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty}\frac{x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots$$

This is true for any $x \neq 0$, and it shows that we should define the function to be 1 when $x$ equals zero.

**Derivatives and Integrals:** We can differentiate and integrate series term by term. This will not change the radius of convergence, though the behavior at the endpoints may change.[2] I hinted at this above when I used the series for $1/(1+x)$ to get a series for $\ln(1+x)$.

Here's a more interesting, and famous, example. The derivative of $f(x) = \arctan(x)$ is $f'(x) = \dfrac{1}{1+x^2}$. We can find a series for the derivative by starting from a geometric series. Here we go:

$$\frac{1}{1-u} = 1 + u + u^2 + \dots + u^n + \dots \qquad \text{if } |x| < 1.$$

Replacing $u$ by $-x^2$,

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 + \dots + (-1)^n x^{2n} + \dots,$$

valid if $|-x^2| < 1$, which just says $|x| < 1$ again. Now take the antiderivative:

$$\arctan(x) = C + x - \frac{x^3}{3} + \frac{x^5}{5} + \dots + (-1)^n\frac{x^{2n+1}}{2n+1} + \dots,$$

still valid when $|x| < 1$. Notice that computing antiderivatives always gives an arbitrary constant, but we can find the right constant by setting $x = 0$ on both sides: since $\arctan(0) = 0$, we must have $C = 0$. So finally

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} + \dots + (-1)^n\frac{x^{2n+1}}{2n+1} + \dots \qquad \text{if } |x| < 1.$$

This is one bit of evidence for a general rule: the arc tangent function behaves much better than the tangent function.

[2] This is the only case when the behavior at the endpoints can change.

In this example, we see a change in the behavior at the endpoints. The series for $1/(1+x^2)$ does not converge when $x = \pm 1$, but the series for $\arctan(x)$ does.

**Composition:** If we have the series for two functions $f(x)$ and $g(x)$ and $g(0) = 0$, then we can find a series for $f(g(x))$ by plugging in one series into another. This is a messy process, but the resulting series will converge to the composed function in some reasonable interval.

**Behavior at the endpoints:** Suppose we have a series for $f(x)$ that converges when $|x| < R$. As we know, we can't be sure what happens at the endpoints: the series might converge or not. But suppose it does converge: if we plug in $x = R$, do we get $f(R)$?

The answer is given by *Abel's theorem*. It says that as long as the function is continuous at $x = R$ and the series converges when $x = R$, then the sum will be $f(R)$. (We need both things: a continuous function and a convergent series.)

For example, look at the logarithm series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

We know this series converges to $\ln(1+x)$ when $|x| < 1$. But if we plug in $x = 1$ on either side everything works: $\ln(1+1) = \ln(2)$ and the function is continuous at that point, while the series becomes alternating:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Since we know the series converges and the function is continuous, Abel's theorem says that we can conclude that

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

On the other hand, plugging in $-1$ doesn't work for the function or the series.

Another fun example is the series for $\arctan(x)$. Since $\arctan(1) = \pi/4$ and the function is continuous everywhere, we get

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots$$

This series for $\pi/4$ was first discovered by Leibniz.

**What does it converge to?:** One of the nice things about all this is that these manipulations do not just preserve convergence. They also preserve what the series converges to. Since we have a set of series whose sums we know (in our sleep), we can use them to discover what other series converge to.

Here's a famous example that we looked at before, the series

$$f(x) = \sum_{n=1}^{\infty} nx^{2n} = x^2 + 2x^4 + 3x^6 + \dots + nx^{2n} + \dots$$

We saw above that it converges when $|x| < 1$, but we didn't know what it converges to. I called it $f(x)$ because I don't know what it is. Can we find out?

We can if we can figure out how to start with series we know to get this series. This takes some trickery, but it can be done. I'll actually work backward first, transforming this series into one we know. When I finally get a known function, I can unwind all my transformations to see what the original series is equal to.

I'll omit the many failed attempts to find just the right trick, but you can be sure they existed. (You should always keep this in mind when reading a mathematics book.)

Notice first that our series involves only even powers of $x$. I don't like that, so let's change the variable to $u = x^2$ to get rid of it. This gives some function

$$g(u) = u + 2u^2 + 3u^3 + \dots$$

which converges when $|u| < 1$.

Here's the dastardly trick: divide through by $u$. That won't affect the convergence, and

$$\frac{1}{u}\, g(u) = 1 + 2u + 3u^2 + \dots,$$

which looks like a derivative: the general term is $nu^{n-1}$. If we take the antiderivative we get a function

$$h(u) = C + u + u^2 + u^3 + \dots$$

If we take $C = 1$ this is the geometric series! As we'll see soon, we can choose any $C$ we like, so let's choose the good one. So we set $C = 1$ and we know that for $|u| < 1$,

$$h(u) = 1 + u + u^2 + u3 + \dots = \frac{1}{1-u}.$$

All right, we have a function now.

Now work backward. The derivative of $h(u)$ is

$$h'(u) = \frac{1}{(1-u)^2}.$$

(Since the derivative of a constant $C$ is zero, the answer is the same for any value of the constant $C$. So it doesn't matter that we chose $C = 1$).

But remember that $h'(u) = \frac{1}{u}g(u)$, so as long as $|u| < 1$ we have

$$\frac{1}{u}\, g(u) = \frac{1}{(1-u)^2}.$$

Multiply by $u$ to get $g(u)$.

$$g(u) = \frac{u}{(1-u)^2}.$$

We're almost there.

Now remember that we got $g(u)$ by making $u = x^2$ in the series for $f(x)$. So make $u = x^2$ in the answer too:

$$f(x) = \frac{x^2}{(1-x^2)^2}.$$

All of this is true when $|x| < 1$, since none of our moves changed the region of convergence. So now we know what our series converges to!

If $|x| < 1$,

$$x^2 + 2x^4 + 3x^6 + \cdots + nx^{2n} + \cdots = \frac{x^2}{(1-x^2)^2}.$$

Isn't that cool? Just try figuring that series out by taking derivatives...

(Did all that manipulation make you nervous? It did me. We can always check, however:

```
sage: taylor(x^2/(1-x^2)^2,x,0,10)
5*x^10 + 4*x^8 + 3*x^6 + 2*x^4 + x^2
```

Wooo!)

To end with a moral: the fact that we can manipulate series means that we hardly ever need to use Taylor's formula to find a power series for a standard function. Instead, we start from series we know and manipulate them to find the series we want. Maybe I should make this stronger:

> **Warning:** Never try to find a power series representation by taking higher derivatives! Instead, use the series you know in your sleep. This is especially true on tests!

That's a bit of an overstatement, of course, since we did use derivatives to find the series we know in our sleep. But that's the point: we did that because for those functions it can be done. For almost all other functions, taking many higher derivatives is hopeless.

Here are a few examples.

a. $g(x) = \displaystyle\int \frac{\sin(x)}{x}\,dx$

Let's do this using summation notation. First, as we saw above,

$$\frac{\sin(x)}{x} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!}.$$

Now we integrate term by term. The antiderivative of $x^k$ is $\dfrac{x^{k+1}}{k+1}$, and we apply this to each term to get

$$\int \frac{\sin(x)}{x}\,dx = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)\,(2n+1)!},$$

everything valid for all $x$.

b. $f(x) = \dfrac{x}{2+x^2}$

The closest series we know in our sleep is the geometric series

$$\frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots + u^n \ldots$$

We can take care of the $x$ in the numerator by just factoring it out, but we need a $1-u$ in the denominator. The trick is to also factor out a 2:

$$\frac{x}{2+x^2} = \frac{x}{2}\,\frac{1}{1+x^2/2}.$$

Now let $u = -x^2/2$ in the geometric series to get

$$\frac{1}{1+x^2/2} = 1 + (-x^2/2) + (-x^2/2)^2 + (-x^2/2)^3 + \cdots + (-x^2/2)^n + \ldots$$

which simplifies to

$$\frac{1}{1+x^2/2} = 1 - \frac{x^2}{2} + \frac{x^4}{2^2} - \frac{x^6}{2^3} + \cdots + (-1)^n \frac{x^{2n}}{2^n} + \ldots$$

This is valid when $|u| < 1$, so when $|x^2/2| < 1$, which translates to $|x^2| < 2$ and then to $|x| < \sqrt{2}$. Finally, multiply by $x/2$ to get

$$\frac{x}{2+x^2} = \frac{x}{2} - \frac{x^3}{2^2} + \frac{x^5}{2^3} + \frac{x^7}{2^4} + \cdots + (-1)^n \frac{x^{2n+1}}{2^{n+1}} + \ldots$$

valid when $|x| < \sqrt{2}$.

### 4.3.1 Problems

**Problem 4.3.1:** What is the series for the derivative of $\dfrac{1}{1-x}$? Where does it converge?

**Problem 4.3.2:** Check with series that the derivative of $e^x$ is $e^x$.

**Problem 4.3.3:** Check with series that the derivative of $\ln(1+x)$ is $\dfrac{1}{1+x}$.

**Problem 4.3.4:** Find the Taylor series for $x^2 e^x$. Where does it converge?

**Problem 4.3.5:** Find series for these functions. Remember that "find the series" means find the general term and also the region of validity.

a. $\dfrac{1}{1+5x^2}$

b. $e^{-3x^2}$

c. $\dfrac{1-\cos(x)}{x^2}$

d. $\ln(1+x^2)$

e. $\ln(2+x)$ (factor out a 2...)

f. $(x+1)\sin x$

g. $\dfrac{3}{2-x}$

**Problem 4.3.6:** In each case, give an expression in closed form for the infinite series, which are minor modifications of series you should (still!) know in your sleep.

a. $1 - x + \dfrac{x^2}{2!} - \dfrac{x^3}{3!} + \dfrac{x^4}{4!} - \ldots$

b. $1 - x^3 + x^6 - x^9 + \ldots$

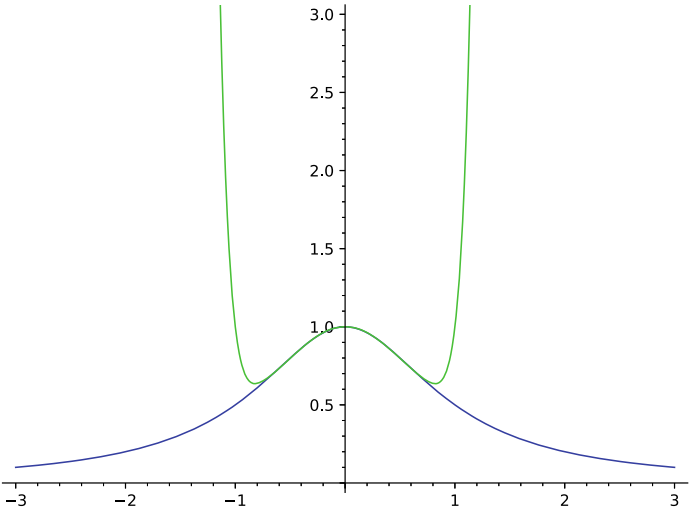
c. $\dfrac{x^2}{2} - \dfrac{x^3}{3\cdot 2} + \dfrac{x^4}{4\cdot 3} - \dfrac{x^5}{5\cdot 4} + \ldots$

**Problem 4.3.7:** Let

$$f(x) = \frac{1}{1+x^2}.$$

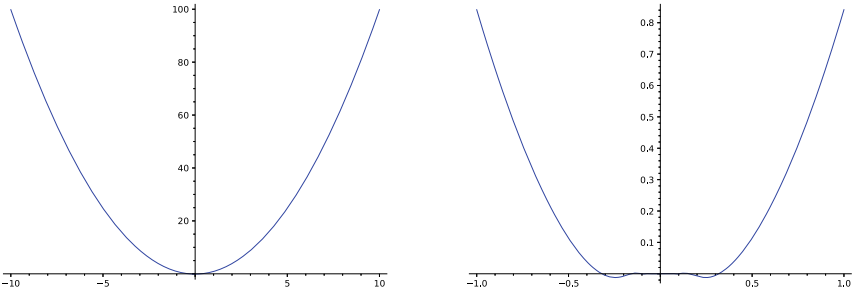
a. Find the Taylor series for $f(x)$ centered at $a = 0$.

b. The graph on the next page shows a graph of $f(x)$ and its Taylor polynomial of degree 12 (that is, the part of its Taylor series up to degree 12). Explain why the graph looks the way itdoes.




**Problem 4.3.8:** On a large scale, the graph of $y = x^3 \sin(\frac{1}{x})$ looks like a parabola. You can see that clearly in the graph on the left below. Use the Taylor series for $\sin(x)$ (centered at zero, as usual) to explain why.

(The first graph shows the region $-10 \leq x \leq 10$, a standard calculator window. In the graph on the right, we see the region $-1 \leq x \leq 1$, where it is clear that the graph is not really a parabola.)




Two graphs of $x^3 \sin(1/x)$

**Problem 4.3.9:** The electric potential generated by a uniformly charged disk of total charge $Q$ and radius $R$ is given by

$$F(y) = \frac{2Q}{R^2}\left(\sqrt{y^2 + R^2} - y\right),$$

where $y$ is the distance from the disk and $F(y)$ is the potential at distance $y$. We want to understand what happens when $y$ is very large, so do this:

a. Change variables from $y$ to $x = 1/y$, so that when $y$ is big, $x$ is close to zero.

b. Find a Taylor polynomial centered at $x = 0$ for the potential in terms of $x$. (Try to find ways to make your life easier. You can get the series from the ones you know in your sleep, but you need to fiddle a bit with the function first.)

c. Use this to justify the following claim: "at points far from the disk, the potential is approximately $Q/y$."

**Problem 4.3.10:** As you may remember, Einstein's general theory of relativity predicts that a light ray passing close to the sun will be bent by a small amount. To test that prediction, we need to know exactly how much bending to expect. Doing that requires solving an equation similar to

$$\sin\theta + b(1 + \cos\theta + \cos^2\theta) = 0.$$

We want to solve for $\theta$ in terms of $b$, but that is very hard to do.

In this situation we know that $b$ is a small positive number. If $b = 0$, then $\theta = 0$ is a solution, so it seems likely that there is a solution $\theta$ (depending on $b$) that is close to 0. Since $\theta$ is close to 0 we should be able to approximate $\sin\theta$ and $\cos\theta$ using a few terms of their Taylor series. Do this and find an approximate solution for $\theta$ in terms of $b$. (The first thing to decide is how many terms to use; base your decision on what equations you can solve.)

Finally, a few challenges for the brave:

**Challenge:** Check that $e^{x+y} = e^x e^y$ by manipulating the series directly. (You'll need to know how to expand $(x + y)^n$ to do this.)

**Challenge:** Find the first six terms of the series for $e^x \sin(x)$.

**Challenge:** Find the first few terms of the series for $e^{\sin(x)}$.

**Challenge:** Find the first few terms of the series for $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$. (This is very hard to do by hand no matter how you go about it.)

## 4.4 New functions

So far, we have started with a function $f(x)$, usually some "black-box" function which we cannot compute by hand, such as the sine or the exponential, and then found a power series representation for it.

What if we reverse the process? We might not be able to know what the resulting function is, but it will give us a function defined on the region of convergence. So, for example, we can invent a function called Charlie. (Why not?)

$$\text{Charlie}(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!}.$$

It's actually fairly easy to figure out a formula for Charlie($x$), but let's not. How much do we know just from the series?

First of all, we can use the ratio test to find the values of $x$ for which Charlie($x$) is defined. I'll leave it to you: check that the radius of convergence is infinite, that is, Charlie($x$) is defined for all $x$.

Next, it's clear that Charlie(0) $= 1$. That's the one value we can be sure about. Notice also that all the terms involve even powers of $x$ and there are no minus signs, so Charlie($x$) $\geq 0$ for all $x$.

For the derivative, we can also find a series by differentiating:

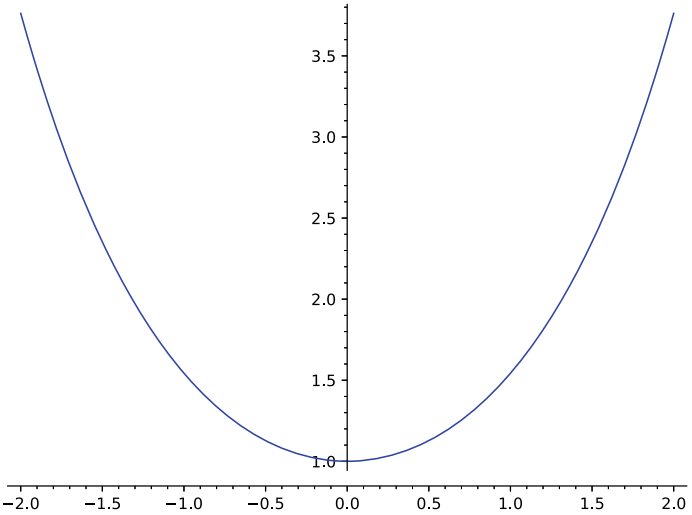
$$\text{Charlie}'(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots,$$

also convergent for every $x$. When $x > 0$ this is clearly positive, and when $x < 0$ it is negative, so Charlie($x$) is decreasing when $x < 0$ and increasing when $x > 0$. That means there is a minimum point at $x = 0$.

I'll leave it to you to check that Charlie$''(x) =$ Charlie($x$). That puts Charlie somewhere between the exponential (which is its own derivative) and sine/cosine (which are equal to their fourth derivatives).

Finally, can we plot it? Well sure; if we tell the computer to plot a long enough chunk of Charlie's series, we'll get a good picture:

```
sage: f(x)=sum(x^(2*n)/factorial(2*n),n,0,10)
sage: plot(f(x),(-2,2))
```




Of course, a natural question here would be who needs that function? If Charlie($x$) is important, why isn't there a Charlie button on my calculator? (Well, there is, but bear with me.)

The answer is that this function answers a real-world question: it gives the shape of a hanging chain when you hold it by its endpoints. People found that out because they knew some properties of the derivative of that curve, and the series can be derived from those properties.

The next two sections will show you two examples of how series are used in the real world, sparing you the gory details but trying to make the main point: you're learning this stuff because it is useful.

### 4.4.1 Problems

**Problem 4.4.1:** Figure out a formula for Charlie($x$) in terms of standard functions you know. Is there a "Charlie button" on your calculator? Is there a SageMath command?

**Problem 4.4.2:** The function Si($x$) is defined as

$$\mathrm{Si}(x) = \int_0^x \frac{\sin(t)}{t}\, dt.$$

("Si" stands for "sine integral.")

Notice that we do not know an antiderivative for $\sin(x)/x$. It can be proved that the antiderivative cannot be expressed in terms of the usual list of functions we know, so that in fact Si($x$) is a new function.

a. Find the Taylor series centered at $a = 0$ for Si($x$).

b. I want to use the series to compute Si(1) with an error smaller than $5\times10^{-6}$. Of course, I want to use as few terms of the series as possible. How many terms of the series do I need to use?

**Problem 4.4.3:** Another example of a function that does not have an antiderivative that can be written in terms of standard functions is $e^{-x^2}$. But we can always make an antiderivative by integrating: if

$$F(x) = \int_0^x e^{-t^2}\, dt,$$

then the fundamental theorem of calculus tells us that $F'(x) = e^{-x^2}$. Find a power series expansion centered around 0 for $F(x)$.

## 4.5 Doing things with power series 1

Many problems in applied mathematics lead to *differential equations*, that is, equations involving a function and its derivatives. In physics, for example, we often express the position as a function of time: $y = f(x)$. If so, $y'$ represents velocity and

$y''$ represents acceleration. Since force is equal to mass times acceleration, we often end up with an equation connecting $y$, $y'$, and $y''$.

Sometimes the resulting equation is easy to solve. The simple harmonic oscillator, for example, leads to the equation $y'' + \omega^2 y = 0$ and it's not too hard to see that both $y = \sin(\omega x)$ and $y = \cos(\omega x)$ are solutions. But sometimes we get something much more difficult to solve. When that happens, series can sometimes be used to find solutions.

This section is about the equation $xy'' + y' + xy = 0$, which also comes up in physics. It turns out that there is no easy way to find the right function $y = f(x)$ that makes this be true, but we can do it by using series. I'll work only on the case where we also know the initial value $f(0) = 1$.

Suppose $y = f(x)$. Then we are trying to find a function $f(x)$ that satisfies $f(0) = 1$ and

$$xf''(x) + f'(x) + xf(x) = 0$$

for all $x$. Since there is no obvious function that does this, we will find a series representation instead. The computation takes several steps, so gird up your loins.

Suppose

$$f(x) = C_0 + C_1 x + C_2 x^2 + \cdots + C_n x^n + \dots,$$

where the coefficients $C_n$ are still to be determined. We will try to find the coefficients by requiring that the function satisfy our differential equation.

**Step 1:** Since we know $f(0) = 1$, it is easy to determine $C_0$: just plug in zero on both sides. We get $C_0 = 1$.

**Step 2:** Since we are allowed to take derivatives of power series, we know that

$$f'(x) = C_1 + 2C_2 x + 3C_3 x^2 \cdots + nC_n x^{n-1} + \dots$$

and

$$f''(x) = 2C_2 + 3 \cdot 2C_3 x + 4 \cdot 3C_4 x^2 \cdots + n(n-1)C_n x^{n-2} + \dots$$

**Step 3:** We can now use our answers to write down the series for $xf''(x) + f'(x) + xf(x)$. The algebra is annoying but not awful. We have the three series for $f(x)$, $f'(x)$, and $f''(x)$. Multiplying by $x$ just changes the powers of $x$.

It's convenient to line up the series so that the terms involving the same power $x^n$ appear above each other. If you do that, you get

$$\begin{aligned}
xf(x) &= C_0 x + C_1 x^2 + C_2 x^3 + \dots && + C_{n-1} x^n + \dots \\
f'(x) &= C_1 + 2C_2 x + 3C_3 x^2 + 3C_4 x^3 + \dots && + (n+1)C_{n+1} x^n + \dots \\
xf''(x) &= 2C_2 x + 3 \cdot 2C_3 x^2 + 4 \cdot 3C_4 x^3 \dots && + (n+1)(n)C_{n+1} x^n + \dots
\end{aligned}$$

So the sum, which is the series we want to equal zero, has $n$-th term equal to

$$C_{n-1} x^n + (n+1)C_{n+1} x^n + (n+1)nC_{n+1} x^n = [C_{n-1} + (n+1)^2 C_{n+1}] x^n.$$

This doesn't work for $n = 0$ (since there's no $C_{-1}$, after all), but it's easy to see that the term of degree zero is just equal to $C_1$. So the series is

$$xf''(x) + f'(x) + xf(x) = C_1 + \sum_{n=1}^{\infty} [[C_{n-1} + (n+1)^2 C_{n+1}]x^n.$$

**Step 4:** Since we want $xf''(x) + f'(x) + xf(x) = 0$, this last series must have all of its coefficients equal to zero. Let's use this to find $C_1, C_2, C_3, C_4, C_5, C_6$.

The first thing to notice is that $C_1 = 0$. Next, let's look at the other coefficients. We have

$$C_{n-1} + (n+1)^2 C_{n+1} = 0$$

for every $n$. To make this easier, let $k = n + 1$ and rewrite it as

$$C_{k-2} + k^2 C_k = 0$$

or, even better,

$$C_k = -\frac{1}{k^2} C_{k-2}.$$

Now it's easy to use the formula to see that the fact that $C_1 = 0$ makes $C_3 = C_5 = 0$, and in fact that all the odd-numbered coefficients are zero.

For the others, we know $C_0 = 1$, and we have

$$C_2 = -\frac{1}{4} \qquad C_4 = \frac{1}{4 \cdot 16} \qquad C_6 = -\frac{1}{4 \cdot 16 \cdot 36}.$$

**Step 5:** The hard thing is finding a general pattern for the even-numbered coefficients! We know

$$C_k = -\frac{1}{k^2} C_{k-2},$$

however. Rewriting that for $C_{2k}$ gives

$$C_{2k} = -\frac{1}{(2k)^2} C_{2k-2} = \frac{-1}{2^2 k^2} C_{2k-2}.$$

Since we start from $C_0 = 1$ , the denominator for $C_{2k}$ will have $k$ factors of $2^2$ and then

$$k^2(k-1)^2(k-2)^2 \cdots = (k!)^2.$$

And the sign changes at each step. So

$$C_{2k} = (-1)^k \frac{1}{2^{2k}(k!)^2}.$$

Ooof.

Remembering that $C_{2k}$ is the coefficient of $x^{2k}$, the series for $f(x)$ is

$$f(x) = 1 - \frac{1}{4}x^2 + \frac{1}{64}x^4 - \cdots + (-1)^n \frac{1}{2^{2n}(n!)^2} x^{2n} + \ldots$$

Whenever this converges it will define a function that solves our differential equation.

Notice that this is not any of the series we already know, so when it converges the series will define a "new" function, i.e., one that is not yet in our repertoire.

**Step 6:** Using the ratio test, it's easy to show that the radius of convergence is $R = \infty$, so this series converges for every $x$. I'll let you check that this is right.

So in fact we have found a solution of our equation that is valid for every $x$. It is a function we don't know about yet, but in fact it is a standard function that appears in many places. The usual name for the function $f(x)$ is "the Bessel function of the first kind and of order 0." A common notation for it is $J_0$. In SageMath it is called `bessel_J(0,x)`.

Let's investigate $J_0(x)$ a little bit. Some things are easy. We know $J_0(0) = 1$, and since all the terms of the series involve even powers, we also know that $J_0(-x) = J_0(x)$. We can find a series for $J_0'(x)$ by differentiating this one term by term.

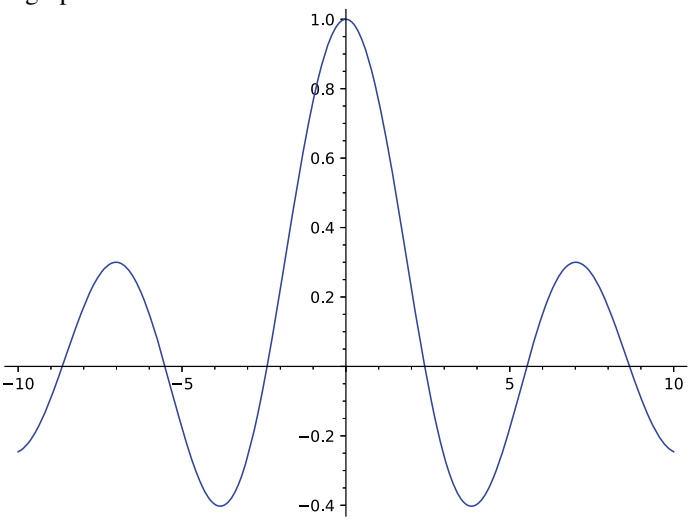
$$J_0'(x) = -\frac{1}{2}x + \frac{1}{16}x^3 - \cdots + (-1)^n \frac{1}{2^{2n-1}n!(n-1)!}x^{2n-1} + \dots,$$

still valid for all $x$. We can get a second derivative the same way. That doesn't tell us a whole lot, but it does allow us to check, for example, that $J_0(x)$ has a maximum at $x = 0$. On the other hand, the series doesn't allow us to find any other critical points, since we do not know how to solve $J_0'(x) = 0$.

We can see what the graph of $J_0(x)$ looks like (at least approximately) by plotting enough terms of the series. We don't really know how many terms are enough, so we need to experiment. Let's try going until $n = 20$, so until the term of degree 40.

```
sage:  ff(x)=sum((-1)^n*x^(2*n)/(2^(2*n)*factorial(n)^2),n,0,20)
sage:  plot(ff(x),(-10,10))
```

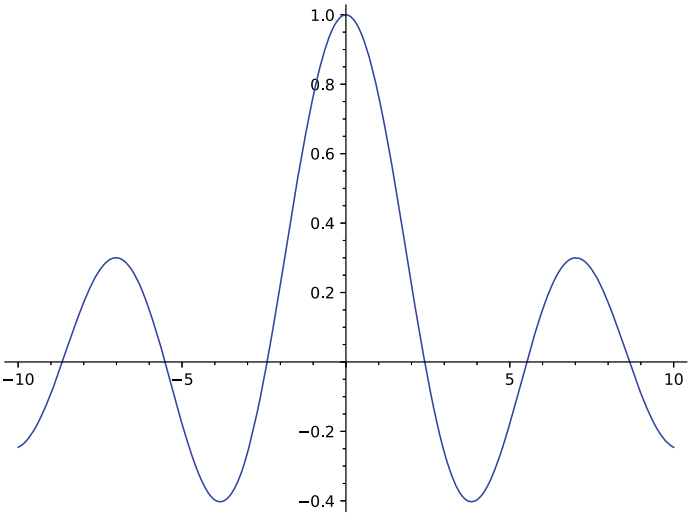
Here is the graph:



This looks something like a cosine function, but the amplitude is decreasing as we move away from $x = 0$. It's hard to see from this graph whether the peaks and troughs are getting father away from each other or not.

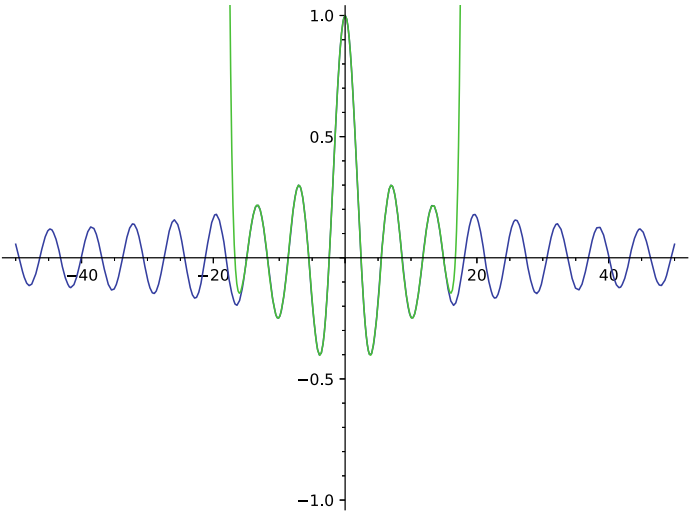
Since SageMath knows $J_0(x)$ itself, we can also ask it to plot that. The result looks a lot like the picture we just saw, not surprisingly.

```
sage: plot(bessel_J(0,x),(-10,10))
```




We should plot both the series and the function at once; unfortunately, between $-10$ and 10 the two look identical! So instead let's look from $-50$ to 50:

```
sage: plot([bessel_J(0,x),ff(x)],(-50,50),ymax=1,ymin=-1)
```



So the approximation is quite good in the middle of the graph, but to get a bigger range of $x$ we would need to work with more than 20 terms.

There is more than you would ever want to know about $J_0$ (and its friends and companions $J_n$) at Wolfram MathWorld. And even more at the National Institute of Standards' database of mathematical functions. (Is it surprising that a government agency has a database of mathematical functions? You might enjoy looking around that site.) But the point for us is just that we have found a solution to the differential equation using series, and it's a useful function in the real world.

## 4.6 Doing things with power series 2

One of the cool things that can be done with power series is to use them to encode sequences of numbers that we happen to be interested in. For example, the power series for $e^x$ encodes the numbers $1/n!$. Here are two examples from real (mathematical) life.

### 4.6.1 Fibonacci numbers

You have probably heard of the Fibonacci numbers:

$$f_0 = 1, \qquad f_1 = 1, \qquad f_n = f_{n-1} + f_{n-2} \qquad \text{for all } n \geq 2.$$

We can find them step by step: each time we add the last two numbers to find the next one:

$$1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21\,,35,\ \ldots$$

This sequence has all sorts of interesting properties and even shows up in nature sometimes.

How can we study these numbers better? Well, one standard trick is to use them to create a new function:

$$F(x) = \sum_{n=0}^{\infty} f_n x^n = 1 + x + 2x^2 + 3x^3 + 5x^4 + 8x^5 + 13x^6 + \ldots$$

If we can figure out things about $F(x)$ then we may get information about the numbers $f_n$.

But it's actually easy to find a formula for $F(x)$. The key thing to notice is that multiplying by $x$ shifts all the coefficients. So

$$xF(x) = \sum_{n=0}^{\infty} f_n x^{n+1} = \sum_{k=1}^{\infty} f_{k-1} x^k = x + x^2 + 2x^3 + 3x^4 + 5x^5 + 8x^6 + 13x^7 + \ldots$$

and

$$x^2 F(x) = \sum_{n=0}^{\infty} f_n x^{n+2} = \sum_{k=2}^{\infty} f_{k-2} x^k = x^2 + x^3 + 2x^4 + 3x^5 + 5x^6 + 8x^7 + 13x^8 + \ldots$$

This sets us up to use the recursion $f_k = f_{k-1} + f_{k-2}$: if we add the last two series we get

$$xF(x) + x^2F(x) = x + \sum_{k=2}^{\infty}(f_{k-1} + f_{k-2})x^k = x + \sum_{k=2}^{\infty} f_k x^k = F(x) - 1.$$

Solving for $F(x)$ gives

$$F(x) - xF(x) - x^2F(x) = 1$$

and so

$$F(x) = \frac{1}{1 - x - x^2}.$$

That's remarkable! The entire Fibonacci sequence is therefore hidden in the function

$$\frac{1}{1 - x - x^2}.$$

It's kind of amazing.

We can do a lot more from here. For example, if we factor $1 - x - x^2$ and then use the "partial fractions" method from integration, we can write $F(x)$ as the sum of two geometric series, whose convergence behavior we know well. We can then use this to get lots of information about the Fibonacci numbers. For example, we can find a non-recursive formula for $f_n$.

### 4.6.2 Generating functions

The process we have just gone through is known as finding a "generating function" for the sequence $f_n$. This idea is used all over mathematics. If we have a sequence we like, say

$$a_0, a_1, a_2, \ldots$$

we can build a generating function by using it as the coefficients of a power series

$$a_0 + a_1x + a_2x^2 + \ldots$$

Sometimes we use the $a_n$ slightly differently, putting factorials in the denominator; this gives the "exponential generating function"

$$a_0 + a_1x + \frac{a_2}{2!}x^2 + \frac{a_3}{3!}x^3 + \frac{a_4}{4!}x^4 + \ldots$$

If we can figure out a formula for either version, we can find out a lot more about the $a_n$.

Here's an example from probability theory. We often study probability distributions via their "moments." There is one moment $M_n$ for each positive integer $n$. Of course we pack all the moments into a generating function, in this case an exponential generating function

$$1 + M_1x + \frac{M_2}{2!}x^2 + \frac{M_3}{3!}x^3 + \frac{M_4}{4!}x^4 + \ldots$$

This is known as the MGF of our probability distribution (for "moment-generating function"), and one proves that if its radius of convergence is nonzero the MGF completely determines the probability distribution. So the MGF becomes a very powerful tool in the theory: if we want to prove that some distribution is what we think it is, it's enough to show that the MGFs are the same.

There is a famous book about all this called Generatingfunctionology, by Herbert S. Wilf. It explains many of the tricks involving generating functions. It's not an easy book, but if you're interested in taking a look it can be found online.




# 5 Distant Mountains

Looking at the horizon we see mountains in the distance. This chapter is a brief invitation to visit the mountains and explore. I have given few of the technical details and no proofs at all. Instead, I point out some interesting bits of scenery.

## 5.1 Complex numbers

As you probably know, complex numbers are numbers of the form $a + bi$ with $a$ and $b$ real numbers and $i$ being a symbol such that $i^2 = -1$. We add and multiply them using the usual rules plus the rule that $i^2 = -1$. For dividing, we use an extra trick: when we multiply $c + di$ by $c - di$ we get $c^2 + d^2$, which is a *real* number. Here are the formulas:

$$\begin{aligned}(a+bi) \pm (c+di) &= (a+c) + (b+d)i \\ (a+bi)(c+di) &= (ac-bd) + (ad+bc)i \\ \frac{a+bi}{c+di} &= \frac{(a+bi)(c-di)}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} - \frac{ad-bc}{c^2+d^2}i.\end{aligned}$$

So, to divide, we first multiply numerator and denominator by $c - di$ and then work out the rest.

A silly remark that may help avoid confusion: since $bi$ and $ib$ are the same thing, we can write $a + ib$ as well. We often do this when $b$ is a complicated symbol. So you will usually see $i\sin(x)$ instead of $\sin(x)i$. A complex *variable* is usually written as $z = x + iy$, with the convention that $x$ and $y$ stand for real numbers.[1]

If we have a complex number $a + bi$, we call $a$ its *real part* and $b$ its *imaginary part*. The usual notation is $\mathrm{Re}(a + bi) = a$ and $\mathrm{Im}(a + bi) = b$. Notice that both real and imaginary parts are real numbers.

An important operation with complex numbers is called *complex conjugation*, denoted by a bar: if $z = a + bi$ then $\overline{z} = a - bi$. One of the reasons conjugation is important is that it interacts well with the operations:

$$\overline{z+w} = \overline{z} + \overline{w} \qquad \text{and} \qquad \overline{zw} = \overline{z}\,\overline{w}.$$

There is also a way of measuring the "size" of a complex number:

$$|a+bi| = \sqrt{a^2+b^2}.$$

[1] I don't know why I tend to write $a+bi$ when it's $a$ and $b$ and $x+iy$ when it is $x$ and $y$, but that's what my fingers want me to do.

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F. Q. Gouvêa, *A Short Book on Long Sums*, Reading in Mathematics,
https://doi.org/10.1007/978-3-031-37557-6_5

I call this the absolute value of $a + bi$, but some people prefer to say "modulus" instead. Notice that if $z = a + bi$ we have

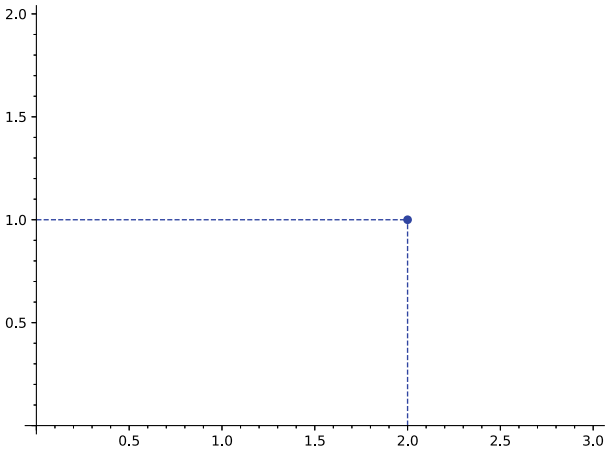
$$z\overline{z} = a^2 + b^2 = |z|^2.$$

In particular, it follows that $|zw| = |z|\,|w|$.

SageMath can work with complex numbers. It uses `I` for our $i$, but otherwise it's just as you would expect. Do these by hand to check that SageMath gets them right:

```
sage: (2+3*I)+(1-8*I)
-5*I + 3
sage: (2+3*I)*(1-8*I)
-13*I + 26
sage: (2+3*I)/(1-8*I)
19/65*I - 22/65
sage: abs(3+2*I)
sqrt(13)
sage: conjugate(3+2*I)
-2*I + 3
```

We can visualize complex numbers on a plane: just label the point $(a, b)$ with the complex number $a + bi$. Let's plot the point $2 + i$.




For obvious reasons, we then call the $x$-axis the "real axis" and the $y$-axis the "imaginary axis."

The basic operations all translate to plane geometry. By the Pythagorean theorem, the absolute value $|a + bi| = \sqrt{a^2 + b^2}$ is just the length of the segment from $(0, 0)$ to $(a, b)$. Adding two complex numbers is just the usual addition of vectors using the parallelogram law. One can interpret the product geometrically as well, but it's a little more complicated.

Since we can add and multiply complex numbers, it makes sense to plug them into a power series! And since we can measure how big a complex number is, we can make sense of convergence too (in fact, using exactly the same words as before). So, for example, we can define something like $e^{i\pi}$ to be what we get when we plug in $i\pi$ into the usual power series for $e^x$.

Let's try it. It helps to work out the powers of $i$ first. Since $i^2 = -1$, we have $i^3 = -i$ and $i^4 = 1$. And then $i^5 = i$ starts the cycle again. So the powers of $i\pi$ will be

$$1,\ i\pi,\ -\pi^2,\ -i\pi^3,\ \pi^4,\ i\pi^5\ \ldots$$

Plugging those in,

$$\begin{aligned} e^{i\pi} &= 1 + i\pi + \frac{(i\pi)^2}{2!} + +\frac{(i\pi)^3}{3!} + +\frac{(i\pi)^4}{4!} + \frac{(i\pi)^5}{5!} + \ldots \\ &= 1 + i\pi - \frac{\pi^2}{2!} - \frac{i\pi^3}{3!} + \frac{\pi^4}{4!} + \frac{i\pi^5}{5!} + \ldots \\ &= \left(1 - \frac{\pi^2}{2!} + \frac{\pi^4}{4!} + \ldots\right) + i\left(\pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!}\right), \end{aligned}$$

where at the end we grouped all the terms without an $i$ in one parenthesis and all the terms with an $i$ in the other. If you stare at that, you'll see that the first parenthesis ends up being what we get when we plug $x = \pi$ in the series for $\cos(x)$, while the second parenthesis is what we get when we plug in $x = \pi$ into $\sin(x)$. So both series converge and we have

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1,$$

which we can also write as

$$e^{i\pi} + 1 = 0.$$

Some people call this "the most beautiful formula in mathematics."

In fact, our computation would have worked just as well for $e^{iy}$, giving us $e^{iy} = \cos(y) + i\sin(y)$. Multiplying both sides by $e^x$ we get a formula for the exponential of any complex number:

$$e^z = e^{x+iy} = e^x(\cos(y) + i\sin(y)).$$

Once complex numbers are in play, then, there is a deep link between exponentials and trigonometric functions, which we discover using power series.

All of our theory works just as well—maybe even better—with complex numbers. In fact, every single one of the theorems we found for real power series is also true for complex power series. And the proofs are pretty much the same as well. The only one of our proofs that depends on working with series of real numbers is the theorem about absolute convergence. That theorem is true for complex numbers too, but the proof has to be different.

One pleasant thing about working with complex power series is that the radius of convergence is really a radius! After all, $|x + iy| < R$ translates to $\sqrt{x^2 + y^2} < R$

and so $x^2 + y^2 < R^2$, which says that $(x, y)$ is inside a circle of radius $R$. The big difference is that where before we had only two "endpoints," namely, $x = -R$ and $x = R$, now we have the entire circle $x^2+y^2 = R^2$, and while the series will converge inside the circle, the behavior at the boundary can be very complicated.

Sometimes the complex world affects what happens with real numbers. Here's an elementary example. We saw that

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \ldots \qquad \text{when } |x| < 1.$$

That's kind of strange, since the left-hand side is well defined for every real number $x$. Why does the series only work for $|x| < 1$? Well, in the *complex* world, if $x = i$ we have $1 + x^2 = 0$, so the function is not defined. Notice that $|i| = 1$. It turns out that a power series will always have *the same radius of convergence* whether we plug in real or complex numbers. So the discontinuity at $x = i$ forces the radius of convergence to be at most $R = 1$ even if we are only working with real numbers. Somehow the real series "knows" about the problem when $x = i$.

There are many very nice things about complex numbers. For example, the equation $e^{iy} = \cos(y) + i\sin(y)$ allows us to move from trigonometric functions to exponentials. Exponentials are much easier to work with! Many trig identities become very easy once we translate them to exponentials.

### 5.1.1 Problems

**Problem 5.1.1:** We pointed out above that if $a$ is some complex number and $R$ is a positive real number the inequality $|z - a| < R$ defines an open disk with center $a$ and radius $R$. Describe the regions defined by the formulas

a. $\mathrm{Re}(z) > 1$,

b. $\mathrm{Im}(z) > 0$,

c. $|z - 1| = |z - i|$,

d. $\overline{z} = z$.

**Problem 5.1.2:** It's fairly easy to see that $\overline{z + w} = \overline{z} + \overline{w}$. Check that $\overline{zw} = \overline{z}\,\overline{w}$. Explain how it follows that $|zw| = |z|\,|w|$.

**Problem 5.1.3:** If $z$ and $w$ are complex numbers, is it still true that $e^{z+w} = e^z e^w$? (Consider the first challenge problem on page 101.)

**Problem 5.1.4:** Use the fact that $e^{i\theta+i\varphi} = e^{i\theta}\, e^{i\varphi}$ to get formulas for $\cos(\theta + \varphi)$ and $\sin(\theta + \varphi)$.

**Problem 5.1.5:** Since adding two complex numbers is just vector addition on the plane, we can try to visualize a series the following way. Take a series

$$c_0 + c_1 + c_2 + \cdots + c_n + \ldots$$

where the $c_i$ are now complex numbers. Start at $(0, 0)$ and draw the segment to the point representing $c_0$. Now draw the vector corresponding to $c_1$ with its tail at the

point $c_0$. The head will be at $c_0 + c_1$, the first partial sum. Keep going. You will get a chain of vectors creating a kind of path in the plane. If the series converges, you will see the path approximate the limit.

a. Try this on the series for $e^{i\pi}$.

b. Try it for the geometric series $f(z) = 1/(1 - z)$ and $z = i/2$.

c. Try it for $f(z) = 1/(1 - z)$ and $z = (1 + i)/2$.

d. What happens if you try it for a value of $x$ for which the series diverges?

### 5.1.2 Notes

The fundamental role of power series in the theory of functions of a complex variable is a major theme of that theory. You can find a reasonably gentle account of this in [19, Chapters 25–27]. My two favorite books on the subject are [8] and [3], but reading them will take serious effort.

## 5.2 Series in general and the idea of uniformity

Power series are very useful, but we could certainly try other kinds of series. Sometimes we can use our theory to understand their behavior, sometimes we need new theory.

Here's an easy one. In problem 2.9.5 we looked at

$$\sum_{n=0}^{\infty} \left(\frac{3}{x}\right)^n = 1 + \frac{3}{x} + \frac{9}{x^2} + \frac{27}{x^3} + \dots$$

This is just a geometric series, since we are adding powers of $3/x$. So we know that it converges when $|3/x| < 1$, which just says $|x| > 3$. Convergence happens if we are far away from zero instead of in an interval centered at zero.

In this case, we also know the sum.

$$1 + \frac{3}{x} + \frac{9}{x^2} + \frac{27}{x^3} + \dots = \frac{1}{1 - 3/x} = \frac{x}{x - 3} \qquad \text{if } |x| > 3.$$

This is a fairly artificial example, since it's just a modification of a known series. But once things get more complicated it is much harder.

It turns out that considering series in general brings up issues that do not trouble us in the case of power series. This section focuses on one particular example of the most important issue.

We will work with this series:

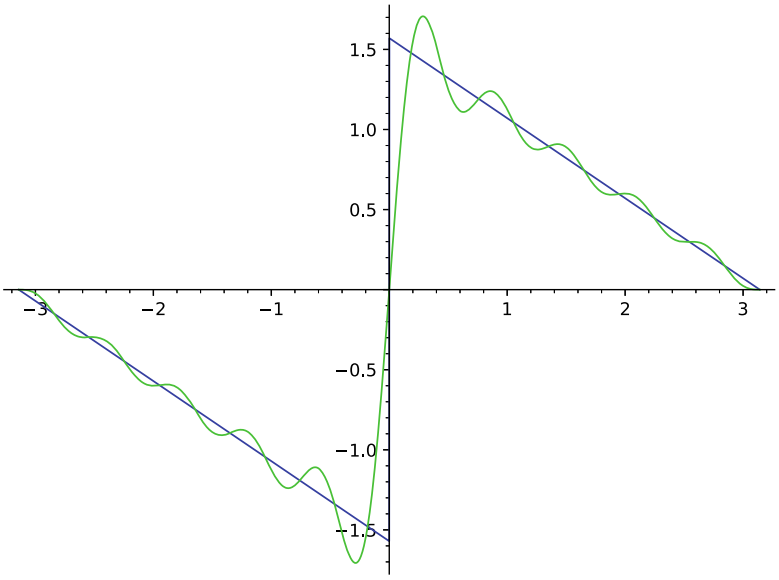
$$f(x) = \sin(x) + \frac{\sin(2x)}{2} + \frac{\sin(3x)}{3} + \frac{\sin(4x)}{4} + \dots$$

This is an example of a Fourier series, which we will discuss in the next section. We can ask the standard questions: does it converge, and if so what is the sum?

For each value of $x$, we get an infinite sum of numbers, which we can then study. When that series converges, we get some value $f(x)$. So the more precise question would be: for which values of $x$ does that series converge and to what?

That is hard to determine with bare hands, but we can use the theory of Fourier series to find the answer. The one obvious fact is that the series converges when $x = 0$, since every term is 0 in that case. (The same happens for all integer multiples of $\pi$.) It's also easy to see that since the terms all involve the sine function, everything will just repeat in intervals of $2\pi$. Once we know what happens between $-\pi$ and $\pi$ we know everything.

It turns out, in fact, that the series converges for every $x$. The sum $f(x)$ has to be defined piecewise: we already know $f(0) = 0$; for $-\pi \leq x < 0$, we have $f(x) = -\frac{1}{2}(x + \pi)$; for $0 < x \leq \pi$ it is $f(x) = \frac{1}{2}(x - \pi)$. For other values of $x$ we have to shift things by multiples of $\pi$. Let me plot that function from $-\pi$ to $\pi$ (blue) together with what we get by summing 10 terms of the series (green).




Several things here are very different from the power series we are used to. The most obvious one is that the sum is not a continuous function! In fact,

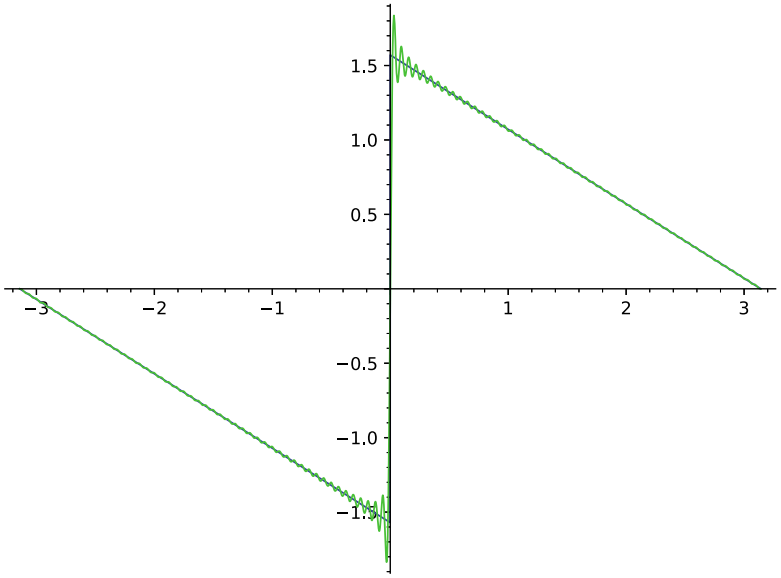
$$\lim_{x\to 0-} f(x) = -\frac{\pi}{2}, \qquad f(0) = 0, \qquad \lim_{(x\to 0+)} f(x) = \frac{\pi}{2}.$$

Every function given by a power series is not only continuous, but in fact infinitely differentiable. So this is very different behavior.

Since the series is made with sine waves, the partial sums are always going to be wavy. But since the limit is a function made up of straight lines, we expect the waviness to get smaller and smaller.

Near zero the function needs to make a big jump from the left-hand limit $-\pi/2$ through 0 to the right-hand limit $\pi/2$. As if it was getting ready to jump, the graph starts to oscillate more and overshoots slightly, dipping below the limit before the jump and reaching above it just after.

To see that something strange is happening let's make a longer partial sum. Here is what we get by adding 100 terms. The jump is so close to vertical that it disappears behind the $y$-axis, but we can see the rest.

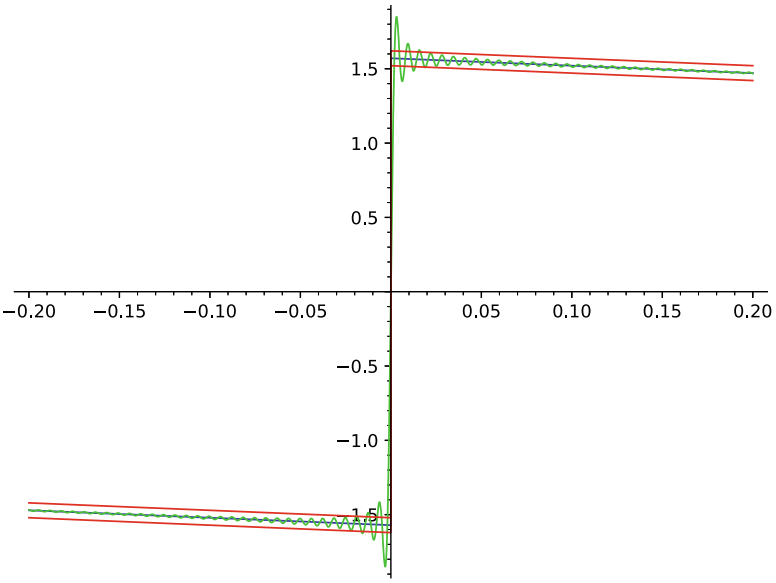



Away from zero we pretty much just see the straight lines. But here's a weird thing: near zero, the wavy part has gotten much narrower, *but the size of the overshoot is almost the same*. We can prove that it never goes away!

Imagine that we are adding more and more terms of the series. You are watching the results from a point near zero, say $x = 0.3$. At first you are in the wavy part, but as we add more and more terms the wavy part gets thinner. Eventually, it gets so thin that it doesn't affect you at $x = 0.3$; from then on you are pretty much on the straight line. So the series converges to $f(0.3)$, as it should.

But no matter how many terms you add the overshoot continues to exist, affecting points much closer to zero. So no matter how many terms we add, at some points the partial sums will not be really close to the function.

One way to describe what is going on is this. Suppose you want to get within 0.05 of the true value of $f(x)$. Let's choose a small window, say $-0.2 \le x \le 0.2$, and plot $f(x) + 0.05$ and $f(x) - 0.05$ in red. We could call the values between $f(x) - 0.05$ and $f(x) + 0.05$ the Goldilocks Zone. In green, we plot a really big partial sum, say 1000 terms. This is what we get near $x = 0$:




For most values of $x$ the partial sum is inside the Goldilocks Zone, but the overshoot still happens for $x$ very near zero. If you want to make the difference less than 0.05 for some chosen very small $x$, you need to add even more terms of the series. But for $x$ even closer to zero the overshoot will still exist. So the closer to zero you get, the more terms you need to get the partial sum to fall within the zone.

With power series, this never happens. If you focus on an interval $-a \le x \le a$ inside the region of convergence, you can force the entire partial sum into a band surrounding the sum. This is described by saying that the convergence is *uniform* in that interval: you can make the difference between partial sums and the final sum small *everywhere at once*. In our series that doesn't happen. The convergence is not uniform in the interval $-\pi \le x \le \pi$. We say that the series converges *pointwise*, for each value of $x$, but not uniformly.

There is a huge difference between uniform convergence and pointwise convergence. For example, if all the terms are continuous functions and the convergence is uniform, then the sum is continuous too. But, as in our example, if we only have pointwise convergence it is perfectly possible for the sum to have jumps. Similar problems appear when we try to integrate or differentiate series. We can do it for power series because the convergence is always uniform in every closed interval. But for more general series we need to check whether the convergence is uniform.

### 5.2.1 Problems

**Problem 5.2.1:** We noticed above that

$$1 + \frac{3}{x} + \frac{9}{x^2} + \frac{27}{x^3} + \cdots = \frac{1}{1 - 3/x} = \frac{x}{x - 3} \qquad \text{if } |x| > 3.$$

What does the Taylor series centered at $a = 0$ for this function look like? Where does it converge?

**Problem 5.2.2:** (A nontrivial example.) For which $x$ does the series

$$\sum_{n=1}^{\infty} \frac{x}{n(1+nx^2)}$$

converge? Is the convergence uniform?

**Challenge:** Here's a strange one. Look at the series

$$\frac{1+x}{1-x} + \frac{2x}{x^2-1} + \frac{2x^2}{x^4-1} + \frac{2x^4}{x^8-1} + \dots$$

where the exponents are powers of two, so the numerators are $2x^{2^n}$ and the denominators are $x^{2^{n+1}} - 1$. If you plot a partial sum, you will see that the series seems to converge to 1 when $|x| < 1$ and to $-1$ when $|x| > 1$. Can you prove that?

### 5.2.2 Notes

The theory of uniform convergence is one of the important topics in any course in real analysis. In particular, you can find careful definitions and proofs in most real analysis textbooks. The "overshoot" we saw in our example is known in the theory of Fourier series as the Gibbs phenomenon.

## 5.3 Periodic functions and fourier series

Power series are the simplest kind of series representation for functions, but it can be argued that the most useful representation is the one given by Fourier series. The example in the previous section is a Fourier series, so we already know that they are harder to work with than power series: the convergence is not always uniform. Since, however, Fourier series are so useful, in this section we give a very brief survey of the main ideas.

This kind of infinite series is named for the French mathematician Joseph Fourier. He was working on the problem of how heat moves through a solid metal plate. The problem led to a complicated differential equation that was easier to solve if a key function looked like $\sin(nx)$ or $\cos(nx)$ for some integer $n$. Solutions could be added (or "superposed") to give a new solution, so the problem could be solved for functions like

$$c_1 \sin(x) + c_2 \cos(2x) + c_3 \sin(3x),$$

where $c_1, c_2, c_3$ are numbers. Fourier decided to think about *infinite* sums of that kind, and asserted that *any* function would be equal to a sum of such a series. Using that, he could solve the problem in general.

But was it true? What kind of functions are included in that "any"? What kind of convergence does one get? If we can write a function as a Fourier series, how do we

find the coefficients? There was a lot to investigate. This sections gives you a quick glimpse of the answers.

We will be working with series whose terms involve sines and cosines, so the results will be periodic. So we start with a function $f(x)$ which is periodic. To simplify things,[2] let's suppose the period is $2\pi$, as for the sine and cosine, so that we are working with functions such that $f(x + 2\pi) = f(x)$ for any $x$. We only need to know what $f(x)$ is for $-\pi \leq x \leq \pi$, since that determines the entire function. We will often just give a formula valid for $-\pi \leq x \leq \pi$ and assume that the rest of the function is determined by the rule $f(x + 2\pi n) = f(x)$ for every integer $n$. This is called "extending by periodicity."

The kind of series we want to find looks like this:

$$f(x) = a_0 + a_1 \cos(x) + b_1 \sin(x) + a_2 \cos(2x) + b_2 \sin(2x) + \ldots$$

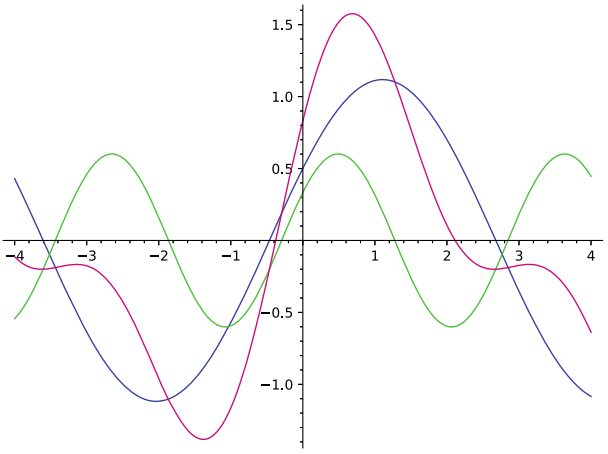
where the $a_n$ and $b_n$ are numbers. Notice the convention of using $a_n$ for the coefficient of the cosine and $b_n$ for the coefficient of the sine. Since $\cos(0) = 1$, the constant term gets called $a_0$. (Warning: In many texts the constant term is $\frac{1}{2}a_0$, for reasons we touch on below.)

We could write the series in summation notation too:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)).$$

So our model functions are sines and cosines and we are trying to express general periodic functions in terms of those.

One way to understand this is to notice that each chunk has a shorter period, and so a higher frequency.




Two waves and their superposition.

[2] It isn't hard to modify the theory to allow for other periods, but we will ignore that issue.

- $a_0$ is constant. In the example above we let $a_0 = 0$.
- $a_1 \cos(x) + b_1 \sin(x)$ is a wave with period $2\pi$.

  The blue graph shows $\frac{1}{2}\cos(x) + \sin(x)$. It has period $2\pi$.
- $a_2 \cos(2x) + b_2 \sin(2x)$ is a wave with period $\pi$.

  The green graph is $\frac{1}{3}\cos(2x) + \frac{1}{2}\sin(2x)$, which has period $\pi$, so higher frequency. The red graph is the sum of blue and green, which has period $2\pi$.
- $a_3 \cos(3x) + b_3 \sin(3x)$ is a wave with period $2\pi/3$.

  I didn't add a wave of period $2\pi/3$ to the graph because it got too messy. But it's easy to play with SageMath if you want to.
- And so on: $a_n \cos(nx) + b_n \sin(nx)$ is a wave with period $2\pi/n$.

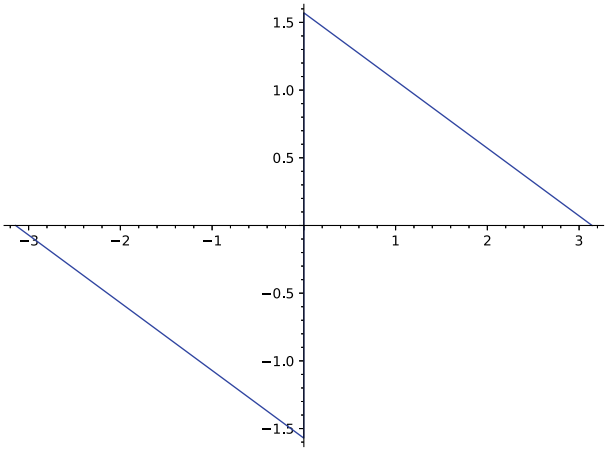
So we can think of each term as being an oscillation with a certain frequency, and we are trying to break up a more complicated oscillation into simpler ones with higher and higher frequency. The more complicated wave $f(x)$ is therefore the sum of many simpler ones. The simple pieces are often called the "harmonics" of the more complicated oscillation.

When the vibration we are studying is a sound wave, all of this is literally true. The overall sound is created by superposing a whole bunch of simple sine and cosine waves. The overall sound of an instrument is determined by how these simple pieces are put together. Sometimes it's even possible to figure out which instruments were used to create a certain sound by studying the harmonics.[3]

We saw an example of this in the previous section: the sum of sines

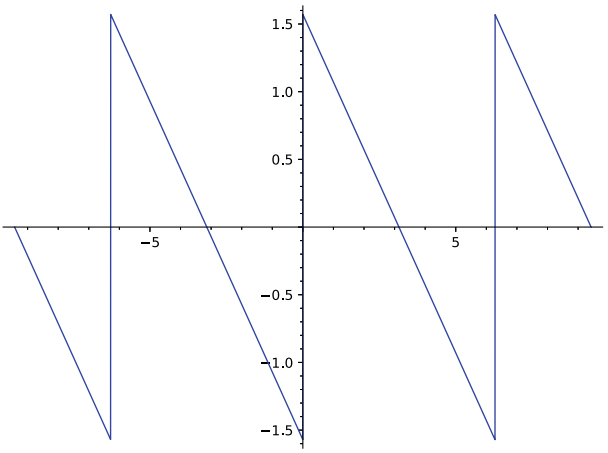
$$\sin(x) + \frac{\sin(2x)}{2} + \frac{\sin(3x)}{3} + \frac{\sin(4x)}{4} + \dots$$

converged to




[3] In 2008 a mathematical analysis of the opening chord of the Beatles' "A Hard Day's Night" made the news. The technique used was a generalization of Fourier series.

That plot only shows the picture from $-\pi$ to $\pi$, but things involving sines are always periodic, so the picture will repeat in intervals of length $2\pi$, so the full picture of the sum is the "sawtooth" function




SageMath draws vertical lines at the jumps; mathematics professors usually prefer dotted lines.

Of course, we want to do this in the opposite direction: from the complicated oscillation $f(x)$ we want to somehow extract the coefficients $a_n$ and $b_n$ that will produce a series converging to $f(x)$. Somehow the sawtooth function should tell us to take $a_n = 0$ and $b_n = 1/n$.

In the case of power series, we did this using derivatives and plugging in zero. That can't work here, since $\cos(0) = 1$: plugging in zero just leaves us with the entire cosine part. No fun. Plus, our functions don't even need to be continuous, so they may not have derivatives for us to work with.

Amazingly, the trick is to use integrals. You may remember that integrating sine and cosine over a whole period always gives zero. Indeed, for any integer $n > 0$ we have

$$\int_{-\pi}^{\pi} \cos(nx)\,dx = \int_{-\pi}^{\pi} \sin(nx) = 0.$$

So if we integrate the equation

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

term by term from $-\pi$ to $\pi$, we get zero for all the terms with $n \geq 1$ and are left with

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} a_0\,dx = 2\pi a_0.$$

(Well, we get that assuming that we are allowed to integrate this kind of series term by term. That may not be true. But let's just plow on and hope for the best.)

If you believe that last formula, you can now see how to get the constant term:

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

How about the other terms? Well, more integrals. It turns out that if $m$ and $n$ are positive integers we have

$$\int_{-\pi}^{\pi} \sin(nx) \sin(mx)\, dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

$$\int_{-\pi}^{\pi} \cos(nx) \cos(mx)\, dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

$$\int_{-\pi}^{\pi} \sin(nx) \cos(mx)\, dx = 0.$$

(If you are good with integrals and trig identities, try to prove those.)

If we multiply $f(x)$ by either $\sin(mx)$ or $\cos(mx)$ we get a series with terms that look like $\sin(nx)\sin(mx)$ and friends. If we then integrate, most of the integrals are zero. Specifically, the only nonzero integrals is the one (only one!) where the trigonometric functions are the same and $n = m$. For example, if we integrate $f(x)\cos(3x)$, then the only term in the sum with a nonzero integral will be $a_3 \cos(3x)\cos(3x)$, with all the others having integral equal to zero.

The upshot (work out the details if you want to) is that, for any integer $n \geq 1$, we have

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx$$

and

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx.$$

If we can do all those integrals (a big if!) then we end up with a candidate for a trigonometric series for our function $f(x)$. It is called the *Fourier series* of $f(x)$.

(Notice that in the formula for $a_0$ we divide by $2\pi$ and in the other formulas we divide by $\pi$. This is why some books prefer to have

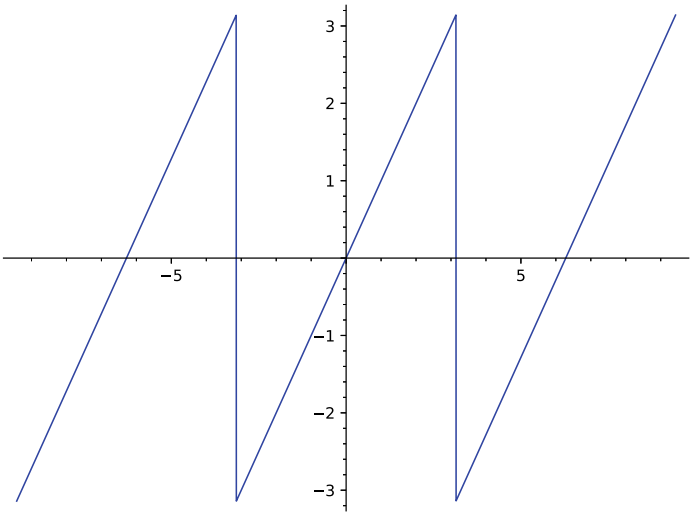
$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx,$$

which forces them to write the series starting with $\frac{1}{2}a_0$.)

Fourier originally claimed that the Fourier series of any function would converge to that function, but it wasn't at all clear what he meant by "any" nor what kind of convergence he had in mind. This helped force mathematicians to give a careful definition of "function," after which it was clear that "any" is wrong. After all, if we can't integrate $f(x)\sin(nx)$ then we can't even find the coefficients.

The eventual conclusion was that the Fourier series of a nice enough function $f(x)$ will converge to $f(x)$ for almost all values of $x$. Sorting out the meaning of that sentence occupied mathematicians for a century after Joseph Fourier. What properties make a function "nice enough"? What does "almost all" mean? When is the convergence uniform? Answers were eventually found, but they required more advanced mathematics. If you want to know more, look at the references we give below. All I will do here is show you an example.

Since we will need to compute integrals, let's choose a really simple function. We'll let $f(x) = x$ for $-\pi < x \leq \pi$ and then extend it periodically using $f(x+2\pi n) = f(x)$. This $f(x)$ is another kind of "sawtooth" function:




Let's compute the $a_n$ and $b_n$ coefficients. The first group is easy. Remember that the integral of an odd function in an interval from $-b$ to $b$ is always zero. Since $f(-x) = -x = -f(x)$ in the interval from $-\pi$ to $\pi$, we have

$$\int_{-\pi}^{\pi} f(x)\,dx = 0,$$

and so $a_0 = 0$. Since $\cos(-x) = \cos(x)$, multiplying our function by $\cos(nx)$ gives another odd function, so we see that $a_n = 0$ for all $n \geq 0$. So all the coefficients of the cosines are zero, i.e., the series will only involve sine functions.

Finding the $b_n$ is a bit harder because we actually need to integrate $x\sin(nx)$. The key is to remember integration by parts. Notice that the antiderivative of $\sin(nx)$ is

$-\frac{1}{n}\cos(nx)$.

$$\begin{aligned}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx &= \int_{-\pi}^{\pi} x\sin(nx)\,dx \\ &= \left[-\frac{x\cos(nx)}{n}\right]_{-\pi}^{\pi} + \frac{1}{n}\int_{-\pi}^{\pi}\cos(nx)\,dx \\ &= -\frac{2\pi}{n}\cos(n\pi) + 0 \\ &= (-1)^{n+1}\frac{2\pi}{n}.\end{aligned}$$

(Notice that $\cos(n\pi) = (-1)^n$ for any integer $n$.) Since we need to divide by $\pi$ to get the $b_n$, we end up with

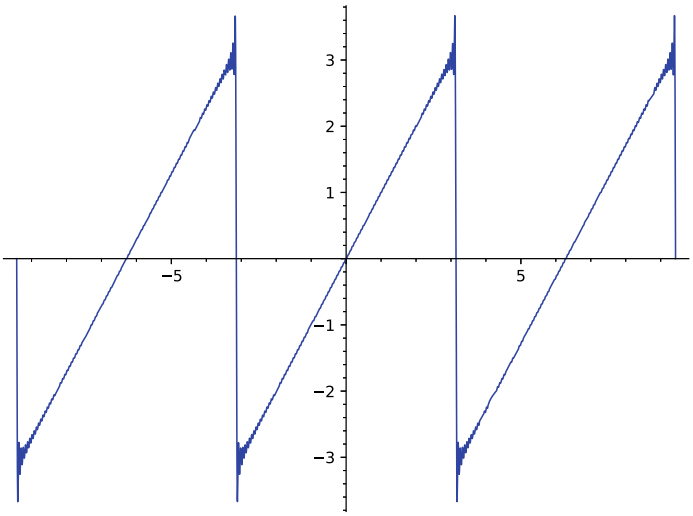
$$b_n = (-1)^{n+1}\frac{2}{n}.$$

This gives the series

$$2\sin(x) - \sin(2x) + \frac{2}{3}\sin(3x) - \frac{2}{4}\sin(4x) + \ldots$$

When $x = \pm\pi$ this is clearly equal to zero while we had $f(x) = \pm\pi$. So there is no convergence at those points.

But for all other values of $x$ the series turns out to converge to $f(x)$. Here's a plot of what we get by adding the first 100 terms:




We see that the issue we noted in the previous section happens here as well, but away from the jumps everything is working as expected. We have decomposed our sawtooth as the sum of infinitely many sine waves.

### 5.3.1 Problems

**Problem 5.3.1:** What happens if you try to find a Fourier series for $f(x) = \sin(x)$?

**Problem 5.3.2:** The "square wave" is the function equal to $-1$ when $-\pi < x < 0$ and equal to 1 when $0 < x < \pi$. Find its Fourier series.

**Problem 5.3.3:** Let $f(x) = x^2$ for $-\pi \leq x \leq \pi$ extended periodically. Find the Fourier series for $f(x)$ and plot a partial sum to see if it looks like $f(x)$.

### 5.3.2 Notes

Fourier series are a huge subject. Many calculus books give a brief introduction to them, but giving a detailed account with proofs requires knowing more. Most books on real analysis provide the basic theory. David Bressoud's *A Radical Approach to Real Analysis* [7] uses the questions raised by Fourier's claims as the main source of motivation. My favorite references for Fourier analysis are [14] and [20], but I'm a mathematician. There are many other options ranging from very applied to very theoretical.

## 5.4 Dirichlet series

This final glimpse of a distant mountain is very different. Power series and Fourier series are fundamental tools of applied mathematics and engineering. Dirichlet series are mostly of interest to mathematicians (and some physicists). They are particularly important in number theory, which is why I love them so much.

You have already met a Dirichlet series. We looked at series whose terms are the reciprocals of powers of $n$.

$$1 + \frac{1}{2^x} + \frac{1}{3^x} + \frac{1}{4^x} + \frac{1}{5^x} + \dots$$

When $x = 1$ this is just the harmonic series, which we know diverges. These series converge when $x > 1$ and diverge when $x \leq 1$. (You solved all the problems in Section 3.7.1, right?) We really only wrote them down for integer values of $x$, but everything works fine for other real numbers $x$ as well, as long as $x > 1$. So we can think of the sum of this series as a function of $x$: for each $x > 1$, we define

$$\zeta(x) = 1 + \frac{1}{2^x} + \frac{1}{3^x} + \frac{1}{4^x} + \frac{1}{5^x} + \dots$$

The squiggle $\zeta$ is a zeta, the Greek letter corresponding to $z$. It looks like a $z$ whose diagonal stroke grew too big.

People have studied $\zeta(x)$ since the eighteenth century. One of Leonhard Euler's first mathematical successes came when he found out that $\zeta(2) = \pi^2/6$. He eventually found a formula for the value of $\zeta(x)$ whenever $x$ was an *even* integer. He tried very hard to get a formula that worked for odd integers, but was never able to.

The zeta function has many interesting and mysterious properties. Let me just mention one surprising thing about it. We can write $\zeta(x)$ as a product:

$$\zeta(x) = \left(1 - \frac{1}{2^x}\right)^{-1} \left(1 - \frac{1}{3^x}\right)^{-1} \left(1 - \frac{1}{5^x}\right)^{-1} \left(1 - \frac{1}{7^x}\right)^{-1} \dots$$

What we have on the right-hand side is an infinite *product*, with one factor for each prime number. (Yes, Virginia, there are infinite products as well as infinite sums.) This means that the zeta function is somehow hiding information about prime numbers inside it. For example, letting $x$ approach 1 in this formula gives a proof that there are infinitely many prime numbers.

To get a general Dirichlet series, we replace the ones in the numerator with some other sequence. In general, a Dirichlet series looks like

$$f(x) = a_1 + \frac{a_2}{2^x} + \frac{a_3}{3^x} + \frac{a_4}{4^x} + \frac{a_5}{5^x} + \dots$$

If we can control how fast the numerators $a_n$ grow, we can decide for which range of $x$ this converges, and so get a new function. The region of convergence is always a half-line $x >$ something.

Here's an easy example. For each positive integer $n$, let $d(n)$ be the number of positive divisors of $n$. So $d(1) = 1$, because 1 is the only divisor of 1, $d(2) = d(3) = 2$ and similarly for every prime number, but $d(4) = 3$ because $1, 2, 4$ are all divisors of 4. Now look at

$$\sum_{n=1}^{\infty} \frac{d(n)}{n^x} = 1 + \frac{2}{2^x} + \frac{2}{3^x} + \frac{3}{4^x} + \frac{2}{5^x} + \frac{4}{6^x} \dots$$

Since $d(n)$ is quite small compared with $n$, it turns out this also converges whenever $x > 1$. The sum is actually $\zeta(x)^2$; can you see why?

The real usefulness of these series, however, requires allowing $x$ to be a *complex* number. In that case the region of convergence is always a half-plane $\mathrm{Re}(x) >$ something. For example, the series for the zeta function converges for all complex numbers $x$ with $\mathrm{Re}(x) > 1$.

It's over the complex numbers that things get really interesting… but it's also hard, and this short book is already getting too long, so I'll leave you with that.

### 5.4.1 Notes

Dirichlet series are discussed in books on analytic number theory, such as [1] and [21]. Most such books are hard going and assume at least some knowledge of complex analysis. There have, however, been several attempts to explain the zeta function. I like [9] and [17]. The zeta function features as one of the "Millennium Problems" (see [10]), surely the hardest way to make a million dollars.

# A SageMath: A (Very) Short Introduction

Mathematical software is part of the standard toolkit of mathematicians. Throughout this book, I will use the program called SageMath [18] (sometimes called *Sage*, but there are various other software systems called "Sage") to do computations. The goal of this appendix is to provide some basic orientation on the program, including where to get it. SageMath is free; it can be used online or installed on your own machine; it is very powerful, and it takes some time to learn. SageMath incorporates the functionality of many other free mathematics programs, including Maxima, Octave, R, GAP, and GP. SageMath runs either in a browser window or a terminal.

Of course, there are other mathematical software tools. *Desmos*, *Geogebra*, *Maple*, *Mathematica*, and MATLAB are well known and quite powerful. The people who make *Mathematica* are also behind Wolfram Alpha, which can do lots of mathematics as well. Many students appear to be familiar with *Desmos* and *Geogebra*. These can all do some of the things we need to do, but I have decided to focus on SageMath.

## A.1 General information

SageMath is an ambitious attempt to create powerful mathematical software that is free and open source. The ultimate ambition, says the Sage home page, is to create "a viable free open source alternative to *Magma*, *Maple*, *Mathematica* and MATLAB." (I'd say that they are very close to that and in some aspects well beyond.) The development approach emphasizes openness: while William Stein is the leader of the team, contributions have come from across the mathematical community.

SageMath can be used through a web interface, without needing to download and install the program. There are two ways to do that: either SageMath Cell or the more elaborate interface based on projects and notebooks offered by *CoCalc*. Most of my students find SageMath Cell is sufficient for everything they need to do.

It is also possible to download and install the program on your own computer. When you do, you can run SageMath in a terminal window or you can run it in your browser. The latter is much like using *CoCalc*, but the program is running on your local machine.

Of the two web interfaces, SageMath Cell is particularly easy to use for small computations. It presents the user with a big blank rectangle where one can type in Sage commands. Below is a button labeled "Evaluate," which does exactly that. The output appears below. The downside is that SageMath just evaluates what is in the

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F. Q. Gouvêa, *A Short Book on Long Sums*, Readings in Mathematics,
https://doi.org/10.1007/978-3-031-37557-6

box. It will not remember what you did, so if you define a symbol and then press the "Evaluate" button, you cannot use it again without repeating the definition. There are two very useful features of SageMath Cell that deserve note. First, it works on a tablet or phone. Second, because SageMath incorporates other open-source mathematical software, SageMath Cell will also allow you to use those: right under the big blank rectangle, there is a "Language" drop-down that you can use to do this.

The SageMath home page offers a link to create a *CoCalc Instant Worksheet*, but if you plan to use it regularly and want to save your work from one session to the next you should create a (free) account. (There are also paid CoCalc accounts if you want/need more features.) Once you log in to your account, you can create projects, and each project can contain many notebooks. Notebooks allow you to enter lines of Sage code, which are evaluated when you hit Shift-Enter. Definitions and results are remembered within each session.

If you are going to use the program a lot, then you might want to download and install it. It takes quite a bit of space, but having it on your own machine avoids connectivity issues.

The central web site for SageMath is www.sagemath.com. You can go there to download the program, find documentation, and so on.

## A.2 First examples

It's time to show you some examples. You can find many more in A Tour of Sage, which is available online. SageMath is built using the Python programming language, so its syntax is similar to Python's.

Everything is done by entering a command and then asking SageMath to evaluate it. In SageMath Cell, you can enter a sequence of commands as well. If you enter

```
3 + 5
```

into SageMath Cell and hit "Evaluate" you will get

```
8
```

in the results box. SageMath Cell typically only prints your last result. If you want to see two results, it's best to ask it explicitly:

```
print(3+5)
print(57.1^100)
```

Then you get two lines:

```
8
4.60904368661396e175
```

(In the last answer, `e175` means $\times 10^{175}$.) You can get some space between the two lines of output by adding `print(" ")` between the two commands.

In the terminal window, SageMath gives you a prompt and you type in your commands and then hit enter. The same example looks like this:

```
sage: 5+3
8
sage: 57.1^100
4.60904368661396e175
```

Most of the examples we give have been done in a terminal window (for one thing, it is easier to cut and paste from the terminal), so you will see prompts and responses.

The CoCalc or Jupyter Notebook interface is like the terminal, but it allows for graphical output. You enter commands and hit shift-enter to execute them. If you define an object with a command like `v=vector([1,2,3])` in CoCalc, the terminal, or a notebook running on your own computer, SageMath will remember what `v` is so that you can use it again later.

SageMath requires you to declare that a letter is a variable before using it that way. The only variable it knows by default is $x$. So, to work with functions of two variables $x$ and $y$ you will need to begin with

```
sage: var('x y')
(x, y)
```

If you use a variable, say $d$, that hasn't been defined, you get an error message. SageMath error messages are usually long and cryptic, but the key information is usually at the bottom; this one will end with something like "name 'd' is not defined." (A brand-new feature of CoCalc allows you to ask ChatGPT what is wrong with your code.)

Here is one way to define a function (everything in SageMath can be done in lots of different ways).

```
sage: f(x)=x^4
sage: f
x |--> x^4
sage: f(x)
x^4
```

The command `f(x)=x^4` produces no output, but SageMath has been told what you mean by `f`. Notice also that SageMath distinguishes the function `f` (the rule that sends $x$ to $x^4$) from `f(x)` (the value of $f$ when you plug in $x$). All the grown-ups do this, but I haven't done it in this book.

You can compute limits.

```
sage: f(x)=(x-2)/(x+1)
sage: limit(f(x),x=1)
-1/2
sage: limit(f(x),x=oo)
1
sage: limit(f(x),x=-1)
Infinity
```

Notice that you can use `oo` (two lowercase os) to mean infinity.

Two ways to compute a derivative:

```
sage: diff(f,x)
x |--> 4*x^3
sage: diff(f(x),x)
4*x^3
```

(On SageMath Cell, you need to define $f$ each time, remember.) The first one is the derivative *function* $f'$, the second is $f'(x)$. Higher derivatives are easy too. Here's the second derivative:

```
sage: diff(f,x,2)
x |--> 12*x^2
sage: diff(f(x),x,2)
12*x^2
```

The command `diff(f,x,x)` is equivalent to `diff(f,x,2)`. SageMath also uses (in fact, prefers) the object-oriented convention:

```
sage: f.diff(x)
x |--> 4*x^3
sage: f(x).diff(x)
4*x^3
sage: f(x).diff(x,4)
24
```

One advantage of this syntax is that either in the terminal window or in a notebook you can write `f.` and then hit the tab key. SageMath will then open a pop-up window telling you the many things you can do to the function $f$.

Here's how to integrate:

```
sage: f(x).integrate(x)
1/5*x^5
sage:
sage: f.integrate(x)
x |--> 1/5*x^5
sage: f(x).integrate(x)
1/5*x^5
sage: integrate(f(x),x)
1/5*x^5
sage: integrate(f(x),x,2,3)
211/5
sage: f(x).integrate(x,2,3)
211/5
```

The last one is the definite integral

$$\int_2^3 f(x)\,dx.$$

Let's try a hard one:

```
sage: g(x)=integrate(sqrt(x)*sqrt(1+x),x)
sage: print(g)
x |--> 1/4*((x + 1)^(3/2)/x^(3/2)
   + sqrt(x + 1)/sqrt(x))/((x + 1)^2/x^2
   - 2*(x + 1)/x + 1) - 1/8*log(sqrt(x + 1)/sqrt(x) + 1)
   + 1/8*log(sqrt(x + 1)/sqrt(x) - 1)
```

The first command produces no output, so I used `print` to tell SageMath to show me the result. The big mess is printed out as a single line, but I have added line breaks here.

Notice that that answer uses the function `log`. That means the *natural logarithm*; SageMath does understand what `ln` means, but it defaults to using `log`, which is what most mathematicians do. Indeed,

```
sage: ln(2)
log(2)
sage: ln(2).n()
0.693147180559945
```

SageMath tries to give you an exact answer when it can, but you can force an approximate answer in decimal form by using the `n` command.

On SageMath Cell or a notebook, you can get better-looking output by using `show` instead of `print`. For example, in SageMath Cell if we enter

```
x = var('x')
g(x)=integrate(sqrt(x)*sqrt(1+x), x)
show(g(x))
```

and hit evaluate, we get something like:

$$\frac{\frac{(x+1)^{\frac{3}{2}}}{x^{\frac{3}{2}}}+\frac{\sqrt{x+1}}{\sqrt{x}}}{4\left(\frac{(x+1)^2}{x^2}-\frac{2(x+1)}{x}+1\right)}-\frac{1}{8}\log\left(\frac{\sqrt{x+1}}{\sqrt{x}}+1\right)+\frac{1}{8}\log\left(\frac{\sqrt{x+1}}{\sqrt{x}}-1\right)$$

There is also `latex(f(x))`, which is what I did to get the LaTeX code to typeset the result. (If you use `show` in the terminal, you get the LaTeX code.)

While `show` produces output that is nicer to look at, the output of `print` is easier to cut and paste. In general, it's best to ask SageMath to either `print` or `show` the outputs you want to see.

Maybe there's a way to simplify that monster? One way is to try

```
sage: g(x).simplify_full()
1/4*(2*x + 1)*sqrt(x + 1)*sqrt(x) - 1/8*log((sqrt(x + 1)
  + sqrt(x))/sqrt(x)) + 1/8*log((sqrt(x + 1) - sqrt(x))/sqrt(x))
```

(I have added a line break.)

If we ask SageMath to `show` that, we get

$$\frac{1}{4}(2x+1)\sqrt{x+1}\sqrt{x} - \frac{1}{8}\log\left(\frac{\sqrt{x+1}+\sqrt{x}}{\sqrt{x}}\right) + \frac{1}{8}\log\left(\frac{\sqrt{x+1}-\sqrt{x}}{\sqrt{x}}\right)$$

I guess that is a little better. There are various different versions of `simplify`:

```
f.simplify()
f.simplify_full()
f.simplify_log()
f.simplify_trig()
f.simplify_rational()
```

The first one doesn't usually help; the others can help depending on the situation. See the manual for more details.

Sometimes you also want to use `factor()` and `expand()` to get formulas to look right. Occasionally, `simplify` and others will tell you that they need more information (e.g., "is $x > 0$?"). In that case, you get a long error message that tells you what to do at the end (most SageMath error messages give their most useful information at the bottom). You can use `assume(x>0)` or similar commands to provide the needed information.

For example, if you wanted SageMath to simplify $\log(\sqrt{x})$ into $\frac{1}{2}\log(x)$, you would need `assume(x>0)`. You might also need to use `simplify_log`. That might help make the formula for $g(x)$ above a little less intimidating.

## A.3 Plotting

SageMath can also plot functions. For example

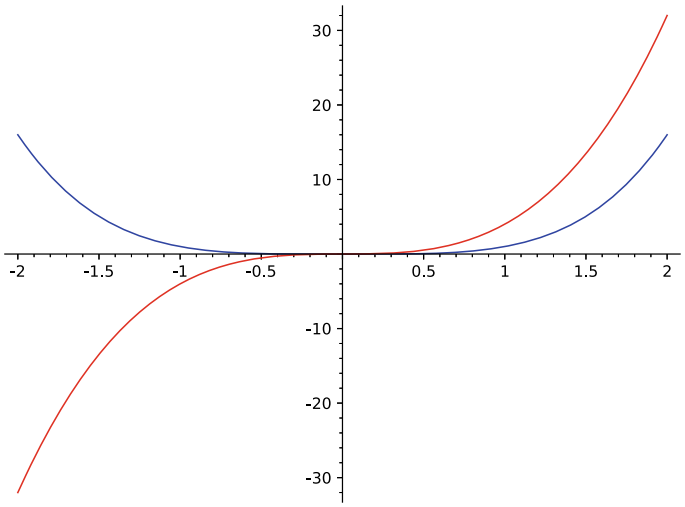
```
sage: plot(f(x),(-2,2))
```

will display a plot of our function. (In a terminal window, SageMath will use your computer's default graphics program to show you the result.) Plotting functions have a zillion options; type `?plot` for documentation and examples. I particularly enjoy playing with the various options related to color.

You can also give your plot a name and then do things to it.

```
sage: P1=plot(f(x),(-2,2),color="blue")
sage: P2=plot(diff(f(x),x),(-2,2),color="red")
sage: P=P1+P2
sage: show(P)
```

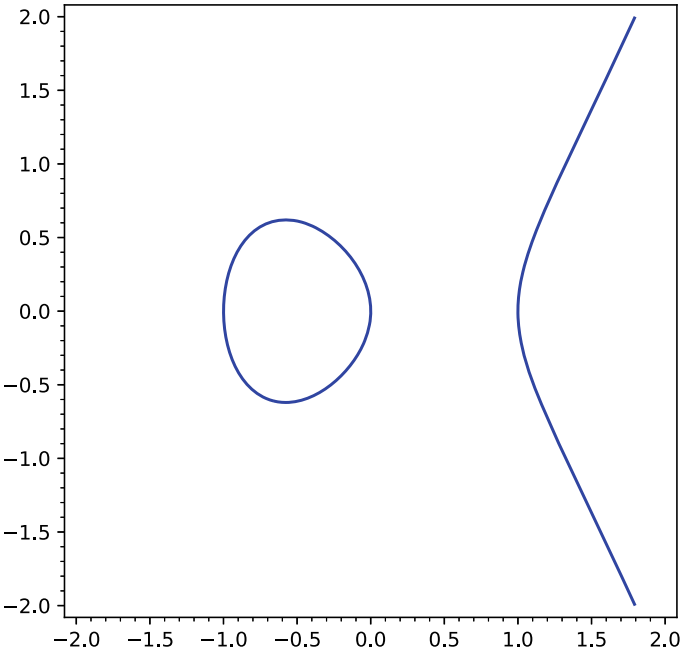
Produces this:




There are lots of other plotting commands. To plot a curve given by an equation like $y^2 = x^3 - x$, do this in the cell server:

```
var('x y')
implicit_plot(y^2==x^3-x,(x,-2,2),(y,-2,2))
```

(Note the double equals sign!) The graph looks like this:



How did I get the plots into this LaTeX file? I used

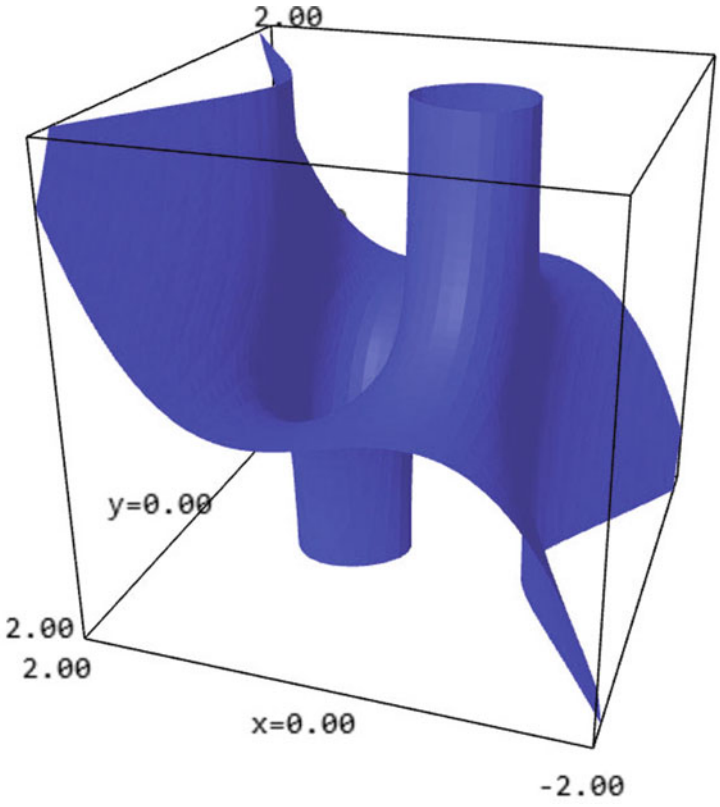
```
sage: P.save('example.eps'}
```

to create an encapsulated postscript (eps) file, which I then included into my file. The `save` command will use the format indicated by your file name, so you can choose the graphics format of your choice.

Sage will make three-dimensional graphs as well. For example, let's look at the equation in three variables $y^2z = x^3 - x$. If we plot the points $(x, y, z)$ that satisfy that equation, we'll get a surface in three-dimensional space. Here's how to get SageMath to draw it for you.

```
var('x y z')
implicit_plot3d(y^2*z==x^3-x,(x,-2,2),(y,-2,2),(z,-2,2))
```

This gives a rotatable graph of the surface defined by that equation. The graph displays in a browser window; if you are using the terminal, it will open a new browser window. It looks something like this:



## A.4 Sums

Since this is a book about sums, it's good to know how to handle them in SageMath. Of course, you can just write them out.

```
sage :1+1/2+1/3+1/4+1/5
137/60
```

That becomes hard to do when you have lots of terms, however. So, let's declare a variable $k$ and use it to describe the general term of the sum instead:

```
sage: var('k')
k
sage: sum(1/k,k,1,5)
137/60
```

(I won't repeat the definition of $k$ again; if you are working with SageMath Cell, it should be done each time.)

Now we can do a larger sum just as easily:

```
sage: sum(1/k,k,1,20)
55835135/1551950
```

SageMath always prefers to give the exact answer, which in this case is a fraction. But we usually want the decimal approximation instead. The easiest way to do this is to make SageMath work with decimals throughout, by replacing 1 with 1.0:

```
sage: sum(1.0/k,k,1,20)
3.597739657143682
```

You can even try an infinite sum:

```
sage: sum(1/k^2,k,1,oo)
1/6*pi^2
```

But that doesn't always work. It certainly fails when the sum does not converge.

# B Why I Do It This Way

The theory of infinite series is a beautiful part of analysis and it is used throughout mathematics. Series are used a lot in applied mathematics, physics, and engineering. Nevertheless, I have heard from many mathematics majors (at Colby and elsewhere) that this part of their calculus courses had made no sense to them. Worse, they said that they did not remember any of it. That convinced me that another approach was needed. Whether they will remember this one, I don't really know, but I hope at least it works a little better.[1] I also hope that it will lead to lasting understanding that will persist beyond the final exam.

The usual approach is the logical one: before we can talk of series of functions, we need to consider numerical sequences, define convergence, define series, and so on. I am convinced, however, that *the logical order is not necessarily the pedagogical order*. So, things got rearranged and some topics were jettisoned.

It seems to me, first of all, that if we start with a numerical series we are essentially answering a question the students do not (yet) want to ask. Why would one want to add infinitely many numbers in the first place? It also seems foreign to what they have learned so far. Isn't calculus supposed to be about functions? Where are the derivatives and integrals we were learning about so far?

Another annoyance with the traditional sequence are the abundant "does it converge?" problems. Students often don't see the point of saying "it converges" without finding what it converges to. And they are right! Knowing that a series converges is not an answer; rather it's the basis for a question: what is the sum? Euler spent a long time thinking about that question for the zeta function.

Clearly, I was looking for a narrative, one that started from things that were recognizably calculus and led to infinite series. Power series provided such a narrative. That gave me an organizing principle: if I don't need this result in the elementary study of power series, then I don't need this result at all.

I decided to start from the idea that the derivative provides the best linear approximation to a (differentiable) function. Since many calculus courses do not emphasize this idea as much as I would like, I dedicated an entire chapter to it.[2] This also serves to set up the rules for studying approximations, introducing reference points, increments, error terms, etc. I later build on this when I define derivatives in the multivariable context, so it's valuable to have.

Trying to improve the linear approximation leads naturally to polynomial approximations and the main theorem on Taylor Polynomials. The simple examples (exponential, sine, cosine, and geometric) then lead to questions of convergence. At this

[1] It works in my calculus classes, but that may just reflect the fact that I like it.

[2] Instructors may be tempted to skip this chapter. My recommendation is not to do this.

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point, I do enough of the theory of convergence to allow students to find the radius of convergence for (simple) power series, and so to define new functions.

This way of doing things is not easier for the students: the idea of approximation remains foreign to most elementary mathematics instruction. Students often ask me about the goal. Why are we doing this? Where are we heading? To answer those questions I introduced the idea of decimal expansions as analogous to power series expansions. Depending on the students, this idea may get more emphasis in class than in the notes.

Throughout the book, I have used "the function $f(x)$" rather than being careful about the distinction between a function and its value. This is the language used in most calculus textbooks and it is the language students understand at this point. It seemed too heavy-handed to insist on the correct language right in the first paragraph of the introduction. Instructors should feel free to point out this "abuse of language" to students, of course. One might use SageMath, which distinguishes between $f$ and $f(x)$, to make the point more vivid.

I believe students should understand that theorems depend on assumptions, but I'm not sure that students at this level should worry about things like "twice differentiable" and friends. I have sometimes given precise assumptions, but at other times relied on more informal language such as "nice enough." I'm sure instructors will be able to provide more precise assumptions if they feel students can benefit from that.

In many places, I have given justifications and arguments for why certain things are true. Students in my target audience are not ready for formal proofs, but I have tried to move them in that direction. My informal rule for informal proofs is that what I say should reflect the content of the proof. In other words, I have tried not to lie.

Like many (most?) teachers of mathematics, I have shamelessly borrowed from others, both in the exposition and in the problem sets. I am sure that I do not remember all my sources. I know I have been influenced by George Welch, Ralph Boas (especially the articles on infinite series in [4]), T. W. Körner's *Calculus for the Ambitious* [13], the first editions of the calculus textbook [12] by Hughes Hallett et. al., and *Calculus Problems for a New Century* [11], ed. by Robert Fraga. I am deeply grateful for their insights and ideas. In the *Instructor's Guide*, I have provided precise credits for the problems whenever I could.

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